

Project Data Page:

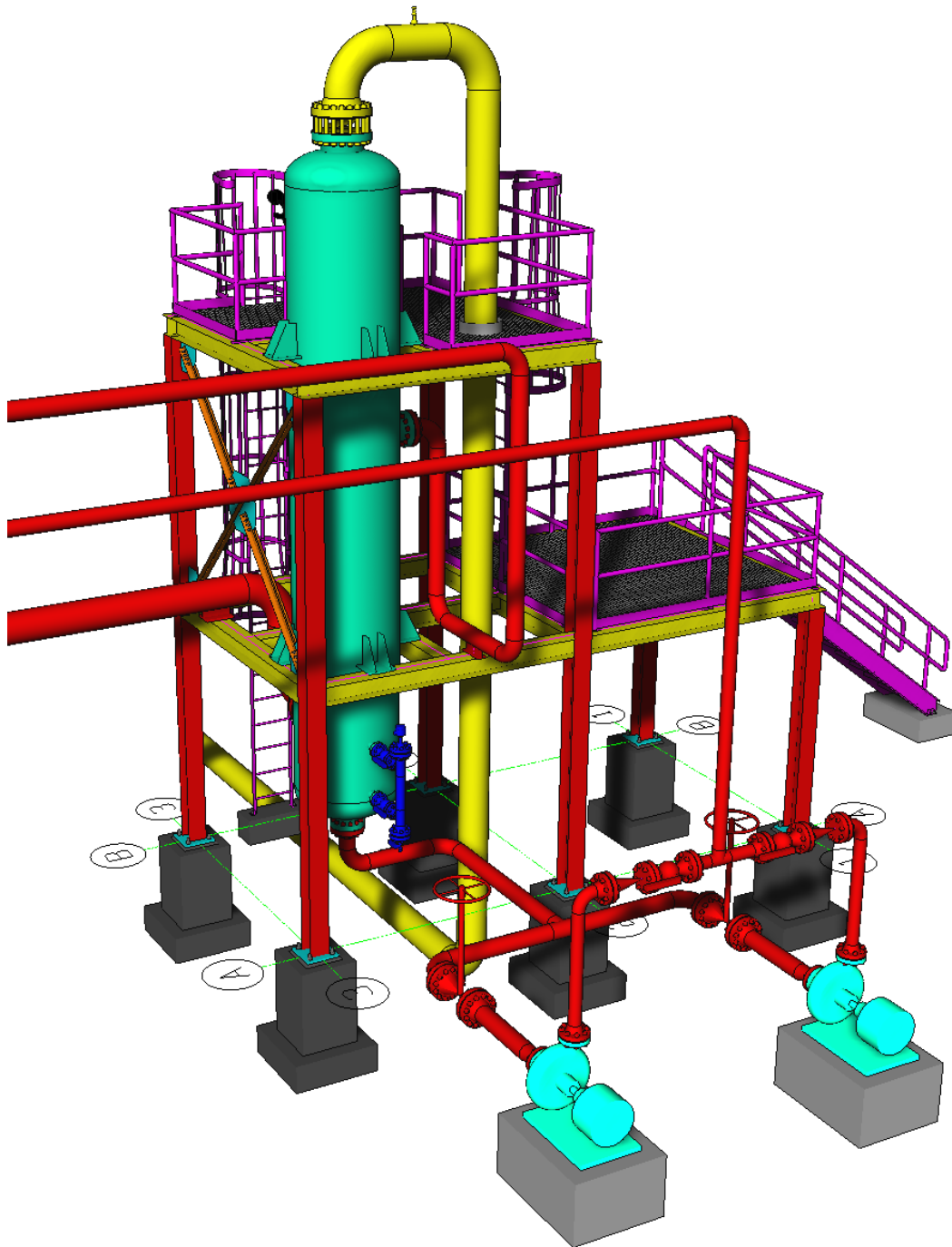




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\410-E-31007Main case.pvdb

Date of Analysis : May 21, 2026 5:33pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1200.000 x 1200.000 x 2300.000 mm ID
 Vessel Design Length, Tangent to Tangent 1094.92 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 80 °C
 Channel Side Design Temperature 80 °C
 Shell Side Design Pressure 23.454 kgf/cm²
 Channel Side Design Pressure 163.158 kgf/cm²
 Shop Shell Side Test Pressure 31.922 kgf/cm²
 Shop Channel Side Test Pressure 220.530 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N
Head	SA-516 70	...	...	K02700	+N
Cone	SA-516 70	...	...	K02700	+N
Flange	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-350 LF2	1	...	K03011	
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Tubes	SA-334 1	...	...	K03008	+IT
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-320 L43	...	<= 100	G43400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	+N+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
TS END COVER	163.158	0.00	167.000	1.5000	N/A	No
TS FLG1	163.158	0.00	165.000	1.5000	N/A	No
CH. SHELL	163.278	0.00	173.000	1.5000	N/A	No
SH.1	23.574	0.00	39.000	1.5000	N/A	No
SH.2	23.684	1.05	24.000	1.5000	N/A	No
SH.3	23.684	1.05	27.000	1.5000	N/A	No
DISH END	23.684	1.05	24.000	1.5000	No	No

Liquid Level: 230.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
--------------	--------------	-------------------	----------------------	-----------------------	----------------------	----------------------	----------	----------



Body Flg	30.6	30.3	311.0	303.0	296.3	...	1.00	1.00
Body Flg	70.4	39.8	298.0	293.0	286.3	...	1.00	1.00
Cylinder	195.9	125.5	68.0	68.0	64.1	...	1.00	1.00
Cylinder	256.9	30.0	16.0	16.0	10.2	...	1.00	1.00
Conical	447.9	191.0	21.0	21.0	20.7	...	1.00	1.00
Cylinder	1089.9	642.0	21.0	21.0	18.1	...	1.00	1.00
Ellipse	1094.9	5.0	21.0	17.0	16.4	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
20	30	303	198.704	...	No Calc
30	40	293	286.283	...	No Calc
40	Tsht	68	No Calc	...	151.005
Tsht	50	68	No Calc	...	No Calc
50	60	16	No Calc	...	31.1031
60	70	21	8.9296	1.05461	10.2094
70	80	21	12.3401	1.05461	4.65354
80	90	17	6.91643	1.05461	8.32114

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
20	30	No Calc	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	125.497	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	30.0032	No Calc	No Calc	No Calc
60	70	191	191	No Calc	No Calc
70	80	666.167	2710.82	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
NOZZLE N1	WRC-297	0.946	Shell	Passed
NOZZLE N2	WRC-297	0.946	Shell	Passed
Noz N4	WRC-297	0.846	Shell	Passed
Noz N5	WRC-297	0.846	Shell	Passed
Noz N3	WRC-297	0.848	Shell	Passed
Noz N6	WRC-297	0.854	Shell	Passed
Noz N7	WRC-297	0.854	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

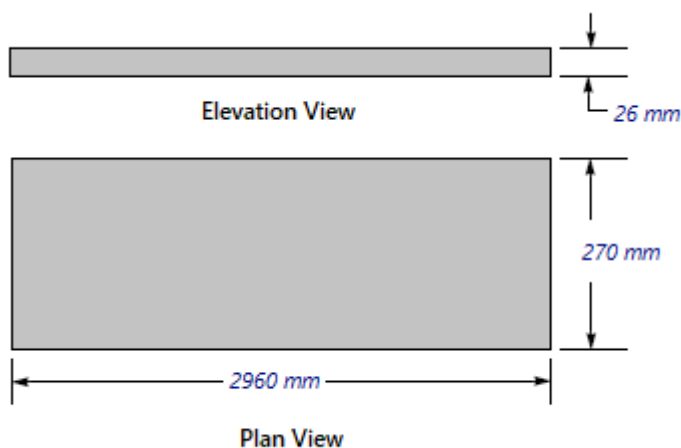
Saddle Width	240.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1500.000 mm
Wear Pad Width	480.000 mm
Wear Pad Thickness	12.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	550.000 mm

Baseplate Length	2960.000 mm
Baseplate Thickness	26.000 mm
Baseplate Width	270.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	22.000 mm



Web Thickness	22.000	mm
Height of Center Web	306.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	74689.02	kgf
Maximum Transverse Saddle Shear Load	3443.48	kgf
Maximum Longitudinal Saddle Shear Load	6886.95	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	70047.52	kgf
Maximum Transverse Saddle Shear Load	774.07	kgf
Maximum Longitudinal Saddle Shear Load	458.91	kgf

Weights:

Fabricated - Bare w/o removable internals	45218.4	kg
Shop Test - Fabricated + water (full)	79058.0	kg
Shipping - Fab. + rem. intls.+ shipping app.	45448.4	kg
Erected - Fab. + rem. intls.+ insul. (etc)	45448.4	kg
Empty - Fab. + intls. + details + wghts.	45448.4	kg
Operating - Empty + operating liquid (no ca)	75733.7	kg

Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
NOZZLE N1	173	...	...	OK	OK	Passed
NOZZLE N2	173	...	...	OK	OK	Passed
Noz N9	241	...	...	OK	OK	Passed
Noz N8	39	...	...	OK	OK	Passed
Noz N4	25	OK	...	OK	OK	Passed
Noz N5	28	OK	...	OK	OK	Passed
Noz N3	28	OK	...	OK	OK	Passed
Noz N6	27	OK	...	OK	OK	Passed
Noz N7	28	OK	...	OK	OK	Passed
Noz K2A	28	OK	...	OK	OK	Passed
Noz K2B	28	OK	...	OK	OK	Passed
Noz K3	28	OK	...	OK	OK	Passed
Noz K4	28	OK	...	OK	OK	Passed
Noz K5	28	OK	...	OK	OK	Passed
Noz K1A	25	OK	...	OK	OK	Passed
Noz K1B	24	OK	...	OK	OK	Passed
MH1	24	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 24 Nozzle : Noz K1B [Shellside]
 Minimum MAWP Nozzles : 173 Nozzle : NOZZLE N2 [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
40	NOZZLE N1	1329.175	90.000	195.175
40	NOZZLE N2	1329.175	-90.000	195.175
40	Noz N9	0.000	0.000	22.925
50	Noz N8	2419.175	-90.000	22.925
70	Noz N4	5079.175	90.000	124.437
70	Noz N5	7679.175	90.000	124.437
70	Noz N3	6579.175	30.823	124.437
70	Noz N6	10579.175	-90.000	24.425
70	Noz N7	6579.175	-58.646	28.400
70	Noz K2A	4614.175	30.823	26.900
70	Noz K2B	4614.175	-30.823	26.900
70	Noz K3	7679.175	30.823	20.550
70	Noz K4	5979.176	90.000	26.900
70	Noz K5	6779.175	90.000	26.900
80	Noz K1A	0.000	90.000	26.900
80	Noz K1B	0.000	270.000	26.900
80	MH1	0.000	0.000	275.363

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite 28 Licensee: SL User
FileName : 410-E-31007_Main case
Nozzle Summary:

Step: 41 5:33pm May 21,2026

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HEXAGON

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Noz K3	1.500 in	Actual	LWN	2.760	16.000	...	...	246.85	300
Noz N9	2.000 in	160	WNF	2.375	8.738	160.32	12.00	237.29	1500
Noz N8	2.000 in	160	WNF	2.375	8.738	160.32	12.00	166.76	300
Noz N6	2.000 in	160	WNF	2.375	8.738	160.32	12.00	291.40	600
Noz N7	2.000 in	Actual	LWN	3.299	16.500	...	...	392.53	600
Noz K2A	2.000 in	Actual	LWN	3.299	16.500	...	...	251.42	300
Noz K2B	2.000 in	Actual	LWN	3.299	16.500	...	...	251.42	300
Noz K4	2.000 in	Actual	LWN	3.299	16.500	...	...	251.76	300
Noz K5	2.000 in	Actual	LWN	3.299	16.500	...	...	251.76	300
Noz K1A	2.000 in	Actual	LWN	3.299	16.500	...	...	232.14	300
Noz K1B	2.000 in	Actual	LWN	3.299	16.500	...	...	232.14	300
Noz N4	10.000 in	80	WNF	10.750	15.088	373.05	12.00	259.13	300
Noz N5	10.000 in	80	WNF	10.750	15.088	473.05	12.00	259.13	300
Noz N3	10.000 in	80	WNF	10.750	15.088	473.05	12.00	352.41	300
NOZZLE N1	18.000 in	120	None	18.000	34.925	...	...	724.01	...
NOZZLE N2	18.000 in	120	None	18.000	34.925	...	...	874.01	...
MH1	24.000 in	80	WNF	24.000	30.937	710.00	12.00	344.41	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Noz K3	SA-106 B	...	10.000	...	...	...
Noz N9	SA-333 6	8.738	10.000	12.000	12.000	...
Noz N8	SA-333 6	8.738	10.000	12.000	12.000	...
Noz N6	SA-333 6	8.738	10.000	12.000	12.000	...
Noz N7	SA-333 6	...	10.000	...	...	...
Noz K2A	SA-106 B	...	10.000	...	...	...
Noz K2B	SA-106 B	...	10.000	...	...	...
Noz K4	SA-106 B	...	10.000	...	...	...
Noz K5	SA-106 B	...	10.000	...	...	...
Noz K1A	SA-106 B	...	10.000	...	...	...
Noz K1B	SA-106 B	...	10.000	...	...	...
Noz N4	SA-333 6	14.275	10.000	12.000	12.000	...
Noz N5	SA-333 6	14.275	10.000	12.000	12.000	...
Noz N3	SA-333 6	14.275	10.000	12.000	12.000	...
NOZZLE N1	SA-350 LF2	68.000	10.000	...	...	...
NOZZLE N2	SA-350 LF2	68.000	10.000	...	...	...
MH1	SA-333 6	21.000	10.000	12.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Noz K3	767.918	30.8	200.00	0.00	SH.3
Noz N9	105.418	0.0	200.00	0.00	NOZZLE N2
Noz N8	241.918	-90.0	150.00	0.00	SH.1



Noz N6	1057.917	-90.0	270.00	0.00	SH.3
Noz N7	657.917	-58.6	270.00	0.00	SH.3
Noz K2A	461.418	30.8	200.00	0.00	SH.3
Noz K2B	461.418	-30.8	200.00	0.00	SH.3
Noz K4	597.918	90.0	230.00	0.00	SH.3
Noz K5	677.918	90.0	230.00	0.00	SH.3
Noz K1A	...	90.0	200.00	0.00	DISH END
Noz K1B	...	270.0	200.00	0.00	DISH END
Noz N4	507.918	90.0	230.00	0.00	SH.3
Noz N5	767.918	90.0	230.00	0.00	SH.3
Noz N3	657.917	30.8	230.00	0.00	SH.3
NOZZLE N1	132.917	90.0	582.00	0.00	CH. SHELL
NOZZLE N2	132.917	-90.0	732.00	0.00	CH. SHELL
MH1	...	0.0	305.00	0.00	DISH END



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 311.0mm THK X 1750.0mm OD	SA-350 LF2
1	WELD NECK FLANGE: 298.0mm THK X 1750.0mm OD	SA-350 LF2
1	CYLINDER: 16.0mm THK X 1200.0mm ID X 30.0cm	SA-516 70
1	ECCENTRIC CONE: 21.0mm THK X 2300.0mm X 1200.0cm ID X 191.0 cm	SA-516 70
1	CYLINDER: 21.0mm THK X 2300.0mm ID X 642.0cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 21.0mm THK X 2300.0mm ID X 5.0cm	SA-516 70
1	INSULATION: 303mm X 70mm THK	
1	INSULATION: 397mm X 70mm THK	
1	INSULATION: 1254mm X 70mm THK	
1	TUBESHEET: 1340.0mm X 310.0mm THK	SA-350 LF2
1066	TUBES: 8297.0mm X 25.0mm DIA X 2.5mm THK	SA-334 1
1	INSULATION: 300mm X 70mm THK	
1	INSULATION: 1910mm X 70mm THK	
2	SADDLE: 240.0mm X 120 DEG	SA-516 70
1	INSULATION: 6420mm X 70mm THK	
1	INSULATION: 625mm X 70mm THK	
1	CLASS 1500 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	GASKET: 1235.0mm OD X 1203.0mm ID	...
24	BODY FLANGE BOLTS: 88.9mm DIA	SA-320 L43
48	NUTS FOR BODY FLANGE BOLTS: 88.9mm DIA	...
1	CYLINDRICAL SEGMENT 68.0mm THK X 3047.3mm X 125.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 68.0mm THK X 722.6mm X 125.5cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
SH.1	[8]	B	6	-12	No	0.557	16.000	1.00
SH.2	[8]	B	10	6	No	0.915	21.000	1.00
SH.3	[8]	B	10	2	No	0.793	21.000	1.00
DISH END	[10]	B	7	2	No	0.895	17.000	1.00
DISH END	[7]	B	10	2	No	0.793	21.000	1.00
Left Saddle	[24]	B	1	1	No	1.000	12.000	1.00
RIGHT SADDLE	[24]	B	1	1	No	1.000	12.000	1.00
Noz N8	[1]	B	1	-8	No	0.762	12.000	1.00
Nozzle Flg	[5]	!	-46	-85	No			
Noz N4	[1]	B	1	1	No	1.000	12.000	1.00
Nozzle Flg	[5]	!	-46	-85	No			
Noz N5	[1]	B	1	1	No	1.000	12.000	1.00
Nozzle Flg	[5]	!	-46	-85	No			
Noz N3	[1]	B	1	1	No	1.000	12.000	1.00
Nozzle Flg	[5]	!	-46	-85	No			
Noz N6	[1]	B	1	1	No	1.000	12.000	1.00
Nozzle Flg	[5]	!	-29	-104	No			
Noz N7	[1]	B	-104	-104	No	0.031	16.500	1.00
Nozzle Flg	[5]	!	-46	-46	No			
Noz K2A	[1]	B	-104	-104	No	0.026	16.500	1.00
Nozzle Flg	[5]	B	-46	-48	No			
Noz K2B	[1]	B	-104	-104	No	0.026	16.500	1.00
Nozzle Flg	[5]	B	-46	-48	No			
Noz K3	[1]	B	-104	-104	No	0.020	16.000	1.00
Nozzle Flg	[5]	B	-46	-48	No			
Noz K4	[1]	B	-104	-104	No	0.026	16.500	1.00
Nozzle Flg	[5]	B	-46	-48	No			
Noz K5	[1]	B	-104	-104	No	0.026	16.500	1.00
Nozzle Flg	[5]	B	-46	-48	No			
Noz K1A	[1]	B	-104	-104	No	0.026	16.500	1.00
Nozzle Flg	[5]	B	-46	-48	No			
Noz K1B	[1]	B	-104	-104	No	0.026	16.500	1.00
Nozzle Flg	[5]	B	-46	-48	No			
MH1	[1]	B	7	2	No	0.886	17.000	1.00
Nozzle Flg	[5]	!	-46	-85	No			
Tubesheet: SS	[13]	B	40	14	No	0.358	77.500	1.00
Warmest MDMT:			40	14				
TS END COVER	[11]	!	-46		No			
TS FLG1	[11]	!	-46		Yes			
CH. SHELL	[8]	B	14	8	Yes	0.874	68.000	1.00
NOZZLE N1	[1]	B	14	8	Yes	0.866	68.000	1.00
NOZZLE N2	[1]	B	14	8	Yes	0.867	68.000	1.00
Noz N9	[1]	!	-46		Yes			
Nozzle Flg	[5]	A	-18	-38	Yes			
Tubesheet: CS	[14]	B	17	11	Yes	0.876	77.500	1.00
Bolting	[21]		-101					
Warmest MDMT:			17	11				
Exchanger Side			Computed MDMT		Required MDMT		Pass/Fail	
Shell			14.0		-28.9		Fail	
Channel/Tube			11.0		-28.9		Fail	

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.



- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (Kgfc/m2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	23.454	kgf/cm ²
Channel Side Design Pressure	163.16	kgf/cm ²
Shell Side Design Temperature	80.0	°C
Channel Side Design Temperature	80.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
-------------	------



0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y

Consider MAP New and Cold in Noz. Design N

Consider External Loads for Nozzle Des. Y

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g_0 is taken as $\min(g_0, \text{attached shell thk})$ Yes

Using Metric Material Databases, ASME II D Yes

Using ASME VIII-1 Appendix 46 No

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	TS END COVER
Distance "FROM" to "TO"	30.3 cm
Flange Outside Diameter	1750 mm
Element Thickness	303 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	311 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	1682.4 kgf/cm ²
Allowable Stress, Operating	1568.8 kgf/cm ²
Allowable Stress, Hydrotest	2421.7 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2349.6 kgf/cm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011



Class / Thickness / Grade	1::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	30.3 cm
Thickness of Insulation	70 mm
Density	0.40000E-04 kg/cm ³

Element#	2/7
Element From Node	30
Element To Node	40
Element Type	Flange
Description	TS FLG1
Distance "FROM" to "TO"	39.8 cm
Flange Inside Diameter	1200 mm
Element Thickness	293 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	298 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	39.8 cm
Thickness of Insulation	70 mm
Density	0.40000E-04 kg/cm ³

Element#	3/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	CH. SHELL
Distance "FROM" to "TO"	125.5 cm
Inside Diameter	1200 mm
Element Thickness	68 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Allowable Stress, Ambient	1784.4 kgf/cm ²
Allowable Stress, Operating	1650.4 kgf/cm ²
Allowable Stress, Hydrotest	2518.6 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2477.8 kgf/cm ²
MDMT Curve	B
External Pressure Chart Name	CS-2



UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 40
 Detail Type Liquid
 Detail ID Liquid: 40
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 120 cm
 Liquid Density 999.54 kg/m³

Element From Node 40
 Detail Type Insulation
 Detail ID Ins: 40
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 125.5 cm
 Thickness of Insulation 70 mm
 Density 0.40000E-04 kg/cm³

Element From Node 40
 Detail Type Nozzle
 Detail ID NOZZLE N1
 Dist. from "FROM" Node / Offset dist 62.5 cm
 Nozzle Diameter 18 in.
 Nozzle Schedule 120
 Nozzle Class None
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 381.44 kgf
 Grade of Attached Flange None
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID NOZZLE N2
 Dist. from "FROM" Node / Offset dist 62.5 cm
 Nozzle Diameter 18 in.
 Nozzle Schedule 120
 Nozzle Class None
 Layout Angle -90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 435.33 kgf
 Grade of Attached Flange None
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID Noz N9
 Dist. from "FROM" Node / Offset dist 35 cm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 1500
 Layout Angle 0.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 26.471 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element# 4/7
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description SH.1
 Distance "FROM" to "TO" 30 cm
 Inside Diameter 1200 mm
 Element Thickness 16 mm



Internal Corrosion Allowance	1.5	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	23.454	kgf/cm ²
Design Temperature Internal Pressure	80	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	37	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	120	cm
Liquid Density	999.54	kg/m ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	30	cm
Thickness of Insulation	70	mm
Density	0.40000E-04	kg/cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz N8	
Dist. from "FROM" Node / Offset dist	15	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.4874	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6	+IT

Element#	5/7	
Element From Node	60	
Element To Node	70	
Element Type	Conical	
Description	SH.2	
Distance "FROM" to "TO"	191	cm
Inside Diameter	1200	mm
Element Thickness	21	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	21	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	23.454	kgf/cm ²
Design Temperature Internal Pressure	80	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	37	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	2300	mm
Design Length of Cone	1910	mm
Half Apex Angle of Cone	29.938337	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	



Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	230	cm
Liquid Density	999.54	kg/m ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	191	cm
Thickness of Insulation	70	mm
Density	0.40000E-04	kg/cm ³

Element#	6/7	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	SH.3	
Distance "FROM" to "TO"	642	cm
Inside Diameter	2300	mm
Element Thickness	21	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	21	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	23.454	kgf/cm ²
Design Temperature Internal Pressure	80	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	37	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +N	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	55	cm
Width of Saddle	240	mm
Height of Saddle at Bottom	1500	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	480	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	70	
Detail Type	Saddle	
Detail ID	RIGHT SADDLE	
Dist. from "FROM" Node / Offset dist	480	cm
Width of Saddle	240	mm
Height of Saddle at Bottom	1500	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	480	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	



Tangent to Tangent dist. (optional)	0	cm
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	230	cm
Liquid Density	999.54	kg/m ³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	642	cm
Thickness of Insulation	70	mm
Density	0.40000E-04	kg/cm ³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N4	
Dist. from "FROM" Node / Offset dist	60	cm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.919	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N5	
Dist. from "FROM" Node / Offset dist	320	cm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.919	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N3	
Dist. from "FROM" Node / Offset dist	210	cm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.919	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N6	
Dist. from "FROM" Node / Offset dist	610	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.5512	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	



Detail Type	Nozzle	
Detail ID	Noz N7	
Dist. from "FROM" Node / Offset dist	210	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	-58.646301	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.579	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K2A	
Dist. from "FROM" Node / Offset dist	13.5	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.6863	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K2B	
Dist. from "FROM" Node / Offset dist	13.5	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	-30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.6863	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K3	
Dist. from "FROM" Node / Offset dist	320	cm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.7689	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K4	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K5	
Dist. from "FROM" Node / Offset dist	230	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	



Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	7/7
Element From Node	80
Element To Node	90
Element Type	Elliptical
Description	DISH END
Distance "FROM" to "TO"	5 cm
Inside Diameter	2300 mm
Element Thickness	17 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	21 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	23.454 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	230 cm
Liquid Density	999.54 kg/m ³

Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 80
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	62.5 cm
Thickness of Insulation	70 mm
Density	0.40000E-04 kg/cm ³

Element From Node	80
Detail Type	Nozzle
Detail ID	Noz K1A
Dist. from "FROM" Node / Offset dist	650 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	80
Detail Type	Nozzle
Detail ID	Noz K1B
Dist. from "FROM" Node / Offset dist	650 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1



Nozzle Matl	SA-106 B
Element From Node	80
Detail Type	Nozzle
Detail ID	MH1
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
TS END COVER		30.6175	...	30.3	...
TS FLG1		70.4175	...	39.8	...
CH. SHELL		195.917	...	125.5	...
SH.1		256.918	...	30	...
SH.2		447.918	...	191	...
SH.3		1089.92	...	642	...
DISH END		1094.92	...	5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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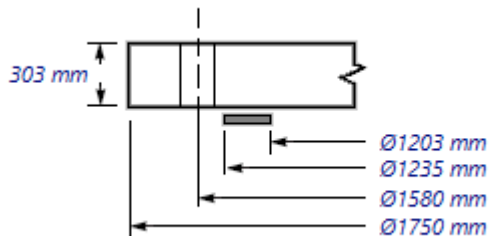


Flange Input Data Values

Description: FLANGE :

TS END COVER

Flange Type		Blind	
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Outside Diameter	A	1750.000	mm
Flange Thickness	t	303.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	1568.83	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1682.44	kgf/cm ²
Bolt Material		SA-320 L43	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	4963.717	mm
Diameter of Bolt Circle	C	1580.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1273.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1235.000	mm
Gasket Inside Diameter	Gi	1203.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1203.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	534.33	kgf/cm ²



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Gasket Contact Width,	$N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width,	$b_o = N/2$	8.000 mm
Effective Gasket Width,	$b = C_b * \text{sqrt}(b_o)$	7.127 mm
Gasket Reaction Diameter,	$G = G_o - 2 * b$	1220.745 mm

Basic Flange and Bolt Loads:



Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1220.7451^2 \cdot 163.158$$

$$= 1909624.625 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 7.1274 \cdot \pi \cdot 1220.7451 \cdot 3.75 \cdot 163.16$$

$$+ 2 \cdot 1203.0 \cdot 5.0 \cdot 3.75 \cdot 163.1581$$

$$= 408089.062 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1909625 + 408089 + 0, 0)$$

$$= 2317713.500 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 534.33 \cdot 7.1274 \cdot \pi \cdot 1220.745 + 534.33 \cdot 5.0 \cdot 1203.0$$

$$= 178195.688 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((2317714 + 0 + 4 \cdot 0/1220.75)/1754, 178196/1754)$$

$$= 1321.525 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1321.525	1354.681	
Radial Distance between Bolts and Edge:	82.550	85.000	
Circ. Spacing between the Bolts:	180.975	206.231	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1354.681 \cdot 1753.82 / (534.33 \cdot \pi (1235.0 + 1203.0))$$

$$= 58.053 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 1354.681$$

$$= 2375862.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 2317713.50 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1580.0 - 1220.7451) / 2$$

$$= 179.6275 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 2317714 \cdot 179.6275 \cdot 1.0 / (303.0^2 \cdot 1220.7451) + 0.3 \cdot 1.0 \cdot 163.16 \cdot 1220.7451^2 / 303.0^2$$

$$= 1500.29 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2375863 \cdot 179.6275 \cdot 1.0 / (303.0^2 \cdot 1220.7451)$$

$$= 723.50 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$



= 2317714/1354.681
= 1710.89 kgf/cm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
= (178196/1354.681)
= 131.54 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1500.29	1568.83	723.50 1682.44
Bolting	1710.89	1753.82	131.54 1753.82

Reqd. Blind Flange Thickness at Center 296.307 mm
Reqd. Blind Flange Thickness at Gasket 203.232 mm

Estimated M.A.W.P. (Operating) 167 kgf/cm²
Estimated Finished Weight of Flange at given Thk. 5648.5 kg
Estimated Unfinished Weight of Forging at given Thk 5648.5 kg

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1500.29	1682.44	723.50 1682.44
Bolting	1710.89	1753.82	131.54 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 286.131 mm

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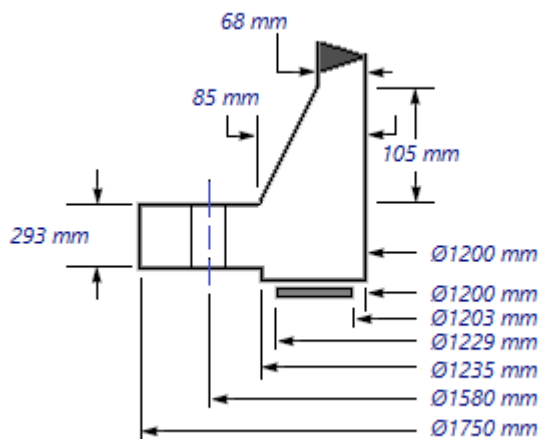


Flange Input Data Values

Description: FLANGE :

TS FLG1

Flange Type	Integral Weld Neck		
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1200.000	mm
Flange Outside Diameter	A	1750.000	mm
Flange Thickness	t	293.0000	mm
Thickness of Hub at Small End	go	68.0000	mm
Thickness of Hub at Large End	gl	85.0000	mm
Length of Hub	h	105.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	1568.83	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1682.44	kgf/cm ²
Bolt Material	SA-320 L43		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	1580.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1235.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1229.000	mm
Gasket Inside Diameter	Gi	1203.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1203.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	534.33	kgf/cm ²





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Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext}) \\ &= (163.158(1200.0 + 1.5) / (1568.835 - 0.5 \cdot 163.158)) + (1.5) \\ &= 67.4870 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= S \cdot \log((2 \cdot T + R) / R) \\ &= 1568.835 \cdot \log((2 \cdot 66.5 + 601.5) / 601.5) \\ &= 164.511 \text{ kgf/cm}^2 \end{aligned}$$

Small End Hub MAPnc:

$$\begin{aligned} &= S_a \cdot \log((2.0 \cdot T + R) / R) \\ &= 1682.443 \cdot \log((2.0 \cdot 68.0 + 600.0) / 600.0) \\ &= 180.625 \text{ kgf/cm}^2 \end{aligned}$$

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	1203.000 mm
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	83.500 mm
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	66.500 mm
Code R Dimension,	$R = ((C - B_{cor}) / 2) - g1_{cor}$	105.000 mm

Gasket Contact Width,	$N = (G_o - G_i) / 2$	13.000 mm
Basic Gasket Width,	$bo = N / 2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1216.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi / 4 \cdot G^2 \cdot P \\ &= \pi / 4 \cdot 1216.1509^2 \cdot 163.158 \\ &= 1895277.875 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P \\ &= 2 \cdot 6.4246 \cdot \pi \cdot 1216.1509 \cdot 3.75 \cdot 163.16 \\ &\quad + 2 \cdot 1203.0 \cdot 5.0 \cdot 3.75 \cdot 163.1581 \\ &= 373970.031 \text{ kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi / 4 \cdot B_{cor}^2 \cdot P \\ &= \pi / 4 \cdot 1203.0^2 \cdot 163.1581 \\ &= 1854510.125 \text{ kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - H_d \\ &= 1895278 - 1854510 \\ &= 40767.828 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(W_o, 1 + W_o, 2 + H'p, 0) \\ &= \max(1895278 + 373970 + 0, 0) \\ &= 2269248.000 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + y_{part} \cdot b_{part} \cdot l_p \\ &= 534.33 \cdot 6.4246 \cdot \pi \cdot 1216.151 + 534.33 \cdot 5.0 \cdot 1203.0 \\ &= 163297.297 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\ &= \max((2269248 + 0 + 4 \cdot 0 / 1216.15) / 1754, 163297 / 1754) \\ &= 1293.891 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 88.9 + 6 \cdot 293.0 / (3.75 + 0.5) \\ &= 591.447 \text{ mm} \end{aligned}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1580.0 \cdot \sin(\pi / 24)$$

$$= 206.231 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{206.231 / (2 \cdot 88.9 + 293.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1293.891	1354.681	
Radial Distance between Hub and Bolts:	104.775	105.000	
Radial Distance between Bolts and Edge:	82.550	85.000	
Circ. Spacing between the Bolts:	180.975	206.231	591.447

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1354.681 \cdot 1753.82 / (534.33 \cdot \pi(1229.0 + 1203.0))$$

$$= 58.196 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 1354.681$$

$$= 2375862.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 2269248 - 1895278$$

$$= 373970.06 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1580.0 - 1216.1509) / 2$$

$$= 181.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1580.0 - 1203.0) / 2 + 181.9246)$$

$$= 185.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1580.0 - 1203.0 - 83.5) / 2$$

$$= 146.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1854510.	146.7500	1.0000	272150.
Face Pressure, Mt	40768.	185.2123	1.0000	7551.
Gasket Load, Mg	373970.	181.9246	1.0000	68034.
Gasket Seating, Matm	2375863.	181.9246	1.0000	432228.

Total Moment for Operation, Mo 347735. kgf-m.
 Total Moment for Gasket seating, Mg 432228. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 282.842 mm
 Hub Ratio, h/h0 = HL / H0 0.371



Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.256

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.883
 Factor V per 4.16.5 0.417
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1750.0/1203.0
 = 1.455

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 = $(0.66845 + (5.7169(1.455^2 \cdot \log_{10}(1.455))) / (1.455^2 - 1)) / (1.455 - 1)$
 = 5.350

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 = $(1.455^2(1 + 8.55246 \cdot \log_{10}(1.455)) - 1) / ((1.0472 + 1.9448 \cdot 1.455^2) \cdot (1.455 - 1))$
 = 1.730

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 = $(1.455^2(1 + 8.55246 \cdot \log_{10}(1.455)) - 1) / (1.36136(1.455^2 - 1) \cdot (1.455 - 1))$
 = 5.879

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
 = $(1.455^2 + 1) / (1.455^2 - 1)$
 = 2.792

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$
 = $5.879 \cdot 66.5^2 \cdot 282.842 / 0.417$
 = 17636692.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.883/282.842
 = 0.003 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
 = $(293.0 \cdot 0.003 + 1) / 1.73 + 293.0^3/17636692$
 = 2.533

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 347735/1203.0) / (2.533 \cdot 83.5^2)$
 = 1636.71 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 432228/1203.0) / (2.533 \cdot 83.5^2)$
 = 2034.41 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 = $(2.2203 \cdot 347735/1203.0) / (2.533 \cdot 293.0^2)$
 = 295.13 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 = $(2.2203 \cdot 432228/1203.0) / (2.533 \cdot 293.0^2)$
 = 366.84 kgf/cm²

Tangential Flange Stress, Operating [STo]:



$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (5.3502 \cdot 347735 / (293.0^2 \cdot 1203.0)) - 2.7919 \cdot 295$$

$$= 977.47 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (5.3502 \cdot 432228 / (293.0^2 \cdot 1203.0)) - 2.7919 \cdot 367$$

$$= 1214.97 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1637 + \max(295, 977)) / 2$$

$$= 1307.09 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (2034 + \max(367, 1215)) / 2$$

$$= 1624.69 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 2269248 / 1354.681$$

$$= 1675.12 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (163297 / 1354.681)$$

$$= 120.54 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1636.71	2353.25	2034.41	2523.66
Radial Flange	295.13	1568.83	366.84	1682.44
Tangential Flange	977.47	1568.83	1214.97	1682.44
Maximum Average	1307.09	1568.83	1624.69	1682.44
Bolting	1675.12	1753.82	120.54	1753.82

Minimum Required Flange Thickness 286.283 mm

Estimated M.A.W.P. (Operating) 165 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 3143.4 kg
 Estimated Unfinished Weight of Forging at given Thk 3930.8 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 432228.4 / 1.0 \cdot 100000.023 \cdot 0.417 / (2.425 \cdot 2052965 \cdot 66.5^2 \cdot 282.842 \cdot 0.3)$$

$$= 0.503 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 347734.9 / 1.0 \cdot 100000.023 \cdot 0.417 / (2.425 \cdot 2018573 \cdot 66.5^2 \cdot 282.842 \cdot 0.3)$$

$$= 0.411 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 432228.4 / 1.0 \cdot 100000.023 \cdot 0.417 / (2.533 \cdot 2052965 \cdot 66.5^2 \cdot 282.842 \cdot 0.3)$$

$$= 0.481 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 347734.9 / 1.0 \cdot 100000.023 \cdot 0.417 / (2.533 \cdot 2018573 \cdot 66.5^2 \cdot 282.842 \cdot 0.3)$$

$$= 0.394 \text{ (should be } \leq 1)$$



Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1631.13	2523.66	2017.16	2523.66
Radial Flange	303.31	1682.44	375.09	1682.44
Tangential Flange	961.93	1682.44	1189.58	1682.44
Maximum Average	1296.53	1682.44	1603.37	1682.44
Bolting	1675.12	1753.82	120.54	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 283.566 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
TS END COVER		163.16	311	1.5	1750	1568.8	1
TS FLG1		163.16	298	1.5	1200	1568.8	2
CH. SHELL		163.28	68	1.5	1200	1650.4	3
SH.1		23.574	16	1.5	1200	1650.4	4
SH.2		23.684	21	1.5	2300	1650.4	5
SH.3		23.684	21	1.5	2300	1650.4	6
DISH END		23.684	21	1.5	2300	1650.4	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
TS END COVER		163.158	167.251	167.252	303	296.307	1
TS FLG1		163.158	164.511	170.824	293	286.283	2
CH. SHELL		163.158	172.944	191.572	68	64.0508	3
SH.1		23.454	39.1933	46.9608	16	10.1534	4
SH.2		23.454	23.8133	28.0158	21	20.7065	5
SH.3		23.454	27.4846	32.2909	21	18.1437	6
DISH END		23.454	24.3763	29.1893	17	16.417	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

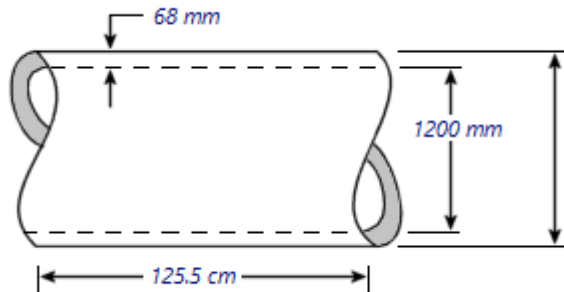
Shell Side MAWP = 24 kgf/cm²
 Shell Side MAPnc = 28 kgf/cm²
 Channel Side MAWP = 165 kgf/cm²
 Channel Side MAPnc = 167 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: CH. SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	163.278 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		125.500 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	1200.000 mm	Actual Stress	1476.9 kgf/cm²	
Spec. Min. Thk t _e or s _a		68.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		52674.5 cm²	SG of Contents	1.000	
Element Volume		1419375.0 cm³	Weight of Contents	1418.7 kgf	
Empty Weight		2634.8 kgf	Operating Weight	4053.5 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1203.0 \cdot (\exp(163.278/(1650.408 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 62.5508 + 1.5 + 0.0 = 64.0508 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1650.408 \cdot 1.0 \cdot \ln((2 \cdot 66.5 + 1203.0) / 1203.0) - 0.12$$

$$= 173.065 - 0.12 = 172.9444 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1784.409 \cdot 1.0 \cdot \ln((2 \cdot 68.0 + 1200.0) / 1200.0)$$

$$= 191.5716 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 163.278 \cdot 1203.0 / (1.0 (1336.0 - 1203.0))$$

$$= 1476.8699 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 163.278 / (1.0 \cdot \ln((2 \cdot 66.5 + 1203.0) / 1203.0))$$
$$= 1557.0823 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 5.363 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : 8 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 14 °C
Temperature reduction per Graph 3.13 : 6 °C
The computed stress ratio is : 0.874

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (58.091 \cdot 1.0) / (68.0 - 1.5 - 0.0)$
 $= 0.874$

Warning: Impact Testing Is Mandatory!

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

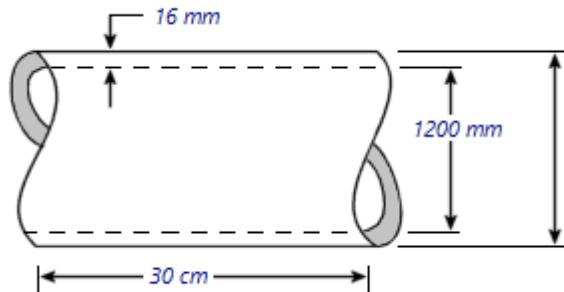
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	164.511 kgf/cm ²
Governing Thk.	68.000 mm	Min. Des. Mtl. Temp.	-29 °C
Stress/Thk. Ratio	0.874	Basic MDMT	14 °C
		Temperature Difference	6 °C
		Computed Min. Temp.	8 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: SH.1

Cylindrical Shell Design Information:					
Design Pressure	P	23.574 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		30.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	1200.000 mm	Actual Stress	977.9 kgf/cm²	
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.	16.000 mm	
Surface Area		11611.3 cm²	SG of Contents	1.000	
Element Volume		182311.0 cm³	Weight of Contents	182.2 kgf	
Empty Weight		142.1 kgf	Operating Weight	324.3 kgf	
			Comp. All. Ope. Stress	1137.6 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1203.0 \cdot (\exp(23.574/(1650.408 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 8.6534 + 1.5 + 0.0 = 10.1534 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1650.408 \cdot 1.0 \cdot \ln((2 \cdot 14.5 + 1203.0) / 1203.0) - 0.12$$

$$= 39.313 - 0.12 = 39.1933 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1784.409 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 1200.0) / 1200.0)$$

$$= 46.9608 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 23.574 \cdot 1203.0 / (1.0 (1232.0 - 1203.0))$$

$$= 977.9203 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 23.574 / (1.0 \cdot \ln((2 \cdot 14.5 + 1203.0) / 1203.0))$$

$$= 989.6604 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.316 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : -12 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 6 °C
 Temperature reduction per Graph 3.12 : 18 °C
 The computed stress ratio is : 0.557

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (8.081 \cdot 1.0) / (16.0 - 1.5 - 0.0)$$

$$= 0.557$$

Warning: Impact Testing Is Mandatory!

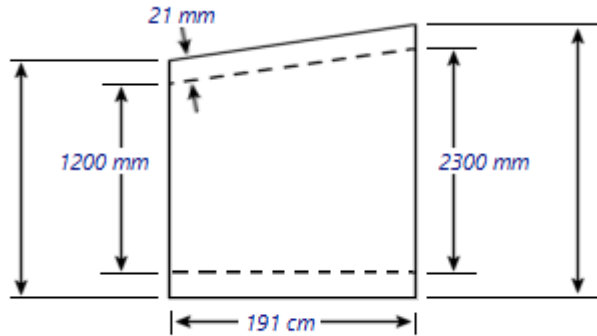
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.813 kgf/cm ²
		Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	16.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.557	Temperature Difference	18 °C
		Computed Min. Temp.	-12 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: SH.2

Conical Head/Section Design Information:					
Design Pressure	P	23.684 kgf/cm²	Int. Design Temperature	80 °C	
Material	SA-516 70		Ext. Design Temperature	37 °C	
Length	191.000 cm		External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	2300.000 mm	Actual Stress	1613.9 kgf/cm²	
Spec. Min. Thk t _e , or sa	21.0000 mm		Spec. Nominal Thk.	21.000 mm	
Surface Area	112301.1 cm²		SG of Contents	1.000	
Element Volume	3745909.5 cm³		Weight of Contents	3744.2 kgf	
Empty Weight	1999.6 kgf		Operating Weight	5743.8 kgf	
			Comp. All. Ope. Stress	1204.4 kgf/cm²	
			To End Diameter	1200.000 mm	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Thickness due to Internal Pressure: Equation 4.3.4.1 [tr]:

$$= 0.5 \cdot D_c / \cos(\alpha) \cdot (\exp(P / (S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2303.0 / \cos(29.938) \cdot (\exp(23.684 / (1650.408 \cdot 1.0)) - 1) + c_i + c_o$$

$$= 19.2065 + 1.5 + 0.0 = 20.7065 \text{ mm}$$

Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}$$

$$= 1650.408 \cdot 1.0 \cdot \ln(2 \cdot 19.5 \cdot \cos(29.938) + 2303.0) / 2303.0$$

$$= 24.0434 \text{ kgf/cm}^2$$

Maximum Allowable Pressure New and Cold [MAPnc]:

$$= S_a \cdot E \cdot \ln(2 \cdot t \cdot \cos(\alpha) + D) / D$$

$$= 1784.409 \cdot 1.0 \cdot \ln(2 \cdot 21.0 \cdot \cos(29.938) + 2300.0) / 2300.0$$

$$= 28.0158 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D) \cos(\alpha))$$

$$= 23.684 \cdot 2303.0 / (1.0 (2342.0 - 2303.0) \cos(29.938))$$

$$= 1613.935 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor})$$



$$= 23.684 / (1.0 \cdot \ln((2 \cdot 19.5 \cdot \cos(29.938) + 2303.0) / 2303.0))$$

$$= 1625.743 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cones formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.715 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 6 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 10 °C
 Temperature reduction per Graph 3.12 : 4 °C
 The computed stress ratio is : 0.915

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (17.852 \cdot 1.0) / (21.0 - 1.5 - 0.0)$
 $= 0.915$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.813 kgf/cm ²
Governing Thk.	21.000 mm	Min. Des. Mtl. Temp.	-29 °C
Stress/Thk. Ratio	0.915	Basic MDMT	10 °C
		Temperature Difference	4 °C
		Computed Min. Temp.	6 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: SH.2

Cone From 60 to 70 Ext. Chart: CS-2 at 37 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2204.11 / \sqrt{(1355.047 \cdot 19.5)}$$

$$= 13.559$$

Compute the Factor [Ch]:

Since $M_x > 13$ and $M_x < 2 \cdot (D_o/t)^{0.94}$, $Ch = 1.12 \cdot M_x^{-1.058}$

$$= 1.12 \cdot 13.559^{-1.058}$$

$$= 0.0710$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.071 \cdot 2057748(19.5 / 2710.094)$$

$$= 1682.177 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 1682.2 / 2671.6$$

$$= 0.630$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2671.6 / 4921.5$$

$$= 0.543$$

Ratio [AE]:
 = Fhe / E
 = 1682.2/2057748
 = 0.000817

Factor [m1]:
 = (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))
 = (ln(0.543) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))
 = 0.133

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:
 = $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$
 = 2671.6(1 + 0.002) / ln(1 + 0.002) 0.133118
 = 6123.286 kgf/cm²

Curve Fitting Exponent for, Ferritic Steel [m2]:
 = m2f(m2f2 - R)
 = 0.6(1.0 - 0.543)
 = 0.274

Curve Fitting Constant [A2]:
 = ($\sigma_{uts} \cdot \exp(m2)$) / (m2 m2)
 = (4921.5 • exp(0.274)) / (0.274 0.274)
 = 9232.343 kgf/cm²

Material Parameter for Stress-Strain Curve Model [K]:
 = 1.5 • R1.5 - 0.5 • R2.5 - R3.5
 = 1.5 • 0.5431.5 - 0.5 • 0.5432.5 - 0.5433.5
 = 0.374

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.156E-05	0.597E-05	-0.564E-05	0.250E+06	0.904E-02
2	1230	-5.430	0.177E-07	0.177E-07	0.954E-06	-0.954E-06	0.192E+07	0.176E-03
3	1845	-3.966	0.249E-06	0.248E-06	0.279E-05	-0.278E-05	0.101E+07	0.101E-02
4	1537	-4.698	0.759E-07	0.759E-07	0.172E-05	-0.172E-05	0.157E+07	0.165E-03
5	1384	-5.064	0.382E-07	0.382E-07	0.130E-05	-0.130E-05	0.178E+07	0.386E-04
6	1461	-4.881	0.543E-07	0.543E-07	0.150E-05	-0.150E-05	0.168E+07	0.521E-04
7	1422	-4.973	0.457E-07	0.457E-07	0.140E-05	-0.140E-05	0.173E+07	0.437E-05
8	1403	-5.018	0.418E-07	0.418E-07	0.135E-05	-0.135E-05	0.175E+07	0.177E-04
9	1413	-4.995	0.437E-07	0.437E-07	0.138E-05	-0.138E-05	0.174E+07	0.680E-05
10	1417	-4.984	0.447E-07	0.447E-07	0.139E-05	-0.139E-05	0.174E+07	0.125E-05
11	1420	-4.978	0.452E-07	0.452E-07	0.139E-05	-0.139E-05	0.173E+07	0.155E-05
12	1419	-4.981	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.147E-06
13	1418	-4.983	0.448E-07	0.448E-07	0.139E-05	-0.139E-05	0.174E+07	0.552E-06
14	1418	-4.982	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.203E-06
15	1418	-4.982	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.277E-07
16	1418	-4.981	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.598E-07
17	1418	-4.981	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.161E-07

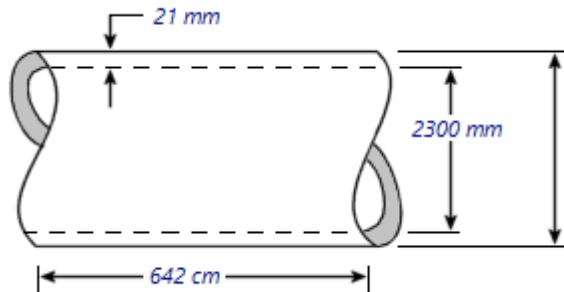
Factor of Safety [FS]:
 Since Fic [1418.9] < 0.55 * 2671.64, FS = 2

Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = 1418.901/2.0
 = 709.451 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:
 = 2 • Fha(t / Do)
 = 2 • 709.451(19.5/2710.094)
 = 10.209 kgf/cm²

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: SH.3

Cylindrical Shell Design Information:					
Design Pressure	P	23.684 kgf/cm²	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		37 °C
Length		642.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	2300.000 mm	Actual Stress		1398.6 kgf/cm²
Spec. Min. Thk t _e , or sa		21.0000 mm	Spec. Nominal Thk.		21.000 mm
Surface Area		472358.6 cm²	SG of Contents		1.000
Element Volume		23150904.0 cm³	Weight of Contents		23140.5 kgf
Empty Weight		7619.0 kgf	Operating Weight		30759.5 kgf
			Comp. All. Ope. Stress		1067.1 kgf/cm²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2303.0 \cdot (\exp(23.684/(1650.408 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 16.6437 + 1.5 + 0.0 = 18.1437 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1650.408 \cdot 1.0 \cdot \ln((2 \cdot 19.5 + 2303.0) / 2303.0) - 0.23$
 $= 27.715 - 0.23 = 27.4846 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1784.409 \cdot 1.0 \cdot \ln((2 \cdot 21.0 + 2300.0) / 2300.0)$
 $= 32.2909 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 23.684 \cdot 2303.0 / (1.0 (2342.0 - 2303.0))$
 $= 1398.5767 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 23.684 / (1.0 \cdot \ln((2 \cdot 19.5 + 2303.0) / 2303.0))$$

$$= 1410.3866 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 0.905 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 2 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 10 °C
 Temperature reduction per Graph 3.12 : 9 °C
 The computed stress ratio is : 0.793

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (15.47 \cdot 1.0) / (21.0 - 1.5 - 0.0)$
 $= 0.793$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.813 kgf/cm ²
Governing Thk.	21.000 mm	Min. Des. Mtl. Temp.	-29 °C
Stress/Thk. Ratio	0.793	Basic MDMT	10 °C
		Temperature Difference	9 °C
		Computed Min. Temp.	2 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: SH.3

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 37 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 6661.666 / \sqrt{1171.0 \cdot 19.5}$$

$$= 44.085$$

Compute the Factor [Ch]:

Since $M_x > 13$ and $M_x < 2 \cdot (D_o/t)^{0.94}$, $Ch = 1.12 \cdot M_x^{-1.058}$

$$= 1.12 \cdot 44.085^{-1.058}$$

$$= 0.0204$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.02 \cdot 2057748 (19.5 / 2342.0)$$

$$= 559.142 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 559.1 / 2671.6$$

$$= 0.209$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2671.6 / 4921.5$$

$$= 0.543$$

Ratio [AE]:
 = Fhe / E
 = 559.1/2057748
 = 0.000272

Factor [m1]:
 = (ln(r) + (epsp - epsys)) / ln(ln(1 + epsp) / ln(1 + epsys))
 = (ln(0.543) + (0.00002 - 0.002)) / ln(ln(1 + 0.00002) / ln(1 + 0.002))
 = 0.133

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:
 = $\sigma_{ys}(1 + \epsilon_{psys}) / \ln(1 + \epsilon_{psys}) m1$
 = 2671.6(1 + 0.002) / ln(1 + 0.002) 0.133118
 = 6123.286 kgf/cm²

Curve Fitting Exponent for, Ferritic Steel [m2]:
 = m2f(m2f2 - R)
 = 0.6(1.0 - 0.543)
 = 0.274

Curve Fitting Constant [A2]:
 = ($\sigma_{uts} \cdot \exp(m2)$) / (m2 m2)
 = (4921.5 • exp(0.274)) / (0.274 0.274)
 = 9232.343 kgf/cm²

Material Parameter for Stress-Strain Curve Model [K]:
 = 1.5 • R1.5 - 0.5 • R2.5 - R3.5
 = 1.5 • 0.5431.5 - 0.5 • 0.5432.5 - 0.5433.5
 = 0.374

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.156E-05	0.597E-05	-0.564E-05	0.250E+06	0.958E-02
2	1230	-5.430	0.177E-07	0.177E-07	0.954E-06	-0.954E-06	0.192E+07	0.370E-03
3	615	-6.894	0.194E-09	0.194E-09	0.152E-06	-0.152E-06	0.206E+07	0.275E-04
4	307	-7.626	0.213E-11	0.213E-11	0.244E-07	-0.244E-07	0.206E+07	0.122E-03
5	461	-7.260	0.299E-10	0.299E-10	0.712E-07	-0.712E-07	0.206E+07	0.475E-04
6	538	-7.077	0.815E-10	0.815E-10	0.107E-06	-0.107E-06	0.206E+07	0.100E-04
7	576	-6.986	0.128E-09	0.128E-09	0.129E-06	-0.129E-06	0.206E+07	0.870E-05
8	557	-7.031	0.102E-09	0.102E-09	0.118E-06	-0.118E-06	0.206E+07	0.678E-06
9	567	-7.009	0.114E-09	0.114E-09	0.123E-06	-0.123E-06	0.206E+07	0.401E-05
10	562	-7.020	0.108E-09	0.108E-09	0.120E-06	-0.120E-06	0.206E+07	0.167E-05
11	559	-7.026	0.105E-09	0.105E-09	0.119E-06	-0.119E-06	0.206E+07	0.494E-06
12	558	-7.029	0.104E-09	0.104E-09	0.118E-06	-0.118E-06	0.206E+07	0.918E-07
13	559	-7.027	0.105E-09	0.105E-09	0.119E-06	-0.119E-06	0.206E+07	0.201E-06
14	559	-7.028	0.104E-09	0.104E-09	0.118E-06	-0.118E-06	0.206E+07	0.547E-07
15	558	-7.028	0.104E-09	0.104E-09	0.118E-06	-0.118E-06	0.206E+07	0.185E-07
16	558	-7.028	0.104E-09	0.104E-09	0.118E-06	-0.118E-06	0.206E+07	0.181E-07
17	558	-7.028	0.104E-09	0.104E-09	0.118E-06	-0.118E-06	0.206E+07	0.226E-09

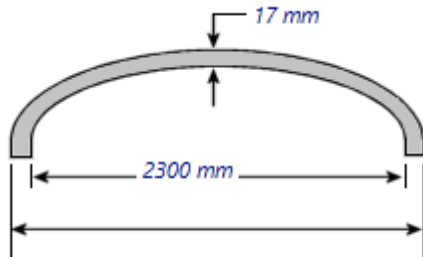
Factor of Safety [FS]:
 Since Fic [558.9] < 0.55 * 2671.64, FS = 2

Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = 558.902/2.0
 = 279.451 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:
 = 2 • Fha(t / Do)
 = 2 • 279.451(19.5/2342.0)
 = 4.654 kgf/cm²

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: DISH END

Elliptical Head Design Information:					
Design Pressure	P	23.684 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		5.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	2300.000 mm	Actual Stress	1588.6 kgf/cm²	
Spec. Min. Thk t _e , or s _a		17.0000 mm	Spec. Nominal Thk.	21.000 mm	
Surface Area		63135.0 cm²	SG of Contents	1.000	
Element Volume		1800398.8 cm³	Weight of Contents	1799.6 kgf	
Empty Weight		1108.7 kgf	Operating Weight	2908.3 kgf	
			Comp. All. Ope. Stress	1067.1 kgf/cm²	
			Aspect Ratio	2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

$$t_r = 14.917 \text{ mm}$$

$$\begin{aligned} t &= t_r + c_i + c_o \\ &= 14.917 + 1.5 + 0.0 \\ &= 16.417 \text{ mm} - \text{see below for the derivation} \end{aligned}$$

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

$$\begin{aligned} h &= D / (2 \cdot \text{Aspect Ratio}) \\ &= 2300.0 / (2 \cdot 2.0) \\ &= 575.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} k &= D / (2 \cdot h) \\ &= 2300.0 / (2 \cdot 575.0) \\ &= 2.000 \end{aligned}$$

$$r = D (0.5/k - 0.08) + c_a$$



$$= 2300.0(0.5/2.0 - 0.08) + 1.5$$

$$= 392.500 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 2300.0(0.44 \cdot 2.0 + 0.02) + 1.5$$

$$= 2071.500 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 14.917 \text{ mm}$, $P = 23.6841 \text{ kgf/cm}^2$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 2303.0 - 392.5)/(2071.5 - 392.5)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(2071.5 \cdot 14.917) / 392.5}$$

$$= 0.4479 \text{ radians - Equation 4.3.9}$$

$\phi Th < \beta Th$, Thus:

$$Rth = (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r$$

$$= (0.5 \cdot 2303.0 - 392.5) / \cos(1.102 - 0.448) + 392.5$$

$$= 1348.7361 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(392.5/2303.0) + 0.605$$

$$= 0.7229 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(392.5/2303.0)$$

$$= 1.0169, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.723 \cdot 2032634 \cdot 14.917^2 / (1.017 \cdot 1348.736(0.5 \cdot 1348.736 - 392.5))$$

$$= 845.82 \text{ kgf/cm}^2, \text{ Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 2477.796 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 2477.796 \cdot 14.917 / (1.017 \cdot 1348.736(0.5 \cdot 1348.736/392.5 - 1))$$

$$= 37.527 \text{ kgf/cm}^2, \text{ Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 845.82/37.527 = 22.539$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 22.539 - 0.20354 \cdot 22.539^2 + 0.019274 \cdot 22.539^3) /$$

$$(1 + 0.19014 \cdot 22.539 - 0.089534 \cdot 22.539^2 + 0.0093965 \cdot 22.539^3)) \cdot 37.527$$

$$= 75.04 \text{ kgf/cm}^2, \text{ Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 75.04/1.5 = 50.03 \text{ kgf/cm}^2 \text{ - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1650.408 \cdot 1.0 / (2071.5/14.917 + 0.5)$$

$$= 23.68 \text{ kgf/cm}^2, \text{ Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(50.026, 23.684)$$

$$= 23.68 \text{ kgf/cm}^2, \text{ Equation 4.3.23: (same as original design pressure)}$$



Allowable pressure in Crown at given thickness [Pac]:

$$= 2 \cdot S \cdot E / (L / t + 0.5)$$

$$= 2 \cdot 1650.408 \cdot 1.0 / (2071.5 / 15.5 + 0.5)$$

$$= 24.606 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 24.6063 - 0.23 = 24.3763 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 29.1893 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 1588.6 \text{ kgf/cm}^2$$

Required Thickness of Straight Flange = 18.144 mm

Percent Fiber Elongation per Table 6.1

$$\text{For Heads formed from Segments } (75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 3.923 \%$$

$$\text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) \quad 20.922 \%$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 63135.0 / 3.14159)$$

$$= 2835.242 \text{ mm}$$

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	2 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	7 °C
Temperature reduction per Graph 3.12	:	5 °C
The computed stress ratio is	:	0.895

The computed thickness ratio [sr]:

$$= (tr \cdot E) / (t_{\text{nom}} - ci - c_{\text{ext}})$$

$$= (13.869 \cdot 1.0) / (17.0 - 1.5 - 0.0)$$

$$= 0.895$$

Warning: Impact Testing Is Mandatory!

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	2 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	10 °C
Temperature reduction per Graph 3.12	:	9 °C
The computed stress ratio is	:	0.793

The computed thickness ratio [sr]:

$$= (tr \cdot E) / (t_{\text{nom}} - ci - c_{\text{ext}})$$

$$= (15.47 \cdot 1.0) / (21.0 - 1.5 - 0.0)$$

$$= 0.793$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.813 kgf/cm ²
Governing Thk.	21.000 mm	Min. Des. Mtl. Temp.	-29 °C
Stress/Thk. Ratio	0.793	Basic MDMT	10 °C
		Temperature Difference	9 °C
		Computed Min. Temp.	2 °C



Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: DISH END

Elliptical Head From 80 to 90 Ext. Chart: CS-2 at 37 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y(t / R_o)$$

$$= 0.075 \cdot 2057748(15.5/2071.95)$$

$$= 1154.532 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 1154.5 / 2671.6$$

$$= 0.432$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2671.6/4921.5$$

$$= 0.543$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 1154.5/2057748$$

$$= 0.000561$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.543) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2671.6(1 + 0.002) / \ln(1 + 0.002) 0.133118$$

$$= 6123.286 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.543)$$

$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4921.5 \cdot \exp(0.274)) / (0.274^{0.274})$$

$$= 9232.343 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.5431.5 - 0.5 \cdot 0.5432.5 - 0.5433.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.156E-05	0.597E-05	-0.564E-05	0.250E+06	0.929E-02
2	1230	-5.430	0.177E-07	0.177E-07	0.954E-06	-0.954E-06	0.192E+07	0.806E-04
3	615	-6.894	0.194E-09	0.194E-09	0.152E-06	-0.152E-06	0.206E+07	0.262E-03



4	922	-6.162	0.273E-08	0.273E-08	0.446E-06	-0.446E-06	0.203E+07	0.108E-03
5	1076	-5.796	0.744E-08	0.744E-08	0.670E-06	-0.670E-06	0.200E+07	0.218E-04
6	1153	-5.613	0.117E-07	0.117E-07	0.804E-06	-0.804E-06	0.196E+07	0.264E-04
7	1115	-5.705	0.935E-08	0.935E-08	0.735E-06	-0.735E-06	0.198E+07	0.169E-05
8	1095	-5.750	0.835E-08	0.835E-08	0.702E-06	-0.702E-06	0.199E+07	0.102E-04
9	1105	-5.727	0.884E-08	0.884E-08	0.719E-06	-0.719E-06	0.199E+07	0.430E-05
10	1110	-5.716	0.909E-08	0.909E-08	0.727E-06	-0.727E-06	0.198E+07	0.131E-05
11	1112	-5.710	0.922E-08	0.922E-08	0.731E-06	-0.731E-06	0.198E+07	0.188E-06
12	1111	-5.713	0.915E-08	0.915E-08	0.729E-06	-0.729E-06	0.198E+07	0.562E-06
13	1112	-5.712	0.919E-08	0.918E-08	0.730E-06	-0.730E-06	0.198E+07	0.187E-06
14	1112	-5.711	0.920E-08	0.920E-08	0.731E-06	-0.731E-06	0.198E+07	0.238E-09

Factor of Safety [FS]:

Since $Fic [1112.32] < 0.55 * 2671.64$, $FS = 2$

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= $1112.322/2.0$

= 556.161 kgf/cm^2

Allowable External Pressure at the given Thickness [Pa]:

= $2 * Fha[t/Ro]$

= $2 * 556.161[15.5/2071.95]$

= 8.321 kgf/cm^2

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Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:

$$= \text{Test Factor} \cdot \text{ShellSide MAWP(St/S)}$$

$$= 1.25 \cdot 23.813(1.072)$$

$$= 31.922 \text{ kgf/cm}^2$$

Hydrotest Pressure Calculation [Channel Side]:

$$= \text{Test Factor} \cdot \text{Channel Side MAWP(St/S)}$$

$$= 1.25 \cdot 164.511(1.072)$$

$$= 220.530 \text{ kgf/cm}^2$$

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
CH. SHELL	2055.3	2518.6	0.816	220.65
SH.1	1217.5	2518.6	0.483	32.04
SH.2	2047.9	2518.6	0.813	32.15
SH.3	1776.7	2518.6	0.705	32.15
DISH END	1965.5	2518.6	0.780	32.15

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
NOZZLE N1	Nozzle	1682.44	1568.83	1.072
NOZZLE N2	Nozzle	1682.44	1568.83	1.072
Noz N9	Nozzle	1641.66	1522.22	1.078
Noz N9	Pad	1784.41	1650.40	1.081
Noz N8	Nozzle	1641.66	1522.22	1.078
Noz N8	Pad	1784.41	1650.40	1.081
Noz N4	Nozzle	1641.66	1522.22	1.078
Noz N4	Pad	1784.41	1650.40	1.081
Noz N5	Nozzle	1641.66	1522.22	1.078
Noz N5	Pad	1784.41	1650.40	1.081
Noz N3	Nozzle	1641.66	1522.22	1.078
Noz N3	Pad	1784.41	1650.40	1.081
Noz N6	Nozzle	1641.66	1522.22	1.078
Noz N6	Pad	1784.41	1650.40	1.081
Noz N7	Nozzle	1641.66	1522.22	1.078
Noz K2A	Nozzle	1641.66	1522.22	1.078
Noz K2B	Nozzle	1641.66	1522.22	1.078
Noz K3	Nozzle	1641.66	1522.22	1.078
Noz K4	Nozzle	1641.66	1522.22	1.078
Noz K5	Nozzle	1641.66	1522.22	1.078
Noz K1A	Nozzle	1641.66	1522.22	1.078
Noz K1B	Nozzle	1641.66	1522.22	1.078
MH1	Nozzle	1641.66	1522.22	1.078
MH1	Pad	1784.41	1650.40	1.081

Minimum

1.072

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
TS END COVER	1682.44	1568.83	1.072
TS FLG1	1682.44	1568.83	1.072
CH. SHELL	1784.41	1650.41	1.081
SH.1	1784.41	1650.41	1.081
SH.2	1784.41	1650.41	1.081



SH.3	1784.41	1650.41	1.081
DISH END	1784.41	1650.41	1.081

Minimum 1.072

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1407.13	1308.09	1.076
Tubesheet Material	1682.44	1568.83	1.072

Minimum 1.072

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
NOZZLE N1	1330.88	2421.70	0.550
NOZZLE N2	1330.88	2421.70	0.550
Noz N9	645.09	2324.83	0.277
Noz N8	93.68	2324.83	0.040
Noz N4	274.55	2324.83	0.118
Noz N5	274.55	2324.83	0.118
Noz N3	274.55	2324.83	0.118
Noz N6	94.00	2324.83	0.040
Noz N7	64.24	2324.83	0.028
Noz K2A	64.24	2324.83	0.028
Noz K2B	64.24	2324.83	0.028
Noz K3	52.73	2324.83	0.023
Noz K4	64.24	2324.83	0.028
Noz K5	64.24	2324.83	0.028
Noz K1A	64.24	2324.83	0.028
Noz K1B	64.24	2324.83	0.028
MH1	300.41	2324.83	0.129

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: U-TUBE SHEET

Shell Data:

Main Shell Description: SH.1

Shell Test Pressure		31.92 kgf/cm ²
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1784.41 kgf/cm ²
Shell Allowable Stress at Ambient		1784.41 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Test Pressure		220.53 kgf/cm ²
Channel Thickness	tc	68.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1784.41 kgf/cm ²
Channel Allowable Stress at Ambient		1784.41 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1066
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	25.0000 mm
Total Straight Tube Length	Lt	8297.00 mm
Straight Tube Length (bet. inner tubsht faces) L		7987.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		1407.13 kgf/cm ²
Tube Allowable Stress At Ambient		1407.13 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2090.31 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	32.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.5000 mm
Groove Weld Leg	ag	2.5000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	584.500 mm
Largest Center-to-Center Tube Distance	Ul	47.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-350 LF2

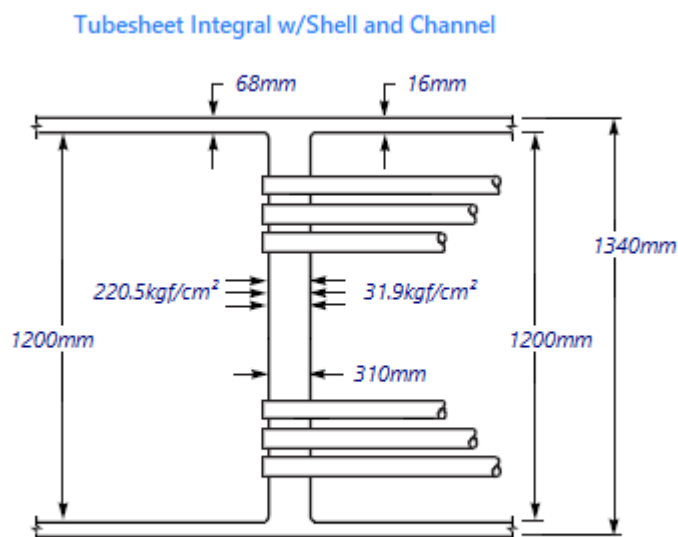
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	1682.44 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1682.44 kgf/cm ²
Thickness of Tubesheet	h	310.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1340.000 mm

Area of the Untubed Lanes	AL	549.4 cm ²
---------------------------	----	-----------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
------------------------------------	---------	---------------------------------



Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 20.0 \cdot (\exp(220.53/(1407.1 \cdot 1.0)) - 1) + 0$
 $= 1.6967 + 0$
 $= 1.6967 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 8571.5 / \sqrt{(12.5 \cdot 2.5)}$
 $= 1533.317$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 / (25.0/2.5)$
 $= 0.0550$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.055 \cdot 2067025 (2.5/25.0)$
 $= 18189.824 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 18189.8/2090.3$
 $= 8.702$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 2090.3/3866.9$
 $= 0.541$

Ratio [AE]:

$= Fhe / E$
 $= 18189.8/2067025$
 $= 0.008800$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.541) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
 $= 0.134$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 2090.3(1 + 0.002) / \ln(1 + 0.002)^{0.134034}$
 $= 4818.264 \text{ kgf/cm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f(m2f2 - R)$
 $= 0.6(1.0 - 0.541)$
 $= 0.276$

Curve Fitting Constant [A2]:

$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 $= (3866.9 \cdot \exp(0.276)) / (0.276^{0.276})$
 $= 7266.839 \text{ kgf/cm}^2$

Material Parameter for Stress-Strain Curve Model [K]:

$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.5411.5 - 0.5 \cdot 0.5412.5 - 0.5413.5$
 $= 0.373$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	1933	-2.474	0.212E-05	0.205E-05	0.770E-05	-0.724E-05	0.196E+06	0.107E-02
2	966	-5.395	0.241E-07	0.241E-07	0.125E-05	-0.125E-05	0.188E+07	0.829E-02
3	1450	-3.934	0.331E-06	0.330E-06	0.361E-05	-0.360E-05	0.866E+06	0.713E-02
4	1691	-3.204	0.896E-06	0.889E-06	0.542E-05	-0.535E-05	0.428E+06	0.485E-02
5	1812	-2.839	0.140E-05	0.138E-05	0.650E-05	-0.632E-05	0.291E+06	0.257E-02
6	1873	-2.656	0.173E-05	0.169E-05	0.708E-05	-0.680E-05	0.239E+06	0.964E-03
7	1903	-2.565	0.192E-05	0.186E-05	0.738E-05	-0.702E-05	0.216E+06	0.525E-05
8	1918	-2.520	0.202E-05	0.195E-05	0.754E-05	-0.713E-05	0.206E+06	0.519E-03
9	1910	-2.542	0.197E-05	0.191E-05	0.746E-05	-0.708E-05	0.211E+06	0.253E-03
10	1907	-2.554	0.194E-05	0.188E-05	0.742E-05	-0.705E-05	0.214E+06	0.123E-03
11	1905	-2.560	0.193E-05	0.187E-05	0.740E-05	-0.704E-05	0.215E+06	0.585E-04
12	1904	-2.562	0.192E-05	0.187E-05	0.739E-05	-0.703E-05	0.216E+06	0.266E-04
13	1903	-2.564	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.107E-04
14	1903	-2.565	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.270E-05
15	1903	-2.565	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.128E-05
16	1903	-2.565	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.708E-06

Factor of Safety [FS]:

Since Fic > 0.55 Fy and Fic < Fy

= 2.407 - 0.741 • Fic / Fy

= 2.407 - 0.741 • 1903.41/2090.31

= 1.732

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 1903.41/1.732

= 1098.805 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 1098.805 (2.5/25.0)

= 219.761 kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.

1.6967 mm

Allowable Internal Pressure at Corroded thickness

313.99 kgf/cm²

Required Internal Design Pressure

220.53 kgf/cm²

Allowable External Pressure at Corroded thickness

219.76 kgf/cm²

Required External Design Pressure

31.92 kgf/cm²

Required Thickness due to Shell Side pressure

0.7049 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= 2 • ro + dt

= 2 • 584.5 + 25.0

= 1194.000 mm

Basic ligament efficiency for shear [μ]:

= (p - dt) / p

= (32.0 - 25.0)/32.0

= 0.219

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

= Ds / Do (Configurations a, b, c)



$$= 1200.0/1194.0$$

$$= 1.0050$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1200.0/1194.0$$

$$= 1.0050$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/310.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2t_t)$$

$$= \max(25.0 - 2 \cdot 2.5 (2067025/2067025) \cdot$$

$$(1407/1682) \cdot (0); 25.0 - 2 \cdot 2.5)$$

$$= 25.0000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 32.0 / \sqrt{ (1 - 4 \cdot \min(549.43; 4 \cdot 1194.0 \cdot 32.0) / (\pi \cdot 1194.0^2)) }$$

$$= 32.8152 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.8152 - 25.0) / 32.8152$$

$$= 0.2382$$

$$h/p = 9.687500; \mu^* = 0.238159; E^* = 416977 \text{ kgf/cm}^2$$

$$E^*/E = 0.201728; \nu^* = 0.409245$$

Note: As h/p (9.688) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ ((1200.0 + 16.0) 16.0) }$$

$$= 0.0130 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.013 \cdot 2067025 \cdot 16.0^3 / (6 (1 - 0.3^2))$$

$$= 20208870.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (1200.0 + 16.0) \cdot 20208870 / 310.0^3 \cdot (1 + 310.0 \cdot 0.013 + (310.0 \cdot 0.013)^2 / 2)$$

$$= 65336.6 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 1200.0^2 / (4 \cdot 2067025 \cdot 16.0) \cdot (1 - 1200.0 \cdot 0.3 / (2 (1200.0 + 16.0)))$$

$$= 0.92739\text{E-}02 \text{ mm/kgf/cm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.005 \cdot 20208870 \cdot 0.013 \cdot 0.009274 (1 + 310.0 \cdot 0.013)$$

$$= 12372.2441 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ ((1200.0 + 68.0) 68.0) }$$

$$= 0.0062 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6 (1 - \nu_s^2))$$



$$= 0.006 \cdot 2067025 \cdot 68.0^3 / (6(1 - 0.3^2))$$

$$= 736922176.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1200.0 + 68.0 / \cos(0)) \cdot 0.73692E+09 / 310.0^3 \cdot (1.0 + 310.0 \cdot 0.006 \cdot (1.0 + 1.0) / 2 + (310.0 \cdot 0.006)^2 / 2 \cdot 1.0)$$

$$= 895927.4 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1200.0^2 / (4 \cdot 2067025 \cdot 68.0) \cdot (1 - 1200.0 \cdot 0.3 / (2(1200.0 + 68.0)))$$

$$= 0.21976E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.005 \cdot 0.73692E+09 [0.002198 (1.0 + 0.006 \cdot 1.0 \cdot 310.0) - 0 (1.0 / 310.0 + 1.0 \cdot 0.006 / 2)]$$

$$= 29413.7266 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1340.0 / 1194.0$$

$$= 1.1223$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 416977 \cdot (65336.55 + 895927 + 2067025 \cdot \ln(1.12))$$

$$= 1.6997$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1194.0^2 / 16 \cdot [(1.005 - 1) \cdot (1.005^2 + 1) \cdot 31.922 - (1.005 - 1) \cdot (1.005^2 + 1) \cdot 220.53]$$

$$= -169750.3281 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -169750.32 + 29413.727 \cdot 220.53 - 12372.244 \cdot 31.922$$

$$= 5921900.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F(P_s - P_t) / (1 + F)$$

$$= ((5921900) - 1194.0^2 / 32 \cdot 1.7 (31.92 - 220.53)) / (1 + 1.7)$$

$$= 7483769.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 7483769 + 1194.0^2 / 64 \cdot (3 + 0.409) \cdot (31.92 - 220.53)$$

$$= -6839619.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|7483769|; |-6839619.5|)$$

$$= 7483769.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$



ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 7483769 / (0.2382 (310.0 - 8.0)^2) \\ &= 2067.2395 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 2549.2 \\ &= 4843.40 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 7521230 / (0.2382 (206.0811 - 8.0)^2) \\ &= 4829.3301 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 206.0811 + 0 + 0 \\ &= 206.0811 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(206.0811; 132.845) \\ &= 206.0811 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |31.92 - 220.53| \\ &= 188.607 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ &= 4 \cdot 0.219 \cdot 310.0 / 1194.0 \cdot 0.8 \cdot 0.95 \cdot 2549.156 \\ &= 440.124 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.219) \cdot (1194.0 / 1937.359) \cdot |31.92 - 220.53| + 0 \\ &= 132.8450 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 16.000mm

Min. Shell len. adjacent to the tubesheet is: 249.415 mm

Note: For a given Channel thickness of: 68.000 mm

Min. Channel len. adjacent to the tubesheet is: 514.183mm

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.0 - 2.5) 2.5 \\ &= 1.767 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.5 (25.0 - 2.5) 1985.793 \\ &= 3509.180 \text{ kgf} \end{aligned}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$



$$= 0.85 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1985.79$$

$$= 3509.1802 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 1985.793/1985.793$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1985.79$$

$$= 2288.1804 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.0)^2 + 1.07 \cdot 2.5 \cdot (25.0 - 2.5) 1.0 \cdot 1.0)} - 0.75 \cdot 25.0 \right)$$

$$= 3.083 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.5416 mm

Minimum Required Groove Weld Leg [agr]: 1.5416 mm

Tube-Tubesheet Jt allowable, 3509.18 is $\geq$ tube strength 3509.18 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	2067.2 $\leq$	4843.4 kgf/cm ²	Ok
Tubesheet shear stress	830.2 $\leq$	1937.4 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	206.081	310.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.542	2.500	Ok
Tube-Tubesheet Groove Weld Leg :	1.542	2.500	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	220.032 ...	2352	4843 971 1937	Ps+Pt D1	Ok
D2uc	92.046 ...	385	4843 141 1937	Ps+Pt D2	Ok
D3uc	206.081 ...	2067	4843 830 1937	Ps+Pt D3	Ok
Max:	220.0315 ... mm	0.486	0.501 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures



[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Shell and Channel Stress Summary:							
-- Integral Cyl. Req'd. Thk.		Shell	Stress	Channel	Stress	Pass	
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	...	3778	...	3778	Ok
D2uc	...	...	...	3778	...	3778	Ok
D3uc	...	...	...	3778	...	3778	Ok
Max	...	...	...	...	...	...	

Note:
 Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	220.032	310.000	Ok
Tube thickness :	1.697	2.500	Ok
Tube-tubesheet fillet weld leg :	1.542	2.500	Ok
Tube-tubesheet groove weld leg :	1.542	2.500	Ok

Given Shell Thickness: 16.0000 mm

Given Channel Thickness: 68.0000 mm

Min Shell length of thk, (16.000) adj. to tubesheet: 249.415 mm

Min Channel length of thk, (68.000) adj. to tubesheet: 514.183 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
20	30	No Calc	...	301.5	No Calc	No Calc
30	40	No Calc	...	291.5	No Calc	No Calc
40	Tsht	125.497	1336	66.5	No Calc	No Calc
Tsht	50	No Calc	...	66.5	No Calc	No Calc
50	60	30.0032	1232	14.5	No Calc	No Calc
60	70	191	2348.47	19.5	No Calc	No Calc
70	80	666.167	2342	19.5	No Calc	No Calc
80	90	No Calc	2334	15.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
20	30	303	198.704	...	No Calc
30	40	293	286.283	...	No Calc
40	Tsht	68	No Calc	...	151.005
Tsht	50	68	No Calc	...	No Calc
50	60	16	No Calc	...	31.1031
60	70	21	8.9296	1.05461	10.2094
70	80	21	12.3401	1.05461	4.65354
80	90	17	6.91643	1.05461	8.32114

Minimum

5

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
20	30	No Calc	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	125.497	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	30.0032	No Calc	No Calc	No Calc
60	70	191	191	No Calc	No Calc
70	80	666.167	2710.82	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
20	30	5797.61	...	5797.61	...	579.76
30	40	3192.81	457471	3175.13	458065	319.281
40	50	2634.78	1419375	2579.71	1426480	263.478
50	60	142.117	182311	128.953	184010	14.2117
60	70	1999.57	3745909	1858.55	3764104	199.957
70	80	7619.04	23150904	7079.41	23220530	761.904
80	90	1108.72	1800398	1029.52	1807181	110.872
Total		22494	30756370.00	21648	30860372.00	2249



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
20	Insl	8.99574	15.15	...	...	Ins: 20
30	Insl	5.38956	19.9	...	...	Ins: 30
40	Liqd	1418.73	62.75	...	...	Liquid: 40
40	Insl	15.5395	62.75	...	...	Ins: 40
40	Nozl	419.579	62.5	79.3675	...	NOZZLE N1
40	Nozl	478.862	62.5	-79.3675	...	NOZZLE N2
40	Nozl	29.1176	35	...	62.1425	Noz N9
50	Liqd	182.229	15	...	...	Liquid: 50
50	Insl	3.43986	15	...	...	Ins: 50
50	Nozl	8.23609	15	-62.1425	...	Noz N8
60	Liqd	3744.21	75.3861	-27.5	...	Liquid: 60
60	Insl	35.7895	95.5	...	...	Ins: 60
70	Sadl	712.769	55	130.85	...	Left Saddle
70	Sadl	712.769	480	130.85	...	RIGHT SADDLE
70	Liqd	23140.5	321	...	...	Liquid: 70
70	Insl	136.371	321	...	...	Ins: 70
70	Nozl	82.4105	60	127.144	...	Noz N4
70	Nozl	82.4105	320	127.144	...	Noz N5
70	Nozl	82.4105	210	65.1463	109.186	Noz N3
70	Nozl	10.5063	610	-117.143	...	Noz N6
70	Nozl	12.7368	210	-100.376	61.1584	Noz N7
70	Nozl	10.6549	13.5	60.2255	100.938	Noz K2A
70	Nozl	10.6549	13.5	-60.2255	100.938	Noz K2B
70	Nozl	8.54578	320	59.9002	100.393	Noz K3
70	Nozl	11.5471	150	117.54	...	Noz K4
70	Nozl	11.5471	230	117.54	...	Noz K5
80	Liqd	1799.59	24.1667	...	...	Liquid: 80
80	Insl	24.4747	31.25	...	...	Ins: 80
80	Nozl	10.6549	52.4342	65	...	Noz K1A
80	Nozl	10.6549	52.4342	-65	...	Noz K1B
80	Nozl	834.359	62.5	...	...	MH1
40	FTsh	2344.2	141	...	...	U-TUBE SHEET
40	Tube	14589.7	540.35	...	...	Tubes

Total Weight of Each Detail Type:

Saddles	1425.5
Liquid	30285.2
Insulation	230.0
Nozzles	2114.9
Exchanger Components	16933.9



Sum of the Detail Weights 50989.6 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	24744.1	24744.1	24744.1	24744.1	24744.1	24744.1
Saddles	1425.5	1425.5	1425.5	1425.5	1425.5	1425.5
Nozzles	2114.9	2114.9	2114.9	2114.9	2114.9	2114.9
Exchanger	16933.9	16933.9	16933.9	16933.9	16933.9	16933.9
Insulation	...	...	230.0	230.0	230.0	230.0
Ope. Liquid	...	...	...	...	...	30285.2
Test Liquid	...	30742.5	...	...	...	...
Tube Tst Lqd	...	3097.1	...	...	...	...
Totals	45218.4	79058.0	45448.4	45448.4	45448.4	75733.7

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	45218.4 kg
Shop Test Wt.	- Fabricated weight + water (full)	79058.0 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	45448.4 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	45448.4 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	45448.4 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	75733.7 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	74803.3 kg

Exchanger Tube Data

Volume of Exchanger tubes :	3098459.2 cm ³
Weight of Ope Liq in tubes :	0.0 kg
Weight of Water in tubes :	3097.1 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
20	30	41150.9	...
30	40	33645.7	15192.7
40	50	52674.5	47312.4
50	60	11611.3	11309.7
60	70	112301	109275
70	80	472359	463888
80	90	63135	60955.7

Total	786877.1	707932.8 cm ²
Total	847.0	762.0 ft ²



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Noz N9	244.39	260.14	80	1500	GR 1.1	...	...	...	...
Noz N8	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz N4	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz N5	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz N3	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz N6	97.77	104.01	80	600	GR 1.1	...	...	...	...
Noz N7	97.77	104.01	80	600	GR 1.1	...	...	...	...
Noz K2A	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K2B	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K3	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K4	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K5	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K1A	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K1B	48.89	52.01	80	300	GR 1.1	...	...	...	...
MH1	48.89	52.01	80	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 48.887 kgf/cm²
Lowest Flange Pressure Rating was (Amb)[ShellSide]: 52.007 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 244.394 kgf/cm²
Lowest Flange Pressure Rating was (Amb)[TubeSide]: 260.136 kgf/cm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Use Gust Response Factor (Dynamic Analysis)	No	

Calculated Values:

Vessel Length to Diameter ratio: 5.499

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 0.9 • 1.0 • 0.6 • 57.5²)

= 1778.7 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.7 • 0.81 • 1778.723

= 990.1 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
20	1.80	1.0000	1.0000	1.0000	57.50	0.8100
30	1.80	1.0000	1.0000	1.0000	57.50	0.8100
40	1.80	1.0000	1.0000	1.0000	57.50	0.8100
50	1.80	1.0000	1.0000	1.0000	57.50	0.8100
60	1.80	1.0000	1.0000	1.0000	57.50	0.8100
70	1.80	1.0000	1.0000	1.0000	57.50	0.8100
80	1.80	1.0000	1.0000	1.0000	57.50	0.8100

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
20	30	180	226.8	6872.04	181.454	111.062
30	40	180	160.8	6399.84	181.454	103.43
40	50	180	177.12	22228.6	181.454	359.244
50	60	180	164.64	4939.2	181.454	79.8243
60	70	180	231.84	44281.4	181.454	715.649
70	80	180	297.84	191213	181.454	3090.27
80	90	180	296.88	14347.6	181.454	231.876

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.0939 • 1.0) • 1.0
= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 8153.09 • 0.0939 • 1.0
= 765.2 kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.0939 • 73378 • 1.0
= 4591.3 kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------



From	To	Height cm	Weight kgf	Ope Load kgf
20	30	87.5	8153.09	765.217
30	40	60	8153.09	765.217
40	50	60	8153.09	765.217
50	60	60	8153.09	765.217
60	70	60	8153.09	765.217
70	Sad1	115	8153.09	765.217
Sad1	80	115	8153.09	765.217
70	80	115	8153.09	765.217
80	90	115	8153.09	765.217

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	715.418 cm
Center of Gravity of the Liquid	702.495 cm
Center of Gravity of the Insulation	644.440 cm
Center of Gravity of the Nozzles	624.306 cm
Center of Gravity of the Tubesheet(s)	211.417 cm
Center of Gravity of the Tubes	610.767 cm
Center of Gravity of Bare Shell New and Cold	376.294 cm
Center of Gravity of Bare Shell Corroded	364.832 cm
Vessel CG in the Operating Condition	559.577 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	466.595 cm
Vessel CG in the Test Condition	561.561 cm

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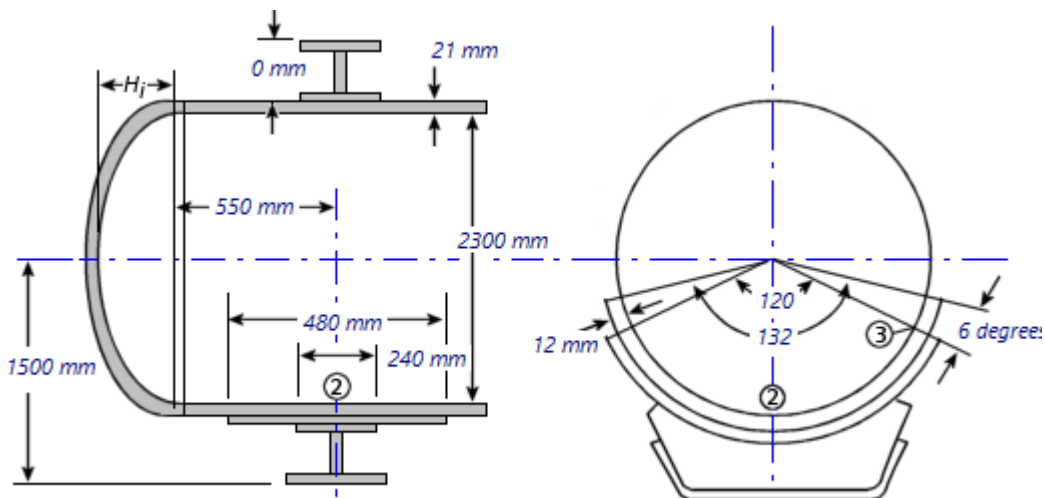


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	868.00	cm
Distance from Saddle to Vessel tangent	a1 or a	550.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1650.86	kgf/cm ²
Shell Allowable Stress used in Calculation		1650.41	kgf/cm ²
Head Allowable Stress used in Calculation		1568.83	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		73976.26	kgf
Pressure used in Analysis	P	23.569	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		549.76	1650.41
Long. Stress at Bottom of Midspan		853.81	1650.41
Long. Stress at Top of Saddles		697.95	1650.41
Long. Stress at Bottom of Saddles		705.62	1650.41
Tangential Shear in Shell		287.45	1320.33
Tangential Shear in Head		19.23	1252.34
Circ. Stress at Horn of Saddle		281.81	2063.01
Circ. Compressive Stress in Shell		37.61	1650.41

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(4691.4/2 + 0) \cdot 1500.0/2049.0164$$

$$= 5151.5 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl.wind; Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(1390.64, 0) \cdot 1500.0/4250.0$$

$$= 490.8 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl.eq; Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(6886.95, 0) \cdot 1500.0/4250.0$$

$$= 2430.7 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(6887/2 + 0) \cdot 1500.0/2049.0164$$

$$= 7562.5 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 66414 + \max(491, 5152, 2431, 7562)$$

$$= 73976.3 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1779.443(0.0 + \pi/4(2.482 \cdot 1.2)^2)$$

$$= 13637.541 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	74689.02	kgf
Transverse Shear Load	3443.48	kgf
Longitudinal Shear Load	6886.95	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1161.25)$$

$$= 2049.016 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -73976 \cdot 55.0 [1 - (1 - 55.0/868.0 + (116.125^2 - 0.0^2)/(2 \cdot 55.0 \cdot 868.0))/(1 + (4 \cdot 0.0)/(3 \cdot 868.0))]$$

$$= 3168.3 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 73976 \cdot 868/4((1 + 2(116^2 - 0^2)/(868^2))/(1 + (4 \cdot 0)/(3 \cdot 868)) - 4 \cdot 55/868)$$

$$= 125588.1 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2t)$$

$$= 23.569 \cdot 1161.25/(2 \cdot 19.5) - 125588.1/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 549.76 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 23.569 \cdot 1161.25/(2 \cdot 19.5) + 125588.1/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 853.81 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 23.569 \cdot 1161.25/(2 \cdot 19.5) - 3168.3/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 697.95 \text{ kgf/cm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 23.569 \cdot 1161.25/(2 \cdot 19.5) + 3168.3/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 705.62 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 73976(868.0 - 2 \cdot 55.0)/(868.0 + (4 \cdot 0.0/3))$$

$$= 64601.4 \text{ kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K3 \cdot Q / (Rm \cdot t)$$

$$= 0.8799 \cdot 73976/(1161.25 \cdot 19.5)$$

$$= 287.45 \text{ kgf/cm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K3 \cdot Q / (Rm \cdot th)$$

$$= 0.8799 \cdot 73976/(1161.25 \cdot 291.5)$$

$$= 19.23 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 55.0)$$

$$= 474.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(1650.865/1650.408, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 73976 \cdot 0.1/(474.75(19.5 + 1.0 \cdot 12.0))$$

$$= -37.61 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q/(4(t+\eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm/(L(t+\eta \cdot tr)^2)$$

$$= -73976/(4(19.5 + 1.0 \cdot 12.0)474.75) -$$

$$12 \cdot 0.013 \cdot 73976 \cdot 1161.25/(868.0(19.5 + 1.0 \cdot 12.0)^2)$$

$$= -281.81 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 4250.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 4250.0(80.0 - 21.1)$$

$$= 2.996 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm



Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)^3) / 12$
 $= 44.104 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 44.104 + 139.184 (113.925 - 9.75)^2$
 $= 15148.791 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.151E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.451E+04
Web	22.0	291.0	177.0	64.0	1133153.4	0.706E+04
BasePlate	270.0	26.0	335.5	70.2	2355209.2	0.345E+05
Totals	...	...	...	331.0	3770946.5	0.612E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 3770947 / 331.004$
 $= 113.925 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 73976.258$
 $= 15055.7725 \text{ kgf}$

Tension Stress, St = (Fh/As) = 78.4892 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1486.6948 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1500.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472$
 $= 547.717 \text{ mm}$

Bending Moment, M = Fh * d = 8246.3125 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 153.4341 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1651.8832 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(73976 + 713)270.0 / (4 \cdot 2960.0 \cdot 1651.883))}$
 $= 17.588 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$



$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 2049.0161 / (4 - 1)$
 $= 683.005 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 683.0054 \cdot 270.0 / 2$
 $= 922.057 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 74689.031 / (2960.0 \cdot 270.0)$
 $= 9.345 \text{ kgf/cm}^2$

Axial Load [P]:
 $= \text{Ap} \cdot \text{Bp}$
 $= 922.1 \cdot 9.35$
 $= 8617.062 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 47.96 + 75.131$
 $= 123.091 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / \text{Ar}$
 $= 8617.1 / 123.0906$
 $= 70.006 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	240.0	...	52.8	...	0.253E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 240.0 - 22.0$
 $= 218.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= \text{AY} / \text{A}$
 $= 0.0 / 123.091$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 240.0 / 2$
 $= 120.000 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$
 $= \sqrt{ (2534.4 / 123.091) }$
 $= 45.376 \text{ mm}$

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1500.0 - \cos(120.0 / 2) (1150.0 + 21.0 + 12.0) - 26.0$
 $= 882.500 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$
 $= \sqrt{ (2 \cdot \pi^2 \cdot 2038903 / 2477.8) }$
 $= 127.446$



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 882.5/45.376$$

$$= 19.449$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 6887.0 / (2 \cdot 2960.0) \cdot 683.005 \cdot 882.5/2$$

$$= 350.602 \text{ kgf-m.}$$

Compressive Allowable, KL/r < Cc (19.4487 < 127.4465) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.45)^2 / (2 \cdot 127.45^2)) 2478 /$$

$$(5/3 + 3 \cdot (19.45) / (8 \cdot 127.45) - (19.45^3) / (8 \cdot 127.45^3))$$

$$= 1421 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 70.01/1420.97 + (350.6 \cdot 120.0/25344000) / 1651.88$$

$$= 0.150$$

AISC Unity Check of Outside Ribs, due to Transverse Force Ft (must be ≤ 1)

This analysis assumes the entire saddle weight load and load due to transverse forces are concentrated on the outer rib + some portion of the web. This generally would not be the case and as such this analysis is very conservative.

Inertia of Saddle, Outer Ribs - Transverse Direction:

	B	D	Y	A	AY	Io
Rib	218.0	22.0	11.0	48.0	52756.0	0.500E+04
Web	22.0	352.0	176.0	77.4	1362944.0	0.111E+05
Totals	...	...	...	125.4	1415700.0	0.161E+05

Web dimension [D]:

$$= 16 \cdot \text{Web Thickness}$$

$$= 16 \cdot 22.0$$

$$= 352.000 \text{ mm}$$

Moment at base of outer Rib [Mbase]:

$$= Ft (\text{Centerline dimension B} - \text{Baseplate thickness})$$

$$= 3443.475 (1500.0 - 26.0)$$

$$= 5075.688 \text{ kgf-m.}$$

Bending Stress due to Transverse Force [$\sigma_{B,base}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5075.688/1424214$$

$$= 356.385 \text{ kgf/cm}^2$$

Vertical Load on the Rib due to the Bending Moment [Fv,mom]:

$$= \text{Bending Moment} / \text{Web Length}$$

$$= 5075.688/2049.016$$

$$= 2477.132 \text{ kgf}$$

Total Vertical Load on the Rib [Fv,total]:

$$= Fv,mom + \text{Vertical Load}$$

$$= 2477.132 + 74689.023$$

$$= 77166.156 \text{ kgf}$$

Compressive Stress on Outer Rib [σ_{cr}]:

$$= Fv,total / \text{Area}$$

$$= 77166.156/125.4$$

$$= 615.361 \text{ kgf/cm}^2$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (16078.6/125.4) }$$

$$= 113.234 \text{ mm}$$



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 882.5/113.234$$

$$= 7.794$$

Compressive Allowable, $KL/r < C_c$ (7.7936 < 127.4465) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.79)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (7.79) / (8 \cdot 127.45) - (7.79^3) / (8 \cdot 127.45^3))$$

$$= 1464 \text{ kgf/cm}^2$$

Unity Check on Outer Rib [UCheck,outer]:

$$= \sigma_{cr} / Sca + \sigma_{B,base} / Sba$$

$$= 615.361/1463.799 + 356.385/1651.883$$

$$= 0.636$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	218.0	0.0	48.0	0.0	0.253E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	198.2	...	0.259E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/198.221$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0/2$$

$$= 120.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1500.0 - \sqrt{ (1183.0 + 12.0)^2 - (683.005/2)^2 } - 26.0$$

$$= 341.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2593.1/198.221 }$$

$$= 36.169 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341.364/36.169$$

$$= 9.438$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 6886.95 / (2 \cdot 2960.0)$$

$$= 1.163 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.163 \cdot 683.005 \cdot 341.364$$

$$= 271.236 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 271.236/216087.75$$

$$= 125.521 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (9.4381 < 127.4465) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.44)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (9.44) / (8 \cdot 127.45) - (9.44^3) / (8 \cdot 127.45^3))$$

$$= 1458 \text{ kgf/cm}^2$$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 86.5/1458.36 + (271.24 \cdot 120.0/2593.051)/1651.88$$

$$= 0.135$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1650.86 kgf/cm ²
Area Available in a Single Bolt	BlArea	2.9291 cm ²
Saddle Load QO (Weight)	QO	67126.5 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2430.7 kgf
Maximum Transverse Force	Ft	3443.5 kgf
Maximum Longitudinal Force	Fl	6887.0 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6887.0 / (2.929 \cdot 4)$$

$$= 587.8 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3443.5 / (2.929 \cdot 4)$$

$$= 293.9 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 6886.95 / (1101.26 \cdot 4.0)$$

$$= 1.5634 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1500.0 \cdot 3443.48 + 0.0$$

$$= 5165.22 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 5165.22/67126.55$$

$$= 76.95 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 1.5634, 0.0]$$

$$= 1.5634 \text{ cm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

This element is Not Post Weld Heat Treated

$$\text{Computed Minimum Design Metal Temperature} : 1 \text{ } ^\circ\text{C}$$



User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 1 °C
 Temperature reduction per Graph 3.12 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (15.47 \cdot 1.0) / (12.0 - 1.5 - 0.0)$
 $= 1.000$

Warning: Impact Testing Is Mandatory!

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	868.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1650.86	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	58.43	cm
Shell Allowable Stress used in Calculation		1650.41	kgf/cm ²
Head Allowable Stress used in Calculation		1650.41	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		19117.78	kgf
Pressure used in Analysis	P	23.569	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		698.44	1650.41
Long. Stress at Bottom of Midspan		705.13	1650.41
Long. Stress at Top of Saddles		807.04	1650.41
Long. Stress at Bottom of Saddles		643.44	1650.41
Tangential Shear in Shell		50.99	1320.33
Circ. Stress at Horn of Saddle		195.44	2063.01
Circ. Compressive Stress in Shell		9.72	1650.41

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(4691.4/2 + 0) \cdot 1500.0/2049.0164$
 $= 5151.5 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(1390.64, 0) \cdot 1500.0/4250.0$
 $= 490.8 \text{ kgf}$



Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl, eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(6886.95, 0) \cdot 1500.0/4250.0$$

$$= 2430.7 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(6887/2 + 0) \cdot 1500.0/2049.0164$$

$$= 7562.5 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 11555 + \max(491, 5152, 2431, 7562)$$

$$= 19117.8 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1779.443(0.0 + \pi/4(2.482 \cdot 1.2)^2)$$

$$= 13637.541 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	19830.55	kgf
Transverse Shear Load	3443.48	kgf
Longitudinal Shear Load	6886.95	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2300.0/2 + 21.0 + 12.0) \sin(120.0/2), 2 \cdot 1161.25)$$

$$= 2049.016 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -19118 \cdot 190.0 [1 - (1 - 190.0/868.0 + (116.125^2 - 58.425^2) / (2 \cdot 190.0 \cdot 868.0)) / (1 + (4 \cdot 58.43) / (3 \cdot 868.0))]$$

$$= -9269.9 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 19118 \cdot 868/4((1 + 2(116^2 - 58^2)/(868^2)) / (1 + (4 \cdot 58) / (3 \cdot 868)) - 4 \cdot 190/868)$$

$$= 2763.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 23.569 \cdot 1161.25/(2 \cdot 19.5) - 2763.0/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 698.44 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 23.569 \cdot 1161.25/(2 \cdot 19.5) + 2763.0/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 705.13 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 23.569 \cdot 1161.25/(2 \cdot 19.5) - 9269.9/(0.1066 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 807.04 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:



$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 23.569 \cdot 1161.25 / (2 \cdot 19.5) + 9269.9 / (0.1923 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 643.44 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 19118(868.0 - 2 \cdot 190.0) / (868.0 + (4 \cdot 58.43/3))$$

$$= 9863.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 9863.07 / (1161.25 \cdot 19.5)$$

$$= 50.99 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 190.0)$$

$$= 474.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(1650.865/1650.408, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 19118 \cdot 0.1 / (474.75(19.5 + 1.0 \cdot 12.0))$$

$$= -9.72 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -19118 / (4(19.5 + 1.0 \cdot 12.0)474.75) -$$

$$12 \cdot 0.053 \cdot 19118 \cdot 1161.25 / (868.0(19.5 + 1.0 \cdot 12.0)^2)$$

$$= -195.44 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)19.5^3) / 12$$

$$= 44.104 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 44.104 + 139.184(113.925 - 9.75)^2$$

$$= 15148.791 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.151E+05



Wearplate	480.0	12.0	25.5	57.6	146880.0	0.451E+04
Web	22.0	291.0	177.0	64.0	1133153.4	0.706E+04
BasePlate	270.0	26.0	335.5	70.2	2355209.2	0.345E+05
Totals	...	...	...	331.0	3770946.5	0.612E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3770947/331.004$$

$$= 113.925 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 19117.783$$

$$= 3890.8833 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 20.2841 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1486.6948 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1500.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 547.717 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2131.1057 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 39.6522 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1651.8832 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) }$$

$$= \sqrt{ 3(19118 + 713)270.0 / (4 \cdot 2960.0 \cdot 1651.883) }$$

$$= 9.062 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$$

$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2049.0161 / (4 - 1)$$

$$= 683.005 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 683.0054 \cdot 270.0/2$$

$$= 922.057 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 19830.551 / (2960.0 \cdot 270.0)$$

$$= 2.481 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 922.1 \cdot 2.48$$

$$= 2287.901 \text{ kgf}$$



Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 47.96 + 75.131$$

$$= 123.091 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 2287.9/123.0906$$

$$= 18.587 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	240.0	...	52.8	...	0.253E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 240.0 - 22.0$$

$$= 218.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/123.091$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0/2$$

$$= 120.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2534.4/123.091)}$$

$$= 45.376 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1500.0 - \cos(120.0/2) (1150.0 + 21.0 + 12.0) - 26.0$$

$$= 882.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903/2477.8)}$$

$$= 127.446$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 882.5/45.376$$

$$= 19.449$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 6887.0 / (2 \cdot 2960.0) \cdot 683.005 \cdot 882.5/2$$

$$= 350.602 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($19.4487 < 127.4465$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.45)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (19.45) / (8 \cdot 127.45) - (19.45^3) / (8 \cdot 127.45^3))$$

$$= 1421 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 18.59/1420.97 + (350.6 \cdot 120.0/25344000) / 1651.88$$

$$= 0.114$$

AISC Unity Check of Outside Ribs, due to Transverse Force Ft (must be ≤ 1)



This analysis assumes the entire saddle weight load and load due to transverse forces are concentrated on the outer rib + some portion of the web. This generally would not be the case and as such this analysis is very conservative.

Inertia of Saddle, Outer Ribs - Transverse Direction:

	B	D	Y	A	AY	Io
Rib	218.0	22.0	11.0	48.0	52756.0	0.500E+04
Web	22.0	352.0	176.0	77.4	1362944.0	0.111E+05
Totals	...	...	...	125.4	1415700.0	0.161E+05

Web dimension [D]:
 = 16 • Web Thickness
 = 16 • 22.0
 = 352.000 mm

Moment at base of outer Rib [Mbase]:
 = Ft(Centerline dimension B - Baseplate thickness)
 = 3443.475(1500.0 - 26.0)
 = 5075.688 kgf-m.

Bending Stress due to Transverse Force [$\sigma_{B,base}$]:
 = Bending Moment / Section Modulus
 = 5075.688/1424214
 = 356.385 kgf/cm²

Vertical Load on the Rib due to the Bending Moment [Fv,mom]:
 = Bending Moment / Web Length
 = 5075.688/2049.016
 = 2477.132 kgf

Total Vertical Load on the Rib [Fv,total]:
 = Fv,mom + Vertical Load
 = 2477.132 + 19830.551
 = 22307.684 kgf

Compressive Stress on Outer Rib [σ_{cr}]:
 = Fv,total / Area
 = 22307.684/125.4
 = 177.892 kgf/cm²

Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(16078.6/125.4)}$
 = 113.234 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 882.5/113.234
 = 7.794

Compressive Allowable, KL/r < Cc (7.7936 < 127.4465) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (7.79)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (7.79) / (8 \cdot 127.45) - (7.79^3) / (8 \cdot 127.45^3))$
 = 1464 kgf/cm²

Unity Check on Outer Rib [UCheck,outer]:
 = $\sigma_{cr} / Sca + \sigma_{B,base} / Sba$
 = 177.892/1463.799 + 356.385/1651.883
 = 0.337

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	218.0	0.0	48.0	0.0	0.253E+04



Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	198.2	...	0.259E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/198.221$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0/2$$

$$= 120.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1500.0 - \sqrt{(1183.0 + 12.0)^2 - (683.005/2)^2} - 26.0$$

$$= 341.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2593.1/198.221)}$$

$$= 36.169 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341.364/36.169$$

$$= 9.438$$

Unit Force [Force,u]:

$$= F1 / (2 \cdot \text{Baseplate Length})$$

$$= 6886.95/(2 \cdot 2960.0)$$

$$= 1.163 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.163 \cdot 683.005 \cdot 341.364$$

$$= 271.236 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 271.236/216087.75$$

$$= 125.521 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (9.4381 < 127.4465) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.44)^2 / (2 \cdot 127.45^2)) 2478 /$$

$$(5/3 + 3 \cdot (9.44) / (8 \cdot 127.45) - (9.44^3) / (8 \cdot 127.45^3))$$

$$= 1458 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 22.97/1458.36 + (271.24 \cdot 120.0/2593.051) / 1651.88$$

$$= 0.092$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1650.86 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load Q0 (Weight)	Q0	12268.1 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2430.7 kgf
Maximum Transverse Force	Ft	3443.5 kgf



Maximum Longitudinal Force F1 6887.0 kgf
Saddle Bolted to Steel Foundation No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $6887.0 / (2.929 \cdot 4)$
= 587.8 kgf/cm^2 . Must be less than 1101.3 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $3443.5 / (2.929 \cdot 4)$
= 293.9 kgf/cm^2 . Must be less than 1101.3 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $6886.95 / (1101.26 \cdot 4.0)$
= 1.5634 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1500.0 \cdot 3443.48 + 0.0$
= 5165.22 kgf-m .

Eccentricity (e):
= R_{mom} / QO
= $5165.22 / 12268.08$
= $421.03 \text{ mm} < B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 1.5634, 0.0]$
= 1.5634 cm^2

MDMT Calculations for: RIGHT SADDLE, Saddle to Wearplate Junction, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 1 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 1 °C
Temperature reduction per Graph 3.12 : 0 °C
The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(15.47 \cdot 1.0) / (12.0 - 1.5 - 0.0)$
= 1.000

Warning: Impact Testing Is Mandatory!

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (480.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

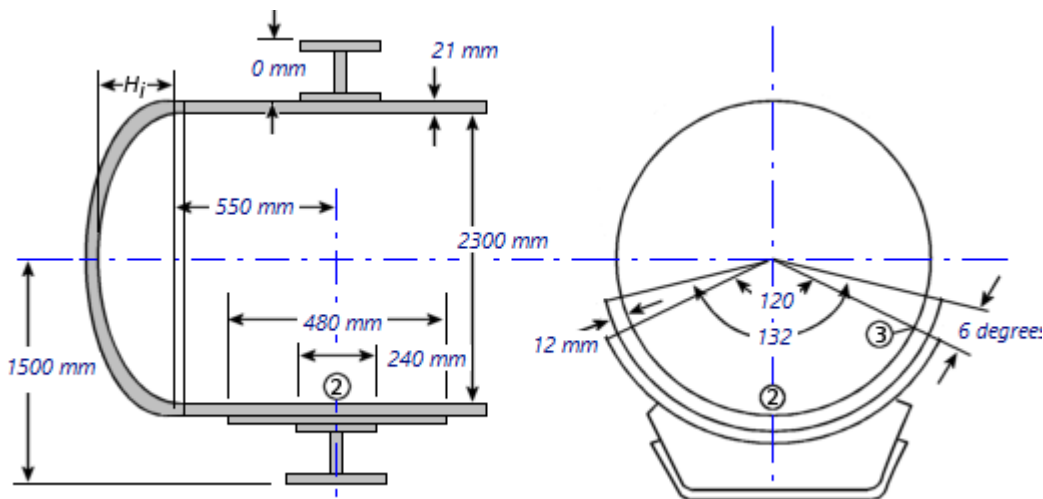
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{1160.5001 \cdot 21.0}, 2 \cdot 550.0)$$

$$= 483.5324 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1160.50	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	868.00	cm
Distance from Saddle to Vessel tangent	a1 or a	550.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		69334.75	kgf
Pressure used in Analysis	P	32.038	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	752.78	2518.57
Long. Stress at Bottom of Midspan	1017.72	2518.57
Long. Stress at Top of Saddles	881.91	2518.57
Long. Stress at Bottom of Saddles	888.58	2518.57
Tangential Shear in Shell	250.34	1413.05
Tangential Shear in Head	17.94	1252.34
Circ. Stress at Horn of Saddle	503.99	3777.85
Circ. Compressive Stress in Shell	51.91	2518.57



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= $F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0(1548.1/2 + 0) \cdot 1500.0/2049.0164$
= 1700.0 kgf

Longitudinal Saddle Reaction Force [Fwl]:
= $\max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
= $\max(458.91, 0) \cdot 1500.0/4250.0$
= 162.0 kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + $\max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
= $67635 + \max(162, 1700, 0, 0)$
= 69334.8 kgf

Longitudinal Wind Force [Fl]:
= $\text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
= $1.1 \cdot 1779.443(0.0 + \pi/4(2.482 \cdot 1.2)^2)$
= 13637.541 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	70047.52	kgf
Transverse Shear Load	774.07	kgf
Longitudinal Shear Load	458.91	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
= $\min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1160.5)$
= 2049.016 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:
= $-Q \cdot a[1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
= $-69335 \cdot 55.0[1 - (1 - 55.0/868.0 + (116.05^2 - 0.0^2)/(2 \cdot 55.0 \cdot 868.0))/(1 + (4 \cdot 0.0)/(3 \cdot 868.0))]$
= 2962.5 kgf-m.

Moment per Equation 4.15.2 [M2]:
= $Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/3L) - 4a/L)$
= $69335 \cdot 868/4((1 + 2(116^2 - 0^2)/(868^2))/(1 + (4 \cdot 0)/(3 \cdot 868)) - 4 \cdot 55/868)$
= 117701.3 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
= $P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
= $32.038 \cdot 1160.5/(2 \cdot 21.0) - 117701.3/(\pi \cdot 1160.5^2 \cdot 21.0)$
= 752.78 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
= $P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
= $32.038 \cdot 1160.5/(2 \cdot 21.0) + 117701.3/(\pi \cdot 1160.5^2 \cdot 21.0)$
= 1017.72 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
= $P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$
= $32.038 \cdot 1160.5/(2 \cdot 21.0) - 2962.5/(\pi \cdot 1160.5^2 \cdot 21.0)$
= 881.91 kgf/cm²



Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= 32.038 \cdot 1160.5 / (2 \cdot 21.0) + 2962.5 / (\pi \cdot 1160.5^2 \cdot 21.0)$$

$$= 888.58 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 69335(868.0 - 2 \cdot 55.0) / (868.0 + (4 \cdot 0.0 / 3))$$

$$= 60548.1 \text{ kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 69335 / (1160.5001 \cdot 21.0)$$

$$= 250.34 \text{ kgf/cm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 69335 / (1160.5001 \cdot 293.0)$$

$$= 17.94 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1160.5 \cdot 21.0)}$$

$$= 121.766 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 69335 \cdot 0.1 / (21.0(240.0 + 121.77 + 121.77))$$

$$= -51.91 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -69335 / (4 \cdot 21.0(240.0 + 121.766 + 121.766)) -$$

$$12 \cdot 0.013 \cdot 69335 \cdot 1160.5 / (868.0 \cdot 21.0^2)$$

$$= -503.99 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1160.5 \cdot 21.0)}, 2 \cdot 55.0)$$

$$= 483.53 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1150.0 \cdot 21.0)} + 480.0) 21.0^3) / 12$$

$$= 55.753 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 55.753 + 151.71(111.295 - 10.5)^2$$

$$= 15468.860 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	722.4	21.0	10.5	151.7	159295.4	0.155E+05
Wearplate	480.0	12.0	27.0	57.6	155520.0	0.410E+04



Web	22.0	291.0	178.5	64.0	1142756.4	0.741E+04
BasePlate	270.0	26.0	337.0	70.2	2365739.0	0.358E+05
Totals	...	...	...	343.5	3823310.8	0.628E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3823311/343.53$$

$$= 111.295 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 69334.75$$

$$= 14111.1250 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 73.5645 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1500.0 - 1150.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 548.958 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 7746.4175 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 137.3284 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(69335 + 713) \cdot 270.0 / (4 \cdot 2960.0 \cdot 1767.415)}$$

$$= 16.466 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$$

$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2049.0161 / (4 - 1)$$

$$= 683.005 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 683.0054 \cdot 270.0/2$$

$$= 922.057 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 70047.516 / (2960.0 \cdot 270.0)$$

$$= 8.765 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 922.1 \cdot 8.76$$

$$= 8081.559 \text{ kgf}$$

Area of the Rib and Web [Ar]:



$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 47.96 + 75.131 \\
 &= 123.091 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Stress [Sc]:} \\
 &= P/Ar \\
 &= 8081.6/123.0906 \\
 &= 65.655 \text{ kgf/cm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	240.0	...	52.8	...	0.253E+04

$$\begin{aligned}
 &\text{Rib dimension [D]:} \\
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 240.0 - 22.0 \\
 &= 218.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Distance to Centroid from Datum [ytot]:} \\
 &= AY / A \\
 &= 0.0/123.091 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Distance to Centroid [C1]:} \\
 &= \text{Saddle Width} / 2 \\
 &= 240.0/2 \\
 &= 120.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Radius of Gyration [r]:} \\
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(2534.4/123.091)} \\
 &= 45.376 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Outer Rib [L]:} \\
 &= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1500.0 - \cos(120.0/2) (1150.0 + 21.0 + 12.0) - 26.0 \\
 &= 882.500 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Intermediate Term [Cc]:} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 2038903/2651.1)} \\
 &= 123.211
 \end{aligned}$$

$$\begin{aligned}
 &\text{Slenderness ratio [KL/r]:} \\
 &= KL/r \\
 &= 1 \cdot 882.5/45.376 \\
 &= 19.449
 \end{aligned}$$

$$\begin{aligned}
 &\text{Bending Moment [Rm]:} \\
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 458.9 / (2 \cdot 2960.0) \cdot 683.005 \cdot 882.5/2 \\
 &= 23.362 \text{ kgf-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Allowable, } KL/r < Cc \text{ (} 19.4487 < 123.2107 \text{) per AISC E2-1 [Sca]:} \\
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (19.45)^2 / (2 \cdot 123.21^2)) 2651 / \\
 &\quad (5/3 + 3 \cdot (19.45) / (8 \cdot 123.21) - (19.45^3) / (8 \cdot 123.21^3)) \\
 &= 1517 \text{ kgf/cm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 65.66/1517.41 + (23.36 \cdot 120.0/25344000) / 1767.42 \\
 &= 0.050
 \end{aligned}$$

AISC Unity Check of Outside Ribs, due to Transverse Force Ft (must be ≤ 1)



This analysis assumes the entire saddle weight load and load due to transverse forces are concentrated on the outer rib + some portion of the web. This generally would not be the case and as such this analysis is very conservative.

Inertia of Saddle, Outer Ribs - Transverse Direction:

	B	D	Y	A	AY	Io
Rib	218.0	22.0	11.0	48.0	52756.0	0.500E+04
Web	22.0	352.0	176.0	77.4	1362944.0	0.111E+05
Totals	...	...	...	125.4	1415700.0	0.161E+05

Web dimension [D]:
 = 16 • Web Thickness
 = 16 • 22.0
 = 352.000 mm

Moment at base of outer Rib [Mbase]:
 = Ft(Centerline dimension B - Baseplate thickness)
 = 774.074(1500.0 - 26.0)
 = 1140.986 kgf-m.

Bending Stress due to Transverse Force [$\sigma_{B,base}$]:
 = Bending Moment / Section Modulus
 = 1140.986/1424214
 = 80.113 kgf/cm²

Vertical Load on the Rib due to the Bending Moment [Fv,mom]:
 = Bending Moment / Web Length
 = 1140.986/2049.016
 = 556.845 kgf

Total Vertical Load on the Rib [Fv,total]:
 = Fv,mom + Vertical Load
 = 556.845 + 70047.516
 = 70604.367 kgf

Compressive Stress on Outer Rib [σ_{cr}]:
 = Fv,total / Area
 = 70604.367/125.4
 = 563.034 kgf/cm²

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{16078.6/125.4}$
 = 113.234 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 882.5/113.234
 = 7.794

Compressive Allowable, KL/r < Cc (7.7936 < 123.2107) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (7.79)^2 / (2 \cdot 123.21^2)) 2651 / (5/3 + 3 \cdot (7.79) / (8 \cdot 123.21) - (7.79^3) / (8 \cdot 123.21^3))$
 = 1565 kgf/cm²

Unity Check on Outer Rib [UCheck,outer]:
 = $\sigma_{cr} / Sca + \sigma_{B,base} / Sba$
 = 563.034/1565.244 + 80.113/1767.415
 = 0.405

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	218.0	0.0	48.0	0.0	0.253E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6



Totals	...	...	...	198.2	...	0.259E+04
--------	-----	-----	-----	-------	-----	-----------

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/198.221$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0/2$$

$$= 120.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1500.0 - \sqrt{(1183.0 + 12.0)^2 - (683.005/2)^2} - 26.0$$

$$= 341.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2593.1/198.221)}$$

$$= 36.169 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341.364/36.169$$

$$= 9.438$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 458.911/(2 \cdot 2960.0)$$

$$= 0.078 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.078 \cdot 683.005 \cdot 341.364$$

$$= 18.074 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 18.074/216087.75$$

$$= 8.364 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (9.4381 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.44)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (9.44) / (8 \cdot 123.21) - (9.44)^3 / (8 \cdot 123.21^3))$$

$$= 1559 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 81.13/1559.19 + (18.07 \cdot 120.0/2593.051) / 1767.42$$

$$= 0.057$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BlArea	2.9291 cm ²
Saddle Load Q0 (Weight)	Q0	68347.5 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	162.0 kgf
Maximum Transverse Force	Ft	774.1 kgf
Maximum Longitudinal Force	Fl	458.9 kgf
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $458.9 / (2.929 \cdot 4)$
 = 39.2 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $774.1 / (2.929 \cdot 4)$
 = 66.1 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $458.91 / (1052.29 \cdot 4.0)$
 = 0.1090 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1500.0 \cdot 774.07 + 0.0$
 = 1161.11 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $1161.11 / 68347.52$
 = 16.99 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.109, 0.0]
 = 0.1090 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (480.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 = $\min(240.0 + 1.56 \cdot \sqrt{ 1160.5001 \cdot 21.0 }, 2 \cdot 1900.0)$
 = 483.5324 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1160.50	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	868.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	58.35	cm
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	



Distance from Saddle Base to Centerline B 1500.00 mm
 Coefficient of Friction mu 0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 11927.71 kgf
 Pressure used in Analysis P 32.038 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	883.30	2518.57
Long. Stress at Bottom of Midspan	887.19	2518.57
Long. Stress at Top of Saddles	946.29	2518.57
Long. Stress at Bottom of Saddles	851.41	2518.57
Tangential Shear in Shell	29.56	1413.05
Circ. Stress at Horn of Saddle	258.71	3777.85
Circ. Compressive Stress in Shell	8.93	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1548.1/2 + 0) \cdot 1500.0/2049.0164$
 $= 1700.0 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(458.91, 0) \cdot 1500.0/4250.0$
 $= 162.0 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 10228 + \max(162, 1700, 0, 0)$
 $= 11927.7 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1779.443 (0.0 + \pi/4 (2.482 \cdot 1.2)^2)$
 $= 13637.541 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 12640.48 kgf
 Transverse Shear Load 774.07 kgf
 Longitudinal Shear Load 458.91 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2300.0/2 + 21.0 + 12.0) \sin(120.0/2), 2 \cdot 1160.5)$
 $= 2049.016 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/(3L))]$
 $= -11928 \cdot 190.0 [1 - (1 - 190.0/868.0 + (116.05^2 - 58.35^2) / (2 \cdot 190.0 \cdot 868.0)) / (1 + (4 \cdot 58.35)/(3 \cdot 868.0))]$
 $= -5782.3 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$



$$= 11928 \cdot 868 / 4 \cdot ((1 + 2(116^2 - 58^2) / (868^2)) / (1 + (4 \cdot 58) / (3 \cdot 868))) - 4 \cdot 190 / 868$$

$$= 1725.9 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 32.038 \cdot 1160.5 / (2 \cdot 21.0) - 1725.9 / (\pi \cdot 1160.5^2 \cdot 21.0)$$

$$= 883.30 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 32.038 \cdot 1160.5 / (2 \cdot 21.0) + 1725.9 / (\pi \cdot 1160.5^2 \cdot 21.0)$$

$$= 887.19 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 32.038 \cdot 1160.5 / (2 \cdot 21.0) - 5782.3 / (0.1066 \cdot \pi \cdot 1160.5^2 \cdot 21.0)$$

$$= 946.29 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 32.038 \cdot 1160.5 / (2 \cdot 21.0) + 5782.3 / (0.1923 \cdot \pi \cdot 1160.5^2 \cdot 21.0)$$

$$= 851.41 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 11928(868.0 - 2 \cdot 190.0) / (868.0 + (4 \cdot 58.35 / 3))$$

$$= 6154.3 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 6154.29 / (1160.5001 \cdot 21.0)$$

$$= 29.56 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1160.5 \cdot 21.0)}$$

$$= 121.766 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 11928 \cdot 0.1 / (21.0(240.0 + 121.77 + 121.77))$$

$$= -8.93 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -11928 / (4 \cdot 21.0(240.0 + 121.766 + 121.766)) - 12 \cdot 0.053 \cdot 11928 \cdot 1160.5 / (868.0 \cdot 21.0^2)$$

$$= -258.71 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1160.5 \cdot 21.0)}, 2 \cdot 190.0)$$

$$= 483.53 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1150.0 \cdot 21.0)} + 480.0)^3) / 12$$

$$= 55.753 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 55.753 + 151.71 (111.295 - 10.5)^2$$

$$= 15468.860 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	722.4	21.0	10.5	151.7	159295.4	0.155E+05
Wearplate	480.0	12.0	27.0	57.6	155520.0	0.410E+04
Web	22.0	291.0	178.5	64.0	1142756.4	0.741E+04
BasePlate	270.0	26.0	337.0	70.2	2365739.0	0.358E+05
Totals	...	...	...	343.5	3823310.8	0.628E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3823311 / 343.53$$

$$= 111.295 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 11927.712$$

$$= 2427.5481 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 12.6554 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1500.0 - 1150.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 548.958 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1332.6224 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 23.6247 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(11928 + 713) \cdot 270.0) / (4 \cdot 2960.0 \cdot 1767.415)}$$

$$= 6.995 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$$

$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2049.0161 / (4 - 1)$$

$$= 683.005 \text{ mm}$$

Baseplate Pressure Area [Ap]:



$$= e \cdot Bpwid / 2$$

$$= 683.0054 \cdot 270.0 / 2$$

$$= 922.057 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 12640.482 / (2960.0 \cdot 270.0)$$

$$= 1.582 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 922.1 \cdot 1.58$$

$$= 1458.364 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 47.96 + 75.131$$

$$= 123.091 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1458.4 / 123.0906$$

$$= 11.848 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	240.0	...	52.8	...	0.253E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 240.0 - 22.0$$

$$= 218.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 123.091$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0 / 2$$

$$= 120.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2534.4 / 123.091) }$$

$$= 45.376 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1500.0 - \cos(120.0 / 2) (1150.0 + 21.0 + 12.0) - 26.0$$

$$= 882.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 2038903 / 2651.1) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 882.5 / 45.376$$

$$= 19.449$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 458.9 / (2 \cdot 2960.0) \cdot 683.005 \cdot 882.5 / 2$$



= 23.362 kgf-m.

Compressive Allowable, $KL/r < Cc$ (19.4487 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.45)^2 / (2 \cdot 123.21^2)) 2651 / (5/3 + 3 \cdot (19.45) / (8 \cdot 123.21) - (19.45^3) / (8 \cdot 123.21^3))$$

$$= 1517 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.85/1517.41 + (23.36 \cdot 120.0/25344000) / 1767.42$$

$$= 0.014$$

AISC Unity Check of Outside Ribs, due to Transverse Force F_t (must be ≤ 1)

This analysis assumes the entire saddle weight load and load due to transverse forces are concentrated on the outer rib + some portion of the web. This generally would not be the case and as such this analysis is very conservative.

Inertia of Saddle, Outer Ribs - Transverse Direction:

	B	D	Y	A	AY	Io
Rib	218.0	22.0	11.0	48.0	52756.0	0.500E+04
Web	22.0	352.0	176.0	77.4	1362944.0	0.111E+05
Totals	...	...	...	125.4	1415700.0	0.161E+05

Web dimension [D]:

$$= 16 \cdot \text{Web Thickness}$$

$$= 16 \cdot 22.0$$

$$= 352.000 \text{ mm}$$

Moment at base of outer Rib [Mbase]:

$$= F_t (\text{Centerline dimension B} - \text{Baseplate thickness})$$

$$= 774.074 (1500.0 - 26.0)$$

$$= 1140.986 \text{ kgf-m.}$$

Bending Stress due to Transverse Force [$\sigma_{B,base}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1140.986 / 1424214$$

$$= 80.113 \text{ kgf/cm}^2$$

Vertical Load on the Rib due to the Bending Moment [$F_{v,mom}$]:

$$= \text{Bending Moment} / \text{Web Length}$$

$$= 1140.986 / 2049.016$$

$$= 556.845 \text{ kgf}$$

Total Vertical Load on the Rib [$F_{v,total}$]:

$$= F_{v,mom} + \text{Vertical Load}$$

$$= 556.845 + 12640.481$$

$$= 13197.326 \text{ kgf}$$

Compressive Stress on Outer Rib [σ_{cr}]:

$$= F_{v,total} / \text{Area}$$

$$= 13197.326 / 125.4$$

$$= 105.242 \text{ kgf/cm}^2$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (16078.6 / 125.4) }$$

$$= 113.234 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 882.5 / 113.234$$

$$= 7.794$$

Compressive Allowable, $KL/r < Cc$ (7.7936 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.79)^2 / (2 \cdot 123.21^2)) 2651 /$$



$$\begin{aligned} & (5/3+3 \cdot (7.79)/(8 \cdot 123.21) - (7.79^3)/(8 \cdot 123.21^3)) \\ & = 1565 \text{ kgf/cm}^2 \end{aligned}$$

Unity Check on Outer Rib [UCheck,outer]:

$$\begin{aligned} & = \sigma_{cr} / Sca + \sigma_{B,base} / Sba \\ & = 105.242/1565.244 + 80.113/1767.415 \\ & = 0.113 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	218.0	0.0	48.0	0.0	0.253E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	198.2	...	0.259E+04

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} & = AY / A \\ & = 0.0/198.221 \\ & = 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} & = \text{Saddle Width} / 2 \\ & = 240.0/2 \\ & = 120.000 \text{ mm} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} & = \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk \\ & = 1500.0 - \sqrt{(1183.0 + 12.0)^2 - (683.005/2)^2} - 26.0 \\ & = 341.364 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} & = \sqrt{\text{Total Inertia} / \text{Total Area}} \\ & = \sqrt{2593.1/198.221} \\ & = 36.169 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} & = KL/r \\ & = 1 \cdot 341.364/36.169 \\ & = 9.438 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} & = FI / (2 \cdot \text{Baseplate Length}) \\ & = 458.911/(2 \cdot 2960.0) \\ & = 0.078 \text{ kgf/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} & = \text{Unit Force} \cdot e \cdot L \\ & = 0.078 \cdot 683.005 \cdot 341.364 \\ & = 18.074 \text{ kgf-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$\begin{aligned} & = \text{Bending Moment} / \text{Section Modulus} \\ & = 18.074/216087.75 \\ & = 8.364 \text{ kgf/cm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ (9.4381 < 123.2107) per AISC E2-1 [Sca]:

$$\begin{aligned} & = (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\ & = (1 - (9.44)^2 / (2 \cdot 123.21^2)) 2651 / \\ & \quad (5/3 + 3 \cdot (9.44) / (8 \cdot 123.21) - (9.44^3) / (8 \cdot 123.21^3)) \\ & = 1559 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} & = Sc/Sca + (Mbase,c \cdot C1/I) / Sba \\ & = 14.64/1559.19 + (18.07 \cdot 120.0/2593.051) / 1767.42 \\ & = 0.014 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:



Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load QO (Weight)	QO	10940.5 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	162.0 kgf
Maximum Transverse Force	Ft	774.1 kgf
Maximum Longitudinal Force	Fl	458.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 458.9 / (2.929 \cdot 4)$$

$$= 39.2 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 774.1 / (2.929 \cdot 4)$$

$$= 66.1 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 458.91 / (1052.29 • 4.0)
 = 0.1090 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1500.0 • 774.07 + 0.0
 = 1161.11 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 1161.11 / 10940.48
 = 106.13 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.109, 0.0]
 = 0.1090 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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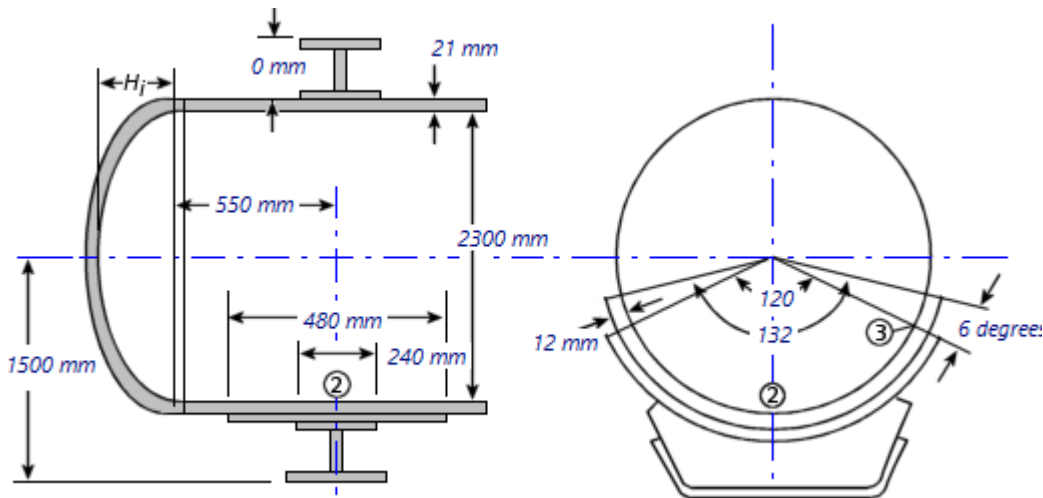


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	868.00	cm
Distance from Saddle to Vessel tangent	a1 or a	550.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1650.86	kgf/cm ²
Shell Allowable Stress used in Calculation		1650.41	kgf/cm ²
Head Allowable Stress used in Calculation		1568.83	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		73976.26	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-183.43	-2159.64
Long. Stress at Top of Midspan		183.43	1650.41
Long. Stress at Bottom of Midspan		120.62	1650.41
Long. Stress at Top of Saddles		-35.24	-2159.64
Long. Stress at Top of Saddles		35.24	1650.41
Long. Stress at Bottom of Saddles		-27.57	-2159.64
Long. Stress at Bottom of Saddles		27.57	1650.41
Tangential Shear in Shell		287.45	1320.33
Tangential Shear in Head		19.23	1252.34
Circ. Stress at Horn of Saddle		281.81	2063.01
Circ. Compressive Stress in Shell		37.61	1650.41



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(4691.4/2 + 0) \cdot 1500.0/2049.0164$
 $= 5151.5\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1390.64, 0) \cdot 1500.0/4250.0$
 $= 490.8\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(6886.95, 0) \cdot 1500.0/4250.0$
 $= 2430.7\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(6887/2 + 0) \cdot 1500.0/2049.0164$
 $= 7562.5\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 66414 + \max(491, 5152, 2431, 7562)$
 $= 73976.3\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1779.443(0.0 + \pi/4(2.482 \cdot 1.2)^2)$
 $= 13637.541\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	74689.02	kgf
Transverse Shear Load	3443.48	kgf
Longitudinal Shear Load	6886.95	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1161.25)$
 $= 2049.016\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/(3L))]$
 $= -73976 \cdot 55.0 [1 - (1 - 55.0/868.0 + (116.125^2 - 0.0^2) / (2 \cdot 55.0 \cdot 868.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 868.0))]$
 $= 3168.3\ kgf \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 73976 \cdot 868 / 4 ((1 + 2(116^2 - 0^2)/(868^2)) / (1 + (4 \cdot 0) / (3 \cdot 868)) - 4 \cdot 55 / 868)$
 $= 125588.1\ kgf \cdot m$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= -1.055 \cdot 1161.25 / (2 \cdot 19.5) - 125588.1 / (\pi \cdot 1161.2^2 \cdot 19.5)$



$$= -183.43 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25 / (2 \cdot 19.5) + 125588.1 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 120.62 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25 / (2 \cdot 19.5) - 3168.3 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -35.24 \text{ kgf/cm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25 / (2 \cdot 19.5) + 3168.3 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -27.57 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 73976(868.0 - 2 \cdot 55.0) / (868.0 + (4 \cdot 0.0 / 3))$$

$$= 64601.4 \text{ kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 73976 / (1161.25 \cdot 19.5)$$

$$= 287.45 \text{ kgf/cm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 73976 / (1161.25 \cdot 291.5)$$

$$= 19.23 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 55.0)$$

$$= 474.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1650.865 / 1650.408, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7603 \cdot 73976 \cdot 0.1 / (474.75(19.5 + 1.0 \cdot 12.0))$$

$$= -37.61 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q / (4(t + \eta \cdot t_r)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -73976 / (4(19.5 + 1.0 \cdot 12.0)474.75) -$$

$$12 \cdot 0.013 \cdot 73976 \cdot 1161.25 / (868.0(19.5 + 1.0 \cdot 12.0)^2)$$

$$= -281.81 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0) 19.5^3) / 12$$

$$= 44.104 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 44.104 + 139.184 (113.925 - 9.75)^2$$

$$= 15148.791 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.151E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.451E+04
Web	22.0	291.0	177.0	64.0	1133153.4	0.706E+04
BasePlate	270.0	26.0	335.5	70.2	2355209.2	0.345E+05
Totals	...	...	...	331.0	3770946.5	0.612E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3770947 / 331.004$$

$$= 113.925 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 73976.258$$

$$= 15055.7725 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 78.4892 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1486.6948 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1500.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 547.717 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 8246.3125 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 153.4341 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1651.8832 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(73976 + 713)270.0 / (4 \cdot 2960.0 \cdot 1651.883))}$$

$$= 17.588 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$$

$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2049.016 / (4 - 1)$$



= 683.005 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 683.0054 • 270.0/2
 = 922.057 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 74689.031/(2960.0 • 270.0)
 = 9.345 kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 922.1 • 9.35
 = 8617.062 kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 47.96 + 75.131
 = 123.091 cm²

Compressive Stress [Sc]:

= P/Ar
 = 8617.1/123.0906
 = 70.006 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	240.0	...	52.8	...	0.253E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 240.0 - 22.0
 = 218.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/123.091
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 240.0/2
 = 120.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{2534.4/123.091}$
 = 45.376 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1500.0 - cos(120.0/2)(1150.0 + 21.0 + 12.0) - 26.0
 = 882.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2038903/2477.8}$
 = 127.446

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 882.5/45.376
 = 19.449



Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 6887.0 / (2 \cdot 2960.0) \cdot 683.005 \cdot 882.5 / 2$$

$$= 350.602 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (19.4487 < 127.4465) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.45)^2 / (2 \cdot 127.45^2)) 2478 /$$

$$(5/3 + 3 \cdot (19.45) / (8 \cdot 127.45) - (19.45^3) / (8 \cdot 127.45^3))$$

$$= 1421 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 70.01 / 1420.97 + (350.6 \cdot 120.0 / 25344000) / 1651.88$$

$$= 0.150$$

AISC Unity Check of Outside Ribs, due to Transverse Force F_t (must be ≤ 1)

This analysis assumes the entire saddle weight load and load due to transverse forces are concentrated on the outer rib + some portion of the web. This generally would not be the case and as such this analysis is very conservative.

Inertia of Saddle, Outer Ribs - Transverse Direction:

	B	D	Y	A	AY	Io
Rib	218.0	22.0	11.0	48.0	52756.0	0.500E+04
Web	22.0	352.0	176.0	77.4	1362944.0	0.111E+05
Totals	...	...	...	125.4	1415700.0	0.161E+05

Web dimension [D]:

$$= 16 \cdot \text{Web Thickness}$$

$$= 16 \cdot 22.0$$

$$= 352.000 \text{ mm}$$

Moment at base of outer Rib [Mbase]:

$$= F_t (\text{Centerline dimension B} - \text{Baseplate thickness})$$

$$= 774.074 (1500.0 - 26.0)$$

$$= 1140.986 \text{ kgf-m.}$$

Bending Stress due to Transverse Force [$\sigma_{B,base}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1140.986 / 1424214$$

$$= 80.113 \text{ kgf/cm}^2$$

Vertical Load on the Rib due to the Bending Moment [Fv,mom]:

$$= \text{Bending Moment} / \text{Web Length}$$

$$= 1140.986 / 2049.016$$

$$= 556.845 \text{ kgf}$$

Total Vertical Load on the Rib [Fv,total]:

$$= F_{v,mom} + \text{Vertical Load}$$

$$= 556.845 + 70047.516$$

$$= 70604.367 \text{ kgf}$$

Compressive Stress on Outer Rib [σ_{cr}]:

$$= F_{v,total} / \text{Area}$$

$$= 70604.367 / 125.4$$

$$= 563.034 \text{ kgf/cm}^2$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(16078.6 / 125.4)}$$

$$= 113.234 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 882.5 / 113.234$$

$$= 7.794$$



Compressive Allowable, $KL/r < C_c$ (7.7936 < 127.4465) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.79)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (7.79) / (8 \cdot 127.45) - (7.79^3) / (8 \cdot 127.45^3))$$

$$= 1464 \text{ kgf/cm}^2$$

Unity Check on Outer Rib [UCheck,outer]:

$$= \sigma_{cr} / S_{ca} + \sigma_{B,base} / S_{ba}$$

$$= 563.034 / 1463.799 + 80.113 / 1651.883$$

$$= 0.433$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	218.0	0.0	48.0	0.0	0.253E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	198.2	...	0.259E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 198.221$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0 / 2$$

$$= 120.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1500.0 - \sqrt{ (1183.0 + 12.0)^2 - (683.005/2)^2 } - 26.0$$

$$= 341.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2593.1 / 198.221 }$$

$$= 36.169 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 341.364 / 36.169$$

$$= 9.438$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 6886.95 / (2 \cdot 2960.0)$$

$$= 1.163 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.163 \cdot 683.005 \cdot 341.364$$

$$= 271.236 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 271.236 / 216087.75$$

$$= 125.521 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (9.4381 < 127.4465) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.44)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (9.44) / (8 \cdot 127.45) - (9.44^3) / (8 \cdot 127.45^3))$$

$$= 1458 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= \sigma_{cr} / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 86.5 / 1458.36 + (271.24 \cdot 120.0 / 2593.051) / 1651.88$$

$$= 0.135$$



ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	868.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1650.86	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	58.43	cm
Shell Allowable Stress used in Calculation		1650.41	kgf/cm ²
Head Allowable Stress used in Calculation		1650.41	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1500.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		19117.78	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-34.75	-2159.64
Long. Stress at Top of Midspan		34.75	1650.41
Long. Stress at Bottom of Midspan		-28.06	-2159.64
Long. Stress at Bottom of Midspan		28.06	1650.41
Long. Stress at Top of Saddles		73.85	1650.41
Long. Stress at Bottom of Saddles		-89.74	-2159.64
Long. Stress at Bottom of Saddles		89.74	1650.41
Tangential Shear in Shell		50.99	1320.33
Circ. Stress at Horn of Saddle		195.44	2063.01
Circ. Compressive Stress in Shell		9.72	1650.41

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (4691.4 / 2 + 0) \cdot 1500.0 / 2049.0164$
 $= 5151.5 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1390.64, 0) \cdot 1500.0 / 4250.0$
 $= 490.8 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(6886.95, 0) \cdot 1500.0 / 4250.0$
 $= 2430.7 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (6887 / 2 + 0) \cdot 1500.0 / 2049.0164$
 $= 7562.5 \text{ kgf}$



Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 11555 + \max(491, 5152, 2431, 7562)$$

$$= 19117.8 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1779.443(0.0 + \pi/4(2.482 \cdot 1.2)^2)$$

$$= 13637.541 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	19830.55	kgf
Transverse Shear Load	3443.48	kgf
Longitudinal Shear Load	6886.95	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1161.25)$$

$$= 2049.016 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/((1 + (4\text{hm})/3L))]$$

$$= -19118 \cdot 190.0 [1 - (1 - 190.0/868.0 + (116.125^2 - 58.425^2)/(2 \cdot 190.0 \cdot 868.0))/((1 + (4 \cdot 58.43)/(3 \cdot 868.0)))]$$

$$= -9269.9 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/((1 + (4\text{hm}^2)/(3L)) - 4a/L))$$

$$= 19118 \cdot 868/4((1 + 2(116^2 - 58^2)/(868^2))/((1 + (4 \cdot 58)/(3 \cdot 868)) - 4 \cdot 190/868))$$

$$= 2763.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) - 2763.0/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -34.75 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) + 2763.0/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -28.06 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot \text{Rm}/(2t) - M1/(K1 \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) - 9269.9/(0.1066 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 73.85 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot \text{Rm}/(2t) + M1/(K1^* \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) + 9269.9/(0.1923 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -89.74 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot \text{hm}/3))$$

$$= 19118(868.0 - 2 \cdot 190.0)/(868.0 + (4 \cdot 58.43/3))$$

$$= 9863.1 \text{ kgf}$$



Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 9863.07 / (1161.25 \cdot 19.5)$$

$$= 50.99 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 190.0)$$

$$= 474.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1650.865/1650.408, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 19118 \cdot 0.1 / (474.75(19.5 + 1.0 \cdot 12.0))$$

$$= -9.72 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -19118 / (4(19.5 + 1.0 \cdot 12.0)474.75) -$$

$$12 \cdot 0.053 \cdot 19118 \cdot 1161.25 / (868.0(19.5 + 1.0 \cdot 12.0)^2)$$

$$= -195.44 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)19.5^3) / 12$$

$$= 44.104 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 44.104 + 139.184(113.925 - 9.75)^2$$

$$= 15148.791 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.151E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.451E+04
Web	22.0	291.0	177.0	64.0	1133153.4	0.706E+04
BasePlate	270.0	26.0	335.5	70.2	2355209.2	0.345E+05
Totals	...	...	...	331.0	3770946.5	0.612E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3770947 / 331.004$$

$$= 113.925 \text{ mm}$$



Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 19117.783 \\ &= 3890.8833 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 20.2841 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1486.6948 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1500.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472 \\ &= 547.717 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2131.1057 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 39.6522 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1651.8832 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{(3(19118 + 713) \cdot 270.0) / (4 \cdot 2960.0 \cdot 1651.883)} \\ &= 9.062 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0) \\ &= 2049.016 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 2049.0161 / (4 - 1) \\ &= 683.005 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 683.0054 \cdot 270.0/2 \\ &= 922.057 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 19830.551 / (2960.0 \cdot 270.0) \\ &= 2.481 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 922.1 \cdot 2.48 \\ &= 2287.901 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 47.96 + 75.131 \\ &= 123.091 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P/Ar \\ &= 2287.9/123.0906 \\ &= 18.587 \text{ kgf/cm}^2 \end{aligned}$$



Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	240.0	...	52.8	...	0.253E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 240.0 - 22.0
 = 218.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/123.091
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 240.0/2
 = 120.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{2534.4/123.091}$
 = 45.376 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1500.0 - cos(120.0/2)(1150.0 + 21.0 + 12.0) - 26.0
 = 882.500 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 2038903/2477.8}$
 = 127.446

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 882.5/45.376
 = 19.449

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 6887.0 / (2 • 2960.0) • 683.005 • 882.5/2
 = 350.602 kgf-m.

Compressive Allowable, KL/r < Cc (19.4487 < 127.4465) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (19.45)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (19.45) / (8 \cdot 127.45) - (19.45^3) / (8 \cdot 127.45^3))$
 = 1421 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 18.59/1420.97 + (350.6 • 120.0/25344000) / 1651.88
 = 0.114

AISC Unity Check of Outside Ribs, due to Transverse Force Ft (must be ≤ 1)

This analysis assumes the entire saddle weight load and load due to transverse forces are concentrated on the outer rib + some portion of the web. This generally would not be the case and as such this analysis is very conservative.

Inertia of Saddle, Outer Ribs - Transverse Direction:

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	218.0	22.0	11.0	48.0	52756.0	0.500E+04
Web	22.0	352.0	176.0	77.4	1362944.0	0.111E+05
Totals	...	...	...	125.4	1415700.0	0.161E+05

Web dimension [D]:
 = 16 • Web Thickness
 = 16 • 22.0
 = 352.000 mm

Moment at base of outer Rib [Mbase]:
 = Ft(Centerline dimension B - Baseplate thickness)
 = 774.074(1500.0 - 26.0)
 = 1140.986 kgf-m.

Bending Stress due to Transverse Force [$\sigma_{B,base}$]:
 = Bending Moment / Section Modulus
 = 1140.986/1424214
 = 80.113 kgf/cm²

Vertical Load on the Rib due to the Bending Moment [Fv,mom]:
 = Bending Moment / Web Length
 = 1140.986/2049.016
 = 556.845 kgf

Total Vertical Load on the Rib [Fv,total]:
 = Fv,mom + Vertical Load
 = 556.845 + 12640.481
 = 13197.326 kgf

Compressive Stress on Outer Rib [σ_{cr}]:
 = Fv,total / Area
 = 13197.326/125.4
 = 105.242 kgf/cm²

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{16078.6/125.4}$
 = 113.234 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 882.5/113.234
 = 7.794

Compressive Allowable, KL/r < Cc (7.7936 < 127.4465) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (7.79)^2 / (2 \cdot 127.45^2)) 2478 / (5/3 + 3 \cdot (7.79) / (8 \cdot 127.45) - (7.79^3) / (8 \cdot 127.45^3))$
 = 1464 kgf/cm²

Unity Check on Outer Rib [UCheck,outer]:
 = $\sigma_{cr} / Sca + \sigma_{B,base} / Sba$
 = 105.242/1463.799 + 80.113/1651.883
 = 0.12

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	218.0	0.0	48.0	0.0	0.253E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	198.2	...	0.259E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/198.221
 = 0.000 mm

Distance to Centroid [C1]:



$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 240.0 / 2 \\
 &= 120.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Inner Rib [L]:} \\
 &= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk} \\
 &= 1500.0 - \sqrt{(1183.0 + 12.0)^2 - (683.005/2)^2} - 26.0 \\
 &= 341.364 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Radius of Gyration [r]:} \\
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{2593.1/198.221} \\
 &= 36.169 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Slenderness ratio [KL/r]:} \\
 &= \text{KL/r} \\
 &= 1 \cdot 341.364 / 36.169 \\
 &= 9.438
 \end{aligned}$$

$$\begin{aligned}
 &\text{Unit Force [Force,u]:} \\
 &= \text{FI} / (2 \cdot \text{Baseplate Length}) \\
 &= 6886.95 / (2 \cdot 2960.0) \\
 &= 1.163 \text{ kgf/mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Moment at base of inner Rib [Mbase,c]:} \\
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 1.163 \cdot 683.005 \cdot 341.364 \\
 &= 271.236 \text{ kgf-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Bending Stress due to Transverse Force and Weight Load } [\sigma_B, \text{base,c}]: \\
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 271.236 / 216087.75 \\
 &= 125.521 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Allowable, } \text{KL/r} < \text{Cc} \text{ (} 9.4381 < 127.4465 \text{) per AISC E2-1 [Sca]:} \\
 &= (1 - (\text{KL/r})^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL/r}) / (8 \cdot \text{Cc}) - (\text{KL/r})^3 / (8 \cdot \text{Cc}^3)) \\
 &= (1 - (9.44)^2 / (2 \cdot 127.45^2)) 2478 / \\
 &\quad (5/3 + 3 \cdot (9.44) / (8 \cdot 127.45) - (9.44^3) / (8 \cdot 127.45^3)) \\
 &= 1458 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{AISC Unity Check of Inside Ribs (must be } \leq 1 \text{)} \\
 &= \text{Sc/Sca} + (\text{Mbase,c} \cdot \text{C1/I}) / \text{Sba} \\
 &= 22.97 / 1458.36 + (271.24 \cdot 120.0 / 2593.051) / 1651.88 \\
 &= 0.092
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Calculations per ASME VIII-2, 4.3.11

Input Values:

SH.2

Design Internal Pressure, Small End	23.684	kgf/cm ²
Design Internal Pressure, Large End	23.684	kgf/cm ²
Temperature for Internal Pressure	80.0	°C
Design External Pressure	1.055	kgf/cm ²
Temperature for External Pressure	37.0	°C

CONE Properties:

Cone Material		SA-516 70 +N
Cone Allowable Stress at Temperature	Ss	1650.41 kgf/cm ²
Cone Allowable Stress At Ambient		1784.41 kgf/cm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00
Circumferential Joint Efficiency of Cone		1.00
Actual Thickness of Cone	Tc	21.0000 mm
Corrosion Allowance for Cone	Cac	1.5000 mm
Diameter Basis for Cone and Cylinders		ID
Diameter of Small End of Cone	Ds	1200.000 mm
Diameter of Large End of Cone	Dl	2300.000 mm
Half Apex Angle for Cone		29.94 degrees
Axial Length of Cone	Lc	1910.000 mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70 +N
Operating Allowable Stress	Ss	1650.41 kgf/cm ²
Ambient Allowable Stress		1784.41 kgf/cm ²
Longitudinal Joint Efficiency	Es,E1	1.00
Actual Thickness	Ts	16.0000 mm
Corrosion Allowance	Cas	1.5000 mm
Axial Length of Attached Element	Ls	300.032 mm

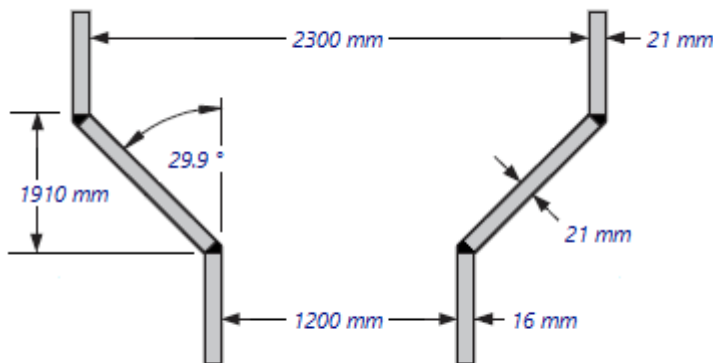
LARGE-END Properties:

Attached Large End Material Specification		SA-516 70 +N
Operating Allowable Stress	Sl	1650.41 kgf/cm ²
Ambient Allowable Stress		1784.41 kgf/cm ²
Longitudinal Joint Efficiency	El,E1	1.00
Actual Thickness	Tl	21.0000 mm
Corrosion Allowance	Cal	1.5000 mm
Axial Length of Attached Element	Ll	6661.666 mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Warning: rtk -1.500mm, must be > Tl, 19.500mm, see 4.3.12



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 37 °C

Elastic Modulus of Cone Material	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Small Cylinder Material	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Large Cylinder Material	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Large End Reinforcement	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Small End Reinforcement	0.206E+07 kgf/cm ² at 37 °C

Elastic Modulus Data from ASME Section II Part D at 80 °C

Elastic Modulus of Cone Material	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Small Cylinder Material	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Large Cylinder Material	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Large End Reinforcement	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Small End Reinforcement	0.203E+07 kgf/cm ² at 80 °C

Axial forces and moments are not computed for Horizontal geometries.
 Compute them manually and enter the loads using the Force/Moment dialog.
 Place the loads at the ends of the cone as needed.

Maximum Centroid Reinforcement Distance Large End	25.4000 mm
Maximum Centroid Reinforcement Distance Small End	25.4000 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Junction Calculations per 4.3.11:

Calculations for the Large End Cone/Shell Junction - Without Knuckle

F_{he} per 4.1.3 or 4.3.11.2 Step 6 [F_{he}]:
 $= 0.4 \cdot E(tc/D)$
 $= 0.4 \cdot 2057748(19.5/2342.0)$
 $= 6853.3 \text{ kgf/cm}^2$

Factor of Safety [FS]:
 Since F_{ic} [2477.8] >= F_y[2477.8], FS = 1.667

Allowable Stress per 4.3.11 Step 6 [F_{ha}]:
 $= F_{ic}/FS$
 $= 2477.7957/1.667$
 $= 1486.380 \text{ kgf/cm}^2$

Large End under Internal Pressure:



Cone Length along Cone surface per 4.3.46 [ConeMinLen]:
 $= 2 \cdot \sqrt{(RL \cdot Tc / \cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc / \cos(\alpha))}}$
 $= 2 \cdot \sqrt{(1151.731 \cdot 19.5 / \cos(29.94)) + 1.4 \cdot \sqrt{(601.731 \cdot 19.5 / \cos(29.94))}}$
 $= 484.854 \text{ mm}$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	699.428	2475.612	Pass
Sigma sm-	699.428	2475.612	Pass
Sigma thetam+	-445.465	2475.612	Pass
Sigma thetam-	-445.465	2475.612	Pass
Sigma sm+ + Sigma	-2674.349	4955.591	Pass
Sigma sm+ - Sigma	4073.204	4955.591	Pass
Sigma sm- + Sigma	-2674.349	4955.591	Pass
Sigma sm- - Sigma	4073.204	4955.591	Pass
Sigma thetam+ +	-1457.598	4955.591	Pass
Sigma thetam+ -	566.668	4955.591	Pass
Sigma thetam- +	-1457.598	4955.591	Pass
Sigma thetam- -	566.668	4955.591	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	699.621	2475.612	Pass
Sigma sm-	699.621	2475.612	Pass
Sigma thetam+	-489.538	2475.612	Pass
Sigma thetam-	-489.538	2475.612	Pass
Sigma sm+ + Sigma	-2674.155	4955.591	Pass
Sigma sm+ - Sigma	4073.398	4955.591	Pass
Sigma sm- + Sigma	-2674.155	4955.591	Pass
Sigma sm- - Sigma	4073.398	4955.591	Pass
Sigma thetam+ +	-1501.671	4955.591	Pass
Sigma thetam+ -	522.595	4955.591	Pass
Sigma thetam- +	-1501.671	4955.591	Pass
Sigma thetam- -	522.595	4955.591	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

Sigma _{smCyl} + + S	699.525	-1191.173	Pass
Sigma _{smCyl} - + S	699.525	-1191.173	Pass
Sigma _{thetamCyl} +	-467.501	-1486.380	Pass
Sigma _{thetamCyl} -	-467.501	-1486.380	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:
 $= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.141 \cdot 1151.731) + 0.0 / (3.142 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Net Axial Force ML -ve [XL-]:
 $= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.142 \cdot 1151.731) - 0.0 / (3.141 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Thickness Ratio [n]:
 $= tc/tL = 19.5/19.5 = 1.0$

Curve Fitting Geometry [H and B]:
 $H = \text{Sqrt}(0.5 \cdot D/tL) = \text{Sqrt}(0.5 \cdot 2303.462/19.5) = 7.6853$
 $B = \tan(29.938) = 0.5759$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: $1.0 < (n=1.0) < 1.25$



Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 = -3.065534 C2 = 3.642747 C3 = 0.810048 C4 = -0.221192
 C5 = -0.081824 C6 = 0.035052 C7 = 0.025775 C8 = -0.015413
 C9 = 0.002102 C10 = -0.005587 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576) \\ X = 3.1672, MsN1 = -\exp(X) = -23.7415$$

Table 4.3.3 [Qn1]:

C1 = -1.983852 C2 = 2.410703 C3 = 0.626443 C4 = -0.119151
 C5 = -0.115841 C6 = 0.122993 C7 = 0.012160 C8 = -0.016987
 C9 = 0.010919 C10 = -0.016653 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576) \\ X = 2.0684, Qn = -\exp(X) = -7.9123$$

Table 4.3.4 MsN2:

C1 = -5.697151 C2 = 0.003838 C3 = 0.476317 C4 = -0.213157
 C5 = 2.233703 C6 = 0.000032 C7 = 0.002506 C8 = -0.001663
 C9 = -0.212965 C10 = 0.000138 C11 = -0.106203

Compute using Table 4.3.4 [MsN2]:

$$X1 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ = C1 + C3 \cdot \ln(7.685^2) + C5 \cdot \ln(0.523) + C7 \cdot (\ln(7.685^2))^2 + C9 \cdot (\ln(0.523))^2 + C11 \cdot \ln(7.685^2) \cdot \ln(0.523) = -0.0531 \\ X2 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ = 1 + C2 \cdot \ln(7.685) + C4 \cdot \ln(0.523) + C6 \cdot (\ln(7.685^2))^2 + C8 \cdot (\ln(0.523))^2 + C10 \cdot \ln(7.69^2) \cdot \ln(0.52) = 0.2743 \\ MsN2 = \exp(X1/X2) = \exp(-0.053/0.274) = -0.8241$$

Table 4.3.4 QN2:

C1 = -4.774616 C2 = 0.000461 C3 = -0.002831 C4 = -0.197117
 C5 = 1.982132 C6 = 0.000069 C7 = -0.000234 C8 = -0.003536
 C9 = -0.202493 C10 = -0.000088 C11 = 0.001365

Compute using Table 4.3.4 [Qn2]:

$$X3 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ = C1 + C3 \cdot \ln(7.685^2) + C5 \cdot \ln(0.523) + C7 \cdot (\ln(7.685^2))^2 + C9 \cdot (\ln(0.523))^2 + C11 \cdot \ln(7.685^2) \cdot \ln(0.523) = -0.0531 \\ X4 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ = 1 + C2 \cdot \ln(7.685) + C4 \cdot \ln(0.523) + C6 \cdot (\ln(7.685^2))^2 + C8 \cdot (\ln(0.523))^2 + C10 \cdot \ln(7.685^2) \cdot \ln(0.523) = 0.2743 \\ Qn2 = \exp(X3/X4) = \exp(-0.373/0.291) = -0.2772$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot MsN1 = 23.684 \cdot 19.5^2 \cdot -23.742 = -2138.1274 \text{ kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = -2138.127 + -0.0 = -2138.1274 \text{ kgf}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = -2138.127 + -0.0 = -2138.1274 \text{ kgf}$$



Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = 23.684 \cdot 19.5 \cdot -7.912 = -36.5422 \text{ kgf/mm}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = -36.542 + -0.0 = -36.5422 \text{ kgf/mm}$$

Compute with XL- [Q-]:

$$= Qp + Qx = -36.542 + -0.0 = -36.5422 \text{ kgf/mm}$$

Compute [βcy]:

$$= 3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25 \\ = (3 \cdot (1 - 0.3^2) / (1151.731^2 \cdot 19.5^2)) \cdot 0.25 = 0.0086 \text{ 1/mm}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = 23.684 \cdot 1151.731/2 + 0.0 = 136.3882 \text{ kgf/mm}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = 23.684 \cdot 1151.731/2 + 0.0 = 136.3882 \text{ kgf/mm}$$

Compute with XL+ [Ntheta+]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = 23.684 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot \\ (-2138.127 \cdot 0.009 + -36.542) = -86.8655 \text{ kgf/mm}$$

Compute with XL- [Ntheta-]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = 23.684 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot \\ (-2138.127 \cdot 0.009 + -36.542) = -86.8655 \text{ kgf/mm}$$

Note: Kpc has been set to 1.0

σSm with XL+ [σsm+]:

$$= Ns / tL \\ = 136.388/19.5 \\ = 699.428 \text{ kgf/cm}^2$$

σSm with XL- [σsm-]:

$$= Ns / tL \\ = 136.388/19.5 \\ = 699.428 \text{ kgf/cm}^2$$

σSb with XL+ [σsb+]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot -2138.127 / (19.5^2 \cdot 1.0) \\ = -3373.776 \text{ kgf/cm}^2$$

σSb with XL- [σsb-]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot -2138.127 / (19.5^2 \cdot 1.0) \\ = -3373.776 \text{ kgf/cm}^2$$

σThetam with XL+ [σthetam+]:

$$= Ntheta / tL \\ = -86.865/19.5 \text{ kgf/cm}^2 \\ = -445.465 \text{ kgf/cm}^2$$

σThetam with XL- [σthetam-]:

$$= Ntheta / tL \\ = -86.865/19.5 \\ = -445.465 \text{ kgf/cm}^2$$

σThetab with XL+ [σthetab+]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc)$$



$$= 6 \cdot 0.3 \cdot -2138.127 / (19.5^2 \cdot 1.0)$$

$$= -1012.133 \text{ kgf/cm}^2$$

σ_{Thetab} with XL- [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc)$$

$$= 6 \cdot 0.3 \cdot -2138.127 / (19.5^2 \cdot 1.0)$$

$$= -1012.133 \text{ kgf/cm}^2$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= Msp = Msp = -2138.1274 \text{ kgf}$$

McsX with XL+ [McsX+]:

$$= MsX = -0.0 \text{ kgf}$$

McsX with XL- [McsX-]:

$$= MsX = -0.0 \text{ kgf}$$

Mcs with XL+ [Mcs+]:

$$= McsP + MsX = -2138.127 + -0.0$$

$$= -2138.127 \text{ kgf}$$

Mcs with XL- [Mcs-]:

$$= McsP + MsX = -2138.127 + -0.0$$

$$= -2138.127 \text{ kgf}$$

Compute with XL+ [McsX+]:

$$= MsX = -0.0 \text{ kgf}$$

Compute with XL- [McsX-]:

$$= MsX = -0.0 \text{ kgf}$$

Compute with XL+ [Qc+]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= -36.542 \cdot \cos(29.938) + 136.388 \cdot \sin(29.938)$$

$$= 36.401 \text{ kgf/mm}$$

Compute with XL- [Qc-]:

$$= Q \cdot \cos(a) - Ns \cdot \sin(a)$$

$$= -36.542 \cdot \cos(29.938) - 136.388 \cdot \sin(29.938)$$

$$= 36.401 \text{ kgf/mm}$$

Compute [Rc]:

$$= RL / \cos(a) = 1151.731 / \cos(29.938) = 1329.0795 \text{ mm}$$

Compute [β -co]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (1329.079^2 \cdot 19.5^2))^{1/4}$$

$$= 0.008 \text{ 1/mm}$$

Compute with XL +ve [Ncs+]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= 136.388 \cdot \cos(29.938) - -36.542 \cdot \sin(29.938)$$

$$= 136.426 \text{ kgf/mm}$$

Compute with XL -ve [Ncs-]:

$$= Ns \cdot \cos(a) + Q \cdot \sin(a)$$

$$= 136.388 \cdot \cos(29.938) + -36.542 \cdot \sin(29.938)$$

$$= 136.426 \text{ kgf/mm}$$

Compute with XL +ve [NcTheta+]:

$$= P \cdot RL / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 23.684 \cdot 1151.731 / \cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot (-2138.127 \cdot 0.008 - 36.401)$$

$$= -95.460 \text{ kgf/mm}$$

Compute with XL -ve [NcTheta-]:

$$= P \cdot RL / \cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 23.684 \cdot 1151.731 / \cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot$$



(--2138.127 • 0.008 - 36.401)
 = -95.460 kgf/mm

Note: Kcpc = 1.0

Compute with XL +ve [σ_{sm+}]:
 = Ncs/tC = 136.426/19.5 = 699.6214 kgf/cm²

Compute with XL -ve [σ_{sm-}]:
 = Ncs/tC = 136.426/19.5 = 699.6214 kgf/cm²

Compute with XL +ve [σ_{sb+}]:
 = 6 • Mcs/(tC² • Kcpc)
 = 6 • -2138.127/(19.5² • 1.0)
 = -3373.776 kgf/cm²

Compute with XL -ve [σ_{sb-}]:
 = 6 • Mcs/(tC² • Kcpc)
 = 6 • -2138.127/(19.5² • 1.0)
 = -3373.776 kgf/cm²

Compute with XL +ve [$\sigma_{Thetam+}$]:
 = NCTheta/tC = -95.46/19.5 = -489.54 kgf/cm²

Compute with XL -ve [$\sigma_{Thetam-}$]:
 = NCTheta/tC = -95.46/19.5 = -489.54 kgf/cm²

Compute with XL +ve [$\sigma_{thetab+}$]:
 = 6 • v • Mcs/(tC² • Kcpc)
 = 6 • 0.3 • -2138.127/(19.5² • 1.0)
 = -1012.133 kgf/cm²

Compute with XL -ve [$\sigma_{thetab-}$]:
 = 6 • v • Mcs / (tC² • Kcpc)
 = 6 • 0.3 • -2138.127/(19.5²•1.0)
 = -1012.133 kgf/cm²

Large End under External Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:
 = $2 \cdot \sqrt{(RL \cdot Tc/\cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc/\cos(\alpha))}}$
 = $2 \cdot \sqrt{(1151.731 \cdot 19.5/\cos(29.94)) + 1.4 \cdot \sqrt{(601.731 \cdot 19.5/\cos(29.94))}}$
 = 484.854 mm

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-31.144	2475.612	Pass
Sigma sm-	31.144	2475.612	Pass
Sigma thetam+	19.836	2475.612	Pass
Sigma thetam-	19.836	2475.612	Pass
Sigma sm+ + Sigma	119.083	4955.591	Pass
Sigma sm+ - Sigma	-181.372	4955.591	Pass
Sigma sm- + Sigma	119.083	4955.591	Pass
Sigma sm- - Sigma	-181.372	4955.591	Pass
Sigma thetam+ +	64.904	4955.591	Pass
Sigma thetam+ -	-25.233	4955.591	Pass
Sigma thetam- +	64.904	4955.591	Pass
Sigma thetam- -	-25.233	4955.591	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-31.153	2475.612	Pass
Sigma sm-	31.153	2475.612	Pass
Sigma thetam+	21.798	2475.612	Pass
Sigma thetam-	21.798	2475.612	Pass
Sigma sm+ + Sigma	119.075	4955.591	Pass



Sigma sm+ - Sig	-181.380	4955.591	Pass
Sigma sm- + Sig	119.075	4955.591	Pass
Sigma sm- - Sig	-181.380	4955.591	Pass
Sigma thetam+ +	66.866	4955.591	Pass
Sigma thetam+ -	-23.270	4955.591	Pass
Sigma thetam- +	66.866	4955.591	Pass
Sigma thetam- -	-23.270	4955.591	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

SigmaσmCyl+ + S	-31.148	-1191.173	Pass
SigmaσmCyl- + S	-31.148	-1191.173	Pass
SigmaThetamCyl+	20.817	-1486.380	Pass
SigmaThetamCyl-	20.817	-1486.380	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:
 $= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.141 \cdot 1151.731) + 0.0 / (3.142 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Net Axial Force ML -ve [XL-]:
 $= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.142 \cdot 1151.731) - 0.0 / (3.141 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Thickness Ratio [n]:
 $= tc/tL = 19.5/19.5 = 1.0$

Curve Fitting Geometry [H and B]:
 $H = \text{Sqrt}(0.5 \cdot D/tL) = \text{Sqrt}(0.5 \cdot 2303.462/19.5) = 7.6853$
 $B = \tan(29.938) = 0.5759$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: $1.0 < (n=1.0) < 1.25$

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 =	-3.065534	C2 =	3.642747	C3 =	0.810048	C4 =	-0.221192
C5 =	-0.081824	C6 =	0.035052	C7 =	0.025775	C8 =	-0.015413
C9 =	0.002102	C10 =	-0.005587	C11 =	0.000000		

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576) \\ X = 3.1672, \text{ MsN1} = -\exp(X) = -23.7415$$

Table 4.3.3 [Qn1]:

C1 =	-1.983852	C2 =	2.410703	C3 =	0.626443	C4 =	-0.119151
C5 =	-0.115841	C6 =	0.122993	C7 =	0.012160	C8 =	-0.016987
C9 =	0.010919	C10 =	-0.016653	C11 =	0.000000		

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576) \\ X = 2.0684, \text{ Qn} = -\exp(X) = -7.9123$$

Table 4.3.4 MsN2:

C1 =	-5.697151	C2 =	0.003838	C3 =	0.476317	C4 =	-0.213157
C5 =	2.233703	C6 =	0.000032	C7 =	0.002506	C8 =	-0.001663
C9 =	-0.212965	C10 =	0.000138	C11 =	-0.106203		

Compute using Table 4.3.4 [MsN2]:



$$\begin{aligned}
 X1 &= C1+C3\cdot\ln(H^2)+C5\cdot\ln(A)+C7\cdot(\ln(H^2))^2+C9\cdot(\ln(A))^2+C11\cdot\ln(H^2)\cdot\ln(A) \\
 &= C1+C3\cdot\ln(7.685^2)+C5\cdot\ln(0.523)+C7\cdot(\ln(7.685^2))^2+C9\cdot(\ln(0.523))^2+ \\
 &\quad C11\cdot\ln(7.685^2)\cdot\ln(0.523) = -0.0531 \\
 X2 &= 1+C2\cdot\ln(H^2)+C4\cdot\ln(AR)+C6\cdot(\ln(H^2))^2+C8\cdot(\ln(AR))^2+C10\cdot\ln(H^2)\cdot\ln(AR) \\
 &= 1+C2\cdot\ln(7.685)+C4\cdot\ln(0.523)+C6\cdot(\ln(7.685^2))^2+C8\cdot(\ln(0.523))^2+ \\
 &\quad C10\cdot\ln(7.685^2)\cdot\ln(0.523) = 0.2743 \\
 MsN2 &= \exp(X1/X2) = \exp(-0.053/0.274) = -0.8241
 \end{aligned}$$

Table 4.3.4 QN2:

$$\begin{aligned}
 C1 &= -4.774616 & C2 &= 0.000461 & C3 &= -0.002831 & C4 &= -0.197117 \\
 C5 &= 1.982132 & C6 &= 0.000069 & C7 &= -0.000234 & C8 &= -0.003536 \\
 C9 &= -0.202493 & C10 &= -0.000088 & C11 &= 0.001365
 \end{aligned}$$

Compute using Table 4.3.4 [Qn2]:

$$\begin{aligned}
 X3 &= C1+C3\cdot\ln(H^2)+C5\cdot\ln(A)+C7\cdot(\ln(H^2))^2+C9\cdot(\ln(A))^2+C11\cdot\ln(H^2)\cdot\ln(A) \\
 &= C1+C3\cdot\ln(7.685^2)+C5\cdot\ln(0.523)+C7\cdot(\ln(7.685^2))^2+C9\cdot(\ln(0.523))^2+ \\
 &\quad C11\cdot\ln(7.685^2)\cdot\ln(0.523) = -0.0531 \\
 X4 &= 1+C2\cdot\ln(H^2)+C4\cdot\ln(AR)+C6\cdot(\ln(H^2))^2+C8\cdot(\ln(AR))^2+C10\cdot\ln(H^2)\cdot\ln(AR) \\
 &= 1+C2\cdot\ln(7.685)+C4\cdot\ln(0.523)+C6\cdot(\ln(7.685^2))^2+C8\cdot(\ln(0.523))^2+ \\
 &\quad C10\cdot\ln(7.685^2)\cdot\ln(0.523) = 0.2743 \\
 Qn2 &= \exp(X3/X4) = \exp(-0.373/0.291) = -0.2772
 \end{aligned}$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL \cdot MsN1 = -1.055 \cdot 19.5^2 \cdot -23.742 = 95.2066 \text{ kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = 95.207 + -0.0 = 95.2066 \text{ kgf}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = 95.207 + -0.0 = 95.2066 \text{ kgf}$$

Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = -1.055 \cdot 19.5 \cdot -7.912 = 1.6272 \text{ kgf/mm}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = 1.627 + -0.0 = 1.6272 \text{ kgf/mm}$$

Compute with XL- [Q-]:

$$= Qp + Qx = 1.627 + -0.0 = 1.6272 \text{ kgf/mm}$$

Compute [βcy]:

$$\begin{aligned}
 &= 3 \cdot (1 - \nu^2) / (RL^2 \cdot tL^2) \cdot 0.25 \\
 &= (3 \cdot (1 - 0.3^2) / (1151.731^2 \cdot 19.5^2)) \cdot 0.25 = 0.0086 \text{ 1/mm}
 \end{aligned}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = -1.055 \cdot 1151.731/2 + 0.0 = -6.0731 \text{ kgf/mm}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = -1.055 \cdot 1151.731/2 + 0.0 = -6.0731 \text{ kgf/mm}$$

Compute with XL+ [Ntheta+]:

$$\begin{aligned}
 &= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\
 &= -1.055 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot \\
 &\quad (-95.207 \cdot 0.009 + 1.627) = 3.8679 \text{ kgf/mm}
 \end{aligned}$$

Compute with XL- [Ntheta-]:



$$\begin{aligned} &= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ &= -1.055 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot \\ &\quad (-95.207 \cdot 0.009 + 1.627) = 3.8679 \text{ kgf/mm} \end{aligned}$$

Note: Kpc has been set to 1.0

$$\begin{aligned} \sigma_{Sm} \text{ with XL+ } [\sigma_{sm+}]: \\ &= N_s / t_L \\ &= -6.073/19.5 \\ &= -31.144 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{Sm} \text{ with XL- } [\sigma_{sm-}]: \\ &= N_s / t_L \\ &= -6.073/19.5 \\ &= -31.144 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{Sb} \text{ with XL+ } [\sigma_{sb+}]: \\ &= 6 \cdot Ms / (t_L^2 \cdot K_{pc}) \\ &= 6 \cdot 95.207 / (19.5^2 \cdot 1.0) \\ &= 150.228 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{Sb} \text{ with XL- } [\sigma_{sb-}]: \\ &= 6 \cdot Ms / (t_L^2 \cdot K_{pc}) \\ &= 6 \cdot 95.207 / (19.5^2 \cdot 1.0) \\ &= 150.228 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{Thetam} \text{ with XL+ } [\sigma_{thetam+}]: \\ &= N_{theta} / t_L \\ &= 3.868/19.5 \text{ kgf/cm}^2 \\ &= 19.836 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{Thetam} \text{ with XL- } [\sigma_{thetam-}]: \\ &= N_{theta} / t_L \\ &= 3.868/19.5 \\ &= 19.836 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{Thetab} \text{ with XL+ } [\sigma_{thetab+}]: \\ &= 6 \cdot v \cdot Ms / (t_L^2 \cdot K_{pc}) \\ &= 6 \cdot 0.3 \cdot 95.207 / (19.5^2 \cdot 1.0) \\ &= 45.068 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{Thetab} \text{ with XL- } [\sigma_{thetab-}]: \\ &= 6 \cdot v \cdot Ms / (t_L^2 \cdot K_{pc}) \\ &= 6 \cdot 0.3 \cdot 95.207 / (19.5^2 \cdot 1.0) \\ &= 45.068 \text{ kgf/cm}^2 \end{aligned}$$

For the Large End Cone Junction per Table 4.3.1

$$\begin{aligned} \text{Compute } [McsP]: \\ &= M_{sp} = M_{sp} = 95.2066 \text{ kgf} \end{aligned}$$

$$\begin{aligned} McsX \text{ with XL+ } [McsX+]: \\ &= M_{sX} = -0.0 \text{ kgf} \end{aligned}$$

$$\begin{aligned} McsX \text{ with XL- } [McsX-]: \\ &= M_{sX} = -0.0 \text{ kgf} \end{aligned}$$

$$\begin{aligned} Mcs \text{ with XL+ } [Mcs+]: \\ &= M_{csP} + M_{sX} = 95.207 + -0.0 \\ &= 95.207 \text{ kgf} \end{aligned}$$

$$\begin{aligned} Mcs \text{ with XL- } [Mcs-]: \\ &= M_{csP} + M_{sX} = 95.207 + -0.0 \\ &= 95.207 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Compute with XL+ } [McsX+]: \\ &= M_{sX} = -0.0 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Compute with XL- } [McsX-]: \\ &= M_{sX} = -0.0 \text{ kgf} \end{aligned}$$



Compute with XL+ [Qc+]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a)$
 $= 1.627 \cdot \cos(29.938) + -6.073 \cdot \sin(29.938)$
 $= -1.621 \text{ kgf/mm}$

Compute with XL- [Qc-]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a)$
 $= 1.627 \cdot \cos(29.938) + -6.073 \cdot \sin(29.938)$
 $= -1.621 \text{ kgf/mm}$

Compute [Rc]:
 $= RL/\cos(a) = 1151.731/\cos(29.938) = 1329.0795 \text{ mm}$

Compute [β -co]:
 $= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$
 $= (3 \cdot (1 - 0.3^2) / (1329.079^2 \cdot 19.5^2))^{1/4}$
 $= 0.008 \text{ 1/mm}$

Compute with XL +ve [Ncs+]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= -6.073 \cdot \cos(29.938) - 1.627 \cdot \sin(29.938)$
 $= -6.075 \text{ kgf/mm}$

Compute with XL -ve [Ncs-]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= -6.073 \cdot \cos(29.938) - 1.627 \cdot \sin(29.938)$
 $= -6.075 \text{ kgf/mm}$

Compute with XL +ve [NcTheta+]:
 $= P \cdot RL/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$
 $= -1.055 \cdot 1151.731/\cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot (-95.207 \cdot 0.008 - -1.621)$
 $= 4.251 \text{ kgf/mm}$

Compute with XL -ve [NcTheta-]:
 $= P \cdot RL/\cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$
 $= -1.055 \cdot 1151.731/\cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot (-95.207 \cdot 0.008 - -1.621)$
 $= 4.251 \text{ kgf/mm}$

Note: $Kcpc = 1.0$

Compute with XL +ve [σ_{sm}]:
 $= Ncs/tC = -6.075/19.5 = -31.1528 \text{ kgf/cm}^2$

Compute with XL -ve [σ_{sm}]:
 $= Ncs/tC = -6.075/19.5 = -31.1528 \text{ kgf/cm}^2$

Compute with XL +ve [σ_{sb}]:
 $= 6 \cdot Mcs/(tC^2 \cdot Kcpc)$
 $= 6 \cdot 95.207/(19.5^2 \cdot 1.0)$
 $= 150.228 \text{ kgf/cm}^2$

Compute with XL -ve [σ_{sb}]:
 $= 6 \cdot Mcs/(tC^2 \cdot Kcpc)$
 $= 6 \cdot 95.207/(19.5^2 \cdot 1.0)$
 $= 150.228 \text{ kgf/cm}^2$

Compute with XL +ve [σ_{Theta+}]:
 $= NTheta/tC = 4.251/19.5 = 21.8 \text{ kgf/cm}^2$

Compute with XL -ve [σ_{Theta-}]:
 $= NTheta/tC = 4.251/19.5 = 21.8 \text{ kgf/cm}^2$

Compute with XL +ve [$\sigma_{thetab+}$]:
 $= 6 \cdot v \cdot Mcs/(tC^2 \cdot Kcpc)$
 $= 6 \cdot 0.3 \cdot 95.207/(19.5^2 \cdot 1.0)$
 $= 45.068 \text{ kgf/cm}^2$



Compute with XL -ve [σ_{thetab}]:

$$= 6 \cdot v \cdot M_{cs} / (tC^2 \cdot K_{cpc})$$

$$= 6 \cdot 0.3 \cdot 95.207 / (19.5^2 \cdot 1.0)$$

$$= 45.068 \text{ kgf/cm}^2$$

Calculations for the Small End Cone/Shell Junction

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:

$$= 0.4 \cdot E(tc/D)$$

$$= 0.4 \cdot 2057748(19.5/1232.0)$$

$$= 13027.9 \text{ kgf/cm}^2$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [2477.8] \geq F_y [2477.8], FS = 1.667$$

Allowable Stress per 4.3.11 Step 6 [Fha]:

$$= F_{ic}/FS$$

$$= 2477.7957/1.667$$

$$= 1486.380 \text{ kgf/cm}^2$$

Small End under Internal Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	491.429	2475.612	Pass
Sigma sm-	491.429	2475.612	Pass
Sigma thetam+	1908.194	2475.612	Pass
Sigma thetam-	1908.194	2475.612	Pass
Sigma sm+ + Sigma	2468.833	4955.591	Pass
Sigma sm- + Sigma	2468.833	4955.591	Pass
Sigma sm+ - Sigma	-1485.975	4955.591	Pass
Sigma sm- - Sigma	-1485.975	4955.591	Pass
Sigma thetam+ +	2501.416	4955.591	Pass
Sigma thetam- +	2501.416	4955.591	Pass
Sigma thetam+ -	1314.973	4955.591	Pass
Sigma thetam- -	1314.973	4955.591	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	361.800	2475.612	Pass
Sigma sm-	361.800	2475.612	Pass
Sigma thetam+	1836.556	2475.612	Pass
Sigma thetam-	1836.556	2475.612	Pass
Sigma sm+ + Sigma	1455.157	4955.591	Pass
Sigma sm- + Sigma	1455.157	4955.591	Pass
Sigma sm+ - Sigma	-731.558	4955.591	Pass
Sigma sm- - Sigma	-731.558	4955.591	Pass
Sigma thetam+ +	2164.563	4955.591	Pass
Sigma thetam- +	2164.563	4955.591	Pass
Sigma thetam+ -	1508.549	4955.591	Pass
Sigma thetam- -	1508.549	4955.591	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:

$$= tc/ts = 19.5/14.5 = 1.3448$$

Compute the ratio equation 4.3.61 [H]:

$$= \sqrt{(R_s/ts)} = \sqrt{(601.731/14.5)} = 6.4419$$

Compute the ratio equation 4.3.62 [B]:

$$= \tan(a) = \tan(29.938) = 0.5759$$

Compute Ms+ from equation 4.3.59 [Xs+]:

$$= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$$



$$= 0.0 / (2 \cdot 3.142 \cdot 601.731) + 0.0 / (3.141 \cdot 601.731^2)$$

$$= 0.0000 \text{ kgf/mm}$$

Compute Ms- from equation 4.3.59 [Xs-]:

$$= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$$

$$= 0.0 / (2 \cdot 3.142 \cdot 601.731) - 0.0 / (3.141 \cdot 601.731^2) + 0.0$$

$$= 0.0000 \text{ kgf/mm}$$

From Table 4.3.5 MsN:

$$\begin{aligned} C1 &= -10.652290 & C2 &= 1.976683 & C3 &= 4.469143 & C4 &= -0.063541 \\ C5 &= -1.143284 & C6 &= -0.067321 & C7 &= 0.004729 & C8 &= 0.124486 \\ C9 &= 0.017859 & C10 &= -0.004846 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.5 [MsN]:

$$= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) + C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$$

$$= \exp[-10.652 + 1.977 \cdot \ln(6.442^2) + 4.469 \cdot \ln(29.938) + 0.064 \cdot (\ln(6.442^2))^2 - 1.143 \cdot \ln(29.938)^2 + 0.067 \cdot \ln(6.442^2) \cdot \ln(29.938) + 0.005 \cdot \ln(6.442^2)^3 + 0.124 \cdot \ln(29.938)^3 + 0.018 \cdot \ln(6.442^2) \cdot \ln(29.938)^2 + 0.005 \cdot \ln(6.442^2)^2 \cdot \ln(29.938)] = 13.9152$$

Compute from Table 4.3.2 [Msp]:

$$= p \cdot tL^2 \cdot M_{sn} = 23.684 \cdot 14.5^2 \cdot 13.91515 = 692.9144 \text{ kgf}$$

From table 4.3.6 MsN:

$$\begin{aligned} C1 &= 0.000941 & C2 &= -0.000215 & C3 &= -0.001147 & C4 &= 0.129940 \\ C5 &= 0.009760 & C6 &= 0.000003 & C7 &= -0.000003 & C8 &= 0.008361 \\ C9 &= 0.004813 & C10 &= 0.000040 & C11 &= 0.207115 \end{aligned}$$

Compute from Table 4.3.6 [MsN]:

$$= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) / (1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$$

$$= (0.00094 + -0.00115 \cdot 6.442 + 0.00976 \cdot 29.938 + -0.6 \cdot 4.42^2 + 0.00481 \cdot 29.938^2 + 0.20711 \cdot 6.442 \cdot 29.938) / (1 + -0.00022 \cdot 6.442 + 0.12994 \cdot 29.938 + 0.6 \cdot 4.42^2 + 0.00836 \cdot 29.938^2 + 0.00004 \cdot 6.442 \cdot 29.938)$$

$$= 0.7144$$

Compute from Table 4.3.2 [Msx+]:

$$= X_s \cdot t_s \cdot M_{sN} = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Msx-]:

$$= X_s \cdot t_s \cdot M_{sN} = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Ms+]:

$$= M_s + M_{sx+} = 692.914 + 0.0 = 692.9144 \text{ kgf}$$

Compute from Table 4.3.2 [Ms-]:

$$= M_s + M_{sx-} = 692.914 + 0.0 = 692.9144 \text{ kgf}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.271511 & C2 &= 0.000001 & C3 &= 0.004638 & C4 &= 0.002846 \\ C5 &= -0.000906 & C6 &= 0.000000 & C7 &= -0.000002 & C8 &= -0.000119 \\ C9 &= -0.000126 & C10 &= 0.000001 & C11 &= -0.004307 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha)$$

$$= (0.27151 + 0.00464 \cdot 6.442^2 + -0.00091 \cdot 29.938 - 0.6 \cdot 4.42^4 + -0.00013 \cdot 29.938^2 + -0.00431 \cdot 6.442^2 \cdot 29.938) / (1 + 0.6 \cdot 4.42^2 + 0.00285 \cdot 29.938 + 0.6 \cdot 4.42^4 + -0.00012 \cdot 29.938^2 + 0.000001 \cdot 6.442^2 \cdot 29.938)$$

$$= -5.136$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot t_s \cdot Q_n = 23.684 \cdot 14.5 \cdot -5.136 = -17.6371 \text{ kgf/mm}$$

From table 4.3.6 Qn:



C1 = -0.438658 C2 = 0.009244 C3 = -0.367518 C4 = -0.001240
 C5 = -0.126278 C6 = 0.002049 C7 = 0.000023 C8 = -0.016622
 C9 = 0.000664 C10 = 0.000018 C11 = 0.000000

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned}
 &= C1+C2\cdot\ln(H)+C3\cdot\ln(B)+C4\cdot\ln(H)^2+C5\cdot\ln(B)^2+C6\cdot\ln(H)\cdot\ln(B)+ \\
 &C7\cdot\ln(H)^3+C8\cdot\ln(B)^3+C9\cdot\ln(H)\cdot\ln(B)^2+C10\cdot\ln(H)^2\cdot\ln(B) \\
 &= -0.43866 + 0.00924\cdot\ln(6.442) + -0.36752\cdot\ln(0.576) + \\
 &-0.00124\cdot\ln(6.442)^2 + -0.12628\cdot(\ln(0.576))^2 + \\
 &0.00205\cdot\ln(6.442)\cdot\ln(0.576) + 0.00002\cdot\ln(6.442)^3 + \\
 &-0.01662\cdot\ln(0.576)^3 + 0.00066\cdot\ln(6.442) \cdot \ln(0.576)^2 + \\
 &0.00002\cdot\ln(6.442)^2\cdot\ln(0.576) \\
 &= -0.2602
 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= Xs+ \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs- [Qx-]:

$$= Xs- \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs+ [Q+]:

$$= Qp + Qx+ = -17.637 + -0.0 = -17.6371 \text{ kgf/mm}$$

Compute with Xs- [Q-]:

$$= Qp + Qx- = -17.637 + -0.0 = -17.6371 \text{ kgf/mm}$$

Compute angle [β_{cy}]:

$$\begin{aligned}
 &= (3\cdot(1-\nu^2)/(Rs^2\cdot ts^2))^{0.25} = (3\cdot(1-0.3^2)/(601.731^2\cdot 14.5^2))^{0.25} \\
 &= 0.0138 \text{ 1/mm}
 \end{aligned}$$

Compute with Xs+ [Ns+]:

$$= P\cdot Rs/2 + Xs+ = 23.684\cdot 601.731/2 + 0.0+ = 71.2571 \text{ kgf/mm}$$

Compute with Xs- [Ns-]:

$$= P\cdot Rs/2 + Xs- = 23.684\cdot 601.731/2 + 0.0- = 71.2571 \text{ kgf/mm}$$

Compute with Xs+ [NTheta+]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= 23.684 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot \\
 &(-692.914 \cdot 0.014 - 1.627) = 276.6878 \text{ kgf/mm}
 \end{aligned}$$

Compute with Xs- [NTheta-]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= 23.684 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot \\
 &(-692.914 \cdot 0.014 - 1.627) = 276.6878 \text{ kgf/mm}
 \end{aligned}$$

Compute Stress with Xs+ [σ_{sm+}]:

$$= Ns(+)/ts = 71.257/14.5 = 491.429 \text{ kgf/cm}^2$$

Compute Stress with Xs- [σ_{sm-}]:

$$= Ns(-)/ts = 71.257/14.5 = 491.429 \text{ kgf/cm}^2$$

Compute stress with Xs+ [σ_{sb+}]:

$$= 6\cdot Ms(+)/(ts^2\cdot Kpc) = 6\cdot 692.914/(14.5^2\cdot 1.0) = 1977.4042 \text{ kgf/cm}^2$$

Compute stress with Xs- [σ_{sb-}]:

$$= 6\cdot Ms(-)/(ts^2\cdot Kpc) = 6\cdot 692.914/(14.5^2\cdot 1.0) = 1977.4042 \text{ kgf/cm}^2$$

Compute stress with Xs+ [$\sigma_{\theta_{etam+}}$]:

$$= N_{\theta_{etam}}(+)/ts = 276.688/14.5 = 1908.1943 \text{ kgf/cm}^2$$

Compute stress with Xs- [$\sigma_{\theta_{etam-}}$]:

$$= N_{\theta_{etam}}(-)/ts = 276.688/14.5 = 1908.1943 \text{ kgf/cm}^2$$

Compute stress with Xs+ [$\sigma_{\theta_{etab+}}$]:

$$= 6\cdot v\cdot Ms(+)/(ts^2\cdot Kpc) = 6\cdot 0.3\cdot 692.914/(14.5^2\cdot 1.0) = 593.2213 \text{ kgf/cm}^2$$

Compute stress with Xs- [$\sigma_{\theta_{etab-}}$]:

$$= 6\cdot v\cdot Ms(-)/(ts^2\cdot Kpc) = 6\cdot 0.3\cdot 692.914/(14.5^2\cdot 1.0) = 593.2213 \text{ kgf/cm}^2$$



Small End Cone Without a Flare

Compute with Xs+ [Qc+]:

$$= Q + \cos(a) + Ns \sin(a) = -17.6 \cos(29.94) + 71.3 \sin(29.94) \\ = 20.2785 \text{ kgf/mm}$$

Compute with Xs- [Qc-]:

$$= Q - \cos(a) + Ns \sin(a) = -17.6 \cos(29.94) + 71.3 \sin(29.94) \\ = 20.2785 \text{ kgf/mm}$$

Compute effective radius [Rc]:

$$= Rs / \cos(a) = 601.731 / \cos(29.938) = 694.3881 \text{ mm}$$

Compute the angle [βco]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tc^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (694.388^2 \cdot 19.5^2))^{0.25} \\ = 0.0110 \text{ 1/mm}$$

Compute with Ms+ [Ncs+]:

$$= Ns \cos(a) - Q \sin(a) \\ = 71.257 \cos(29.94) - 17.637 \sin(29.94) = 70.5508 \text{ kgf/mm}$$

Compute with Ms- [Ncs-]:

$$= Ns \cos(a) - Q \sin(a) \\ = 71.257 \cos(29.94) - 17.637 \sin(29.94) = 70.5508 \text{ kgf/mm}$$

Compute with Ms+ [Nctheta+]:

$$= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc (-Mcs \cdot \beta_{co} + Qc) \\ = 23.684 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot 694.388 \cdot \\ (-692.914 \cdot 0.011 + 20.278) = 358.128 \text{ kgf/mm}$$

Compute with Ms- [Nctheta-]:

$$= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc (-Mcs \cdot \beta_{co} + Qc) \\ = 23.684 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot \\ (-692.914 \cdot 0.011 + 20.278) = 358.128 \text{ kgf/mm}$$

Compute the stress with Ms+ [σsm+]:

$$= Ncs / tc = 70.551 / 19.5 = 361.7998 \text{ kgf/cm}^2$$

Compute the stress with Ms- [σsm-]:

$$= Ncs / tc = 70.551 / 19.5 = 361.7998 \text{ kgf/cm}^2$$

Compute the stress with Ms+ [σsb+]:

$$= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 692.914 / (19.5^2 \cdot 1.0) = 1093.3575 \text{ kgf/cm}^2$$

Compute the stress with Ms- [σsb-]:

$$= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 692.914 / (19.5^2 \cdot 1.0) = 1093.3575 \text{ kgf/cm}^2$$

Compute stress with Ms+ [σthetam+]:

$$= Nctheta / tc = 358.128 / 19.5 = 1836.556 \text{ kgf/cm}^2$$

Compute stress with Ms- [σthetam-]:

$$= Nctheta / tc = 358.128 / 19.5 = 1836.556 \text{ kgf/cm}^2$$

Compute stress with Ms+ [σthetab+]:

$$= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot 692.914 / (19.5^2 \cdot 1.0) = 328.0073 \text{ kgf/cm}^2$$

Compute stress with Ms- [σthetab-]:

$$= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot 692.914 / (19.5^2 \cdot 1.0) = 328.0073 \text{ kgf/cm}^2$$

Small End under External Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-21.882	2475.612	Pass
Sigma sm-	-21.882	2475.612	Pass
Sigma thetam+	-84.968	2475.612	Pass
Sigma thetam-	-84.968	2475.612	Pass
Sigma sm+ + Sigm	-109.932	4955.591	Pass



Sigma sm- + Sigm	-109.932	4955.591	Pass
Sigma sm+ - Sigm	66.168	4955.591	Pass
Sigma sm- - Sigm	66.168	4955.591	Pass
Sigma thetam+ +	-111.383	4955.591	Pass
Sigma thetam- +	-111.383	4955.591	Pass
Sigma thetam+ -	-58.553	4955.591	Pass
Sigma thetam- -	-58.553	4955.591	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-16.110	2475.612	Pass
Sigma sm-	-16.110	2475.612	Pass
Sigma thetam+	-81.778	2475.612	Pass
Sigma thetam-	-81.778	2475.612	Pass
Sigma sm+ + Sigm	-64.795	4955.591	Pass
Sigma sm- + Sigm	-64.795	4955.591	Pass
Sigma sm+ - Sigm	32.575	4955.591	Pass
Sigma sm- - Sigm	32.575	4955.591	Pass
Sigma thetam+ +	-96.384	4955.591	Pass
Sigma thetam- +	-96.384	4955.591	Pass
Sigma thetam+ -	-67.173	4955.591	Pass
Sigma thetam- -	-67.173	4955.591	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

Sigma _{smCyl} + + S	-18.996	-1191.173	Pass
Sigma _{smCyl} - + S	-18.996	-1191.173	Pass
Sigma _{thetamCyl} +	-83.373	-1486.380	Pass
Sigma _{thetamCyl} -	-83.373	-1486.380	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
 $= t_c/t_s = 19.5/14.5 = 1.3448$

Compute the ratio equation 4.3.61 [H]:
 $= \sqrt{(R_s/t_s)} = \sqrt{(601.731/14.5)} = 6.4419$

Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(29.938) = 0.5759$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$
 $= 0.0 / (2 \cdot 3.142 \cdot 601.731) + 0.0 / (3.141 \cdot 601.731^2)$
 $= 0.0000 \text{ kgf/mm}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$
 $= 0.0 / (2 \cdot 3.142 \cdot 601.731) - 0.0 / (3.141 \cdot 601.731^2) + 0.0$
 $= 0.0000 \text{ kgf/mm}$

From Table 4.3.5 MsN:

C1 = -10.652290 C2 = 1.976683 C3 = 4.469143 C4 = -0.063541
 C5 = -1.143284 C6 = -0.067321 C7 = 0.004729 C8 = 0.124486
 C9 = 0.017859 C10 = -0.004846 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
 $= \exp[C1+C2 \cdot \ln(H^2)+C3 \cdot \ln(a)+C4 \cdot (\ln(H^2))^2+C5 \cdot \ln(a)^2+C6 \cdot \ln(H^2) \cdot \ln(a)+$
 $C7 \cdot \ln(H^2)^3+C8 \cdot \ln(a)^3+C9 \cdot \ln(H^2) \cdot \ln(a)^2+C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-10.652+1.977 \cdot \ln(6.442^2)+4.469 \cdot \ln(29.938)+$
 $-0.064 \cdot (\ln(6.442^2))^2+-1.143 \cdot \ln(29.938)^2+$
 $-0.067 \cdot \ln(6.442^2) \cdot \ln(29.938)+ 0.005 \cdot \ln(6.442^2)^3$
 $0.124 \cdot \ln(29.938)^3+ 0.018 \cdot \ln(6.442^2) \cdot \ln(29.938)^2+$
 $-0.005 \cdot \ln(6.442^2)^2 \cdot \ln(29.938)] = 13.9152$

Compute from Table 4.3.2 [Msp]:
 $= p \cdot tL^2 \cdot M_{sn} = -1.055 \cdot 14.5^2 \cdot 13.91515 = -30.8541 \text{ kgf}$



From table 4.3.6 MsN:

$$\begin{aligned} C1 &= 0.000941 & C2 &= -0.000215 & C3 &= -0.001147 & C4 &= 0.129940 \\ C5 &= 0.009760 & C6 &= 0.000003 & C7 &= -0.000003 & C8 &= 0.008361 \\ C9 &= 0.004813 & C10 &= 0.000040 & C11 &= 0.207115 \end{aligned}$$

Compute from Table 4.3.6 [MsN]:

$$\begin{aligned} &= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) / \\ &\quad (1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B) \\ &= (0.00094 + -0.00115 \cdot 6.442 + 0.00976 \cdot 29.938 + \\ &\quad -0.6.442^2 + 0.00481 \cdot 29.938^2 + \\ &\quad 0.20711 \cdot 6.442 \cdot 29.938) / (1 + -0.00022 \cdot 6.442 + \\ &\quad 0.12994 \cdot 29.938 + 0.6.442^2 + 0.00836 \cdot 29.938^2 + \\ &\quad 0.00004 \cdot 6.442 \cdot 29.938) \\ &= 0.7144 \end{aligned}$$

Compute from Table 4.3.2 [Msx+]:

$$= Xs \cdot ts \cdot MsN = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Msx-]:

$$= Xs \cdot ts \cdot MsN = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Ms+]:

$$= Ms + Msx+ = -30.854 + 0.0 = -30.8541 \text{ kgf}$$

Compute from Table 4.3.2 [Ms-]:

$$= Ms + Msx- = -30.854 + 0.0 = -30.8541 \text{ kgf}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.271511 & C2 &= 0.000001 & C3 &= 0.004638 & C4 &= 0.002846 \\ C5 &= -0.000906 & C6 &= 0.000000 & C7 &= -0.000002 & C8 &= -0.000119 \\ C9 &= -0.000126 & C10 &= 0.000001 & C11 &= -0.004307 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$\begin{aligned} &= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / \\ &\quad (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha) \\ &= (0.27151 + 0.00464 \cdot 6.442^2 + -0.00091 \cdot 29.938 \\ &\quad -0.6.442^4 + -0.00013 \cdot 29.938^2 + -0.00431 \cdot 6.442^2 \\ &\quad \cdot 29.938) / (1 + 0.6.442^2 + 0.00285 \cdot 29.938 + \\ &\quad 0.6.442^4 + -0.00012 \cdot 29.938^2 + 0.000001 \cdot 6.442^2 \cdot 29.938) \\ &= -5.136 \end{aligned}$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot ts \cdot Qn = -1.055 \cdot 14.5 \cdot -5.136 = 0.7853 \text{ kgf/mm}$$

From table 4.3.6 Qn:

$$\begin{aligned} C1 &= -0.438658 & C2 &= 0.009244 & C3 &= -0.367518 & C4 &= -0.001240 \\ C5 &= -0.126278 & C6 &= 0.002049 & C7 &= 0.000023 & C8 &= -0.016622 \\ C9 &= 0.000664 & C10 &= 0.000018 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned} &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &\quad C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\ &= -0.43866 + 0.00924 \cdot \ln(6.442) + -0.36752 \cdot \ln(0.576) + \\ &\quad -0.00124 \cdot \ln(6.442)^2 + -0.12628 \cdot (\ln(0.576))^2 + \\ &\quad 0.00205 \cdot \ln(6.442) \cdot \ln(0.576) + 0.00002 \cdot \ln(6.442)^3 + \\ &\quad -0.01662 \cdot \ln(0.576)^3 + 0.00066 \cdot \ln(6.442) \cdot \ln(0.576)^2 + \\ &\quad 0.00002 \cdot \ln(6.442)^2 \cdot \ln(0.576) \\ &= -0.2602 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= Xs+ \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs- [Qx-]:

$$= Xs- \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs+ [Q+]:

$$= Qp + Qx+ = 0.785 + -0.0 = 0.7853 \text{ kgf/mm}$$

Compute with Xs- [Q-]:

$$= Qp + Qx- = 0.785 + -0.0 = 0.7853 \text{ kgf/mm}$$



Compute angle [β_{cy}]:
 $= (3 \cdot (1 - v^2) / (R_s^2 \cdot t_s^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (601.731^2 \cdot 14.5^2))^{0.25}$
 $= 0.0138 \text{ 1/mm}$

Compute with X_s+ [N_s+]:
 $= P \cdot R_s / 2 + X_s+ = -1.055 \cdot 601.731 / 2 + 0.0+ = -3.1729 \text{ kgf/mm}$

Compute with X_s- [N_s-]:
 $= P \cdot R_s / 2 + X_s- = -1.055 \cdot 601.731 / 2 + 0.0- = -3.1729 \text{ kgf/mm}$

Compute with X_s+ [$N_{\theta}+$]:
 $= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q)$
 $= -1.055 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot$
 $(-30.854 \cdot 0.014 - 1.627) = -12.3204 \text{ kgf/mm}$

Compute with X_s- [$N_{\theta}-$]:
 $= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q)$
 $= -1.055 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot$
 $(-30.854 \cdot 0.014 - 1.627) = -12.3204 \text{ kgf/mm}$

Compute Stress with X_s+ [$\sigma_{sm}+$]:
 $= N_s(+)/t_s = -3.173/14.5 = -21.8824 \text{ kgf/cm}^2$

Compute Stress with X_s- [$\sigma_{sm}-$]:
 $= N_s(-)/t_s = -3.173/14.5 = -21.8824 \text{ kgf/cm}^2$

Compute stress with X_s+ [$\sigma_{sb}+$]:
 $= 6 \cdot M_s(+)/(t_s^2 \cdot K_{pc}) = 6 \cdot -30.854/(14.5^2 \cdot 1.0) = -88.0499 \text{ kgf/cm}^2$

Compute stress with X_s- [$\sigma_{sb}-$]:
 $= 6 \cdot M_s(-)/(t_s^2 \cdot K_{pc}) = 6 \cdot -30.854/(14.5^2 \cdot 1.0) = -88.0499 \text{ kgf/cm}^2$

Compute stress with X_s+ [$\sigma_{\theta}+$]:
 $= N_{\theta}(+)/t_s = -12.32/14.5 = -84.9682 \text{ kgf/cm}^2$

Compute stress with X_s- [$\sigma_{\theta}-$]:
 $= N_{\theta}(-)/t_s = -12.32/14.5 = -84.9682 \text{ kgf/cm}^2$

Compute stress with X_s+ [$\sigma_{\theta}+$]:
 $= 6 \cdot v \cdot M_s(+)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot -30.854/(14.5^2 \cdot 1.0) = -26.415 \text{ kgf/cm}^2$

Compute stress with X_s- [$\sigma_{\theta}-$]:
 $= 6 \cdot v \cdot M_s(-)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot -30.854/(14.5^2 \cdot 1.0) = -26.415 \text{ kgf/cm}^2$

Small End Cone Without a Flare

Compute with X_s+ [Q_c+]:
 $= Q \cdot \cos(a) + N_s \cdot \sin(a) = 0.8 \cdot \cos(29.94) + -3.2 \cdot \sin(29.94)$
 $= -0.9030 \text{ kgf/mm}$

Compute with X_s- [Q_c-]:
 $= Q \cdot \cos(a) + N_s \cdot \sin(a) = 0.8 \cdot \cos(29.94) + -3.2 \cdot \sin(29.94)$
 $= -0.9030 \text{ kgf/mm}$

Compute effective radius [R_c]:
 $= R_s / \cos(a) = 601.731 / \cos(29.938) = 694.3881 \text{ mm}$

Compute the angle [β_{co}]:
 $= (3 \cdot (1 - v^2) / (R_c^2 \cdot t_c^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (694.388^2 \cdot 19.5^2))^{0.25}$
 $= 0.0110 \text{ 1/mm}$

Compute with M_s+ [$N_{cs}+$]:
 $= N_s \cdot \cos(a) - Q \cdot \sin(a)$
 $= -3.173 \cdot \cos(29.94) - 0.785 \cdot \sin(29.94) = -3.1415 \text{ kgf/mm}$

Compute with M_s- [$N_{cs}-$]:
 $= N_s \cdot \cos(a) - Q \cdot \sin(a)$
 $= -3.173 \cdot \cos(29.94) - 0.785 \cdot \sin(29.94) = -3.1415 \text{ kgf/mm}$



Compute with Ms+ [Ncttheta+]:
= $P \cdot R_s / \cos(a) + 2 \cdot \beta_{co} \cdot R_c (-Mcs \cdot \beta_{co} + Qc)$
= $-1.055 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot 694.388 \cdot$
 $(-30.854 \cdot 0.011 + -0.903) = -15.9467 \text{ kgf/mm}$

Compute with Ms- [Ncttheta-]:
= $P \cdot R_s / \cos(a) + 2 \cdot \beta_{co} \cdot R_c (-Mcs \cdot \beta_{co} + Qc)$
= $-1.055 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot$
 $(-30.854 \cdot 0.011 + -0.903) = -15.9467 \text{ kgf/mm}$

Compute the stress with Ms+ [σ_{sm+}]:
= $Ncs / tc = -3.141 / 19.5 = -16.1102 \text{ kgf/cm}^2$

Compute the stress with Ms- [σ_{sm-}]:
= $Ncs / tc = -3.141 / 19.5 = -16.1102 \text{ kgf/cm}^2$

Compute the stress with Ms+ [σ_{sb+}]:
= $6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -30.854 / (19.5^2 \cdot 1.0) = -48.6851 \text{ kgf/cm}^2$

Compute the stress with Ms- [σ_{sb-}]:
= $6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -30.854 / (19.5^2 \cdot 1.0) = -48.6851 \text{ kgf/cm}^2$

Compute stress with Ms+ [$\sigma_{thetam+}$]:
= $Ncttheta / tc = -15.947 / 19.5 = -81.778 \text{ kgf/cm}^2$

Compute stress with Ms- [$\sigma_{thetam-}$]:
= $Ncttheta / tc = -15.947 / 19.5 = -81.778 \text{ kgf/cm}^2$

Compute stress with Ms+ [$\sigma_{thetab+}$]:
= $6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -30.854 / (19.5^2 \cdot 1.0) = -14.6055 \text{ kgf/cm}^2$

Compute stress with Ms- [$\sigma_{thetab-}$]:
= $6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -30.854 / (19.5^2 \cdot 1.0) = -14.6055 \text{ kgf/cm}^2$



Input, Nozzle Desc: NOZZLE N1

From: 40

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		132.92	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

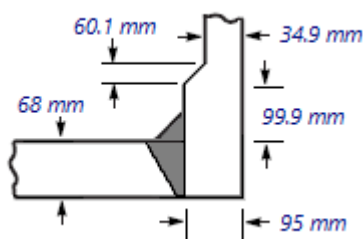
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1568.83	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.44	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	99.9250	mm
Height of Beveled Transition	L'	60.0750	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	582.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: NOZZLE N1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.250	in.
Actual Thickness Used in Calculation	1.375	in.



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 390.35 \cdot (\exp(163.158/(1568.835 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 21.3912 + 1.5 + 0.0 = 22.8912 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1568.835 \cdot 1.0 \cdot \ln((2 \cdot 33.425 + 390.35) / 390.35) - 0.0 \\ &= 247.997 - 0.0 = 247.9969 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 64.056 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 30.556 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 495.2,	Allowable	: 1568.8 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3568.9 kgf/cm ²	Passed
Occasional	: 510.0,	Allowable	: 2086.6 kgf/cm ²	Passed
Shear	: 36.5,	Allowable	: 1098.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1568.8/1650.4, 1] \\ &= 0.951 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 95.0 - 1.5 \\ &= 93.500 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 34.925 - 1.5 \\ &= 33.425 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 1203.0/66.5 \\ &= 18.090 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1200.0/2 + 1.5 \\ &= 601.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{601.5 \cdot 66.5}, 2 \cdot 195.175) \\ &= 199.999 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 66.5, 0.0] + \sqrt{(195.175 \cdot 93.5)}$$
$$= 135.088 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 582.0$$
$$= 582.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(66.5 + 0.0)$$
$$= 532.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[135.088, 582.0, 532.0] + 66.5$$
$$= 201.588 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 66.5 + ((0.0 \cdot 1.0) / 200.0)$$
$$= 66.500 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 195.18 + 93.5) / (\sqrt{(1203.0 + 66.5) \cdot 66.5}))$$
$$= 1.665$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 66.5 \cdot 200.0 \cdot \max((1.665/5)^{0.85}, 1)$$
$$= 133.000 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$
$$= 155.607 + 26.695$$
$$= 182.303 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 133.0 + 0.951(182.303 + 0.0) + 0.5 + 0.0 + 0.0$$
$$= 306.792 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 93.5 / \ln[1 + 93.5/195.175]$$
$$= 238.883 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 66.5 / \ln[1 + 66.5/601.5]$$
$$= 634.169 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 163.158 \cdot 238.883 \cdot 201.588$$
$$= 78570.4 \text{ kgf}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 163.158 \cdot 634.169(200.0 + 93.5) \\ &= 303682.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 163.158 \cdot 634.169 \cdot 195.175 \\ &= 201946.9 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (303682.9 + 201946.9 + 78570.4)/163.158 \\ &= 3580.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.612 / (2 \cdot 3580.582/306.792 - 634.169/66.5) \\ &= 179.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.408[66.5/634.169] \\ &= 173.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(179.318, 173.065) \\ &= 173.065 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (78570.375 + 303682.875 + 201946.922)/306.792 \\ &= 1904.226 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 163.158 \cdot 634.169/66.5 \\ &= 1555.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1904.226 - 1555.937, 1555.937) \\ &= 2252.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2252.51	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	173.06	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (195.175 + 93.5)/195.175 \\ &= 1.479 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (195.175 + 93.5) \\ &= 453.450 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 93.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(201947 \cdot 1.48, 1.5 \cdot 1568.8(182.303 + 0.0), \pi/4 \cdot 163.2 \cdot 195.18^2 \cdot 1.48^2)$$

$$= 106786.508 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds}/(L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 106786.508/(453.45(0.49 \cdot 7.071 + 0.6 \cdot 66.5 + 0.49 \cdot 0.0))$$

$$= 543.064 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	8 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	14 °C
Temperature reduction per Graph 3.13	:	7 °C
The computed stress ratio is	:	0.866

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*)/(t_{nom} - c_i - c_{ext})$$

$$= (57.591 \cdot 1.0)/(68.0 - 1.5 - 0.0)$$

$$= 0.866$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 8 °C

The Drop for this Nozzle is : 74.0088 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 724.0088 mm

Input Echo, WRC297 Item 1, Description: NOZZLE N1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1200.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.41 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.41 kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	387.350	mm
Nozzle Thickness	95.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	1682.44 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1568.83 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-5736.17	kgf
Circumferential Shear	Vc	4588.95	kgf
Longitudinal Shear	Vl	4588.95	kgf
Circumferential Moment	Mc	6.19	kgf-m.
Longitudinal Moment	Ml	6.19	kgf-m.
Torsional Moment	Mt	7.22	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

*Warning - The ratio, Dv/Tv (19.647) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (6.077) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 577.35 + 2 \cdot 1.65 \cdot \sqrt{(634.75(68.0 - 1.5))}$
 $= 1255.344 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1269.500 mm
 Nozzle Outside Diameter (d) = 577.350 mm
 Vessel Thickness used (T) = 66.500 mm
 Nozzle Thickness used (t) = 93.500 mm
 T / t = 0.711
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.987$$

Nr/P = 0.076
 Mr/P = 0.132
 M0/P = 0.040
 N0/P = 0.101
 MrD/Mc = 0.269
 NrDL/Mc = 0.118
 M0d/Mc = 0.088
 N0DL/Mc = 0.063
 MrD/Ml = 0.113
 NrDL/Ml = 0.068
 M0D/Ml = 0.039
 N0DL/Ml = 0.126

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	31	-31	31	-31
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1398	1561	1398	1561
Outplane Stress Summary	1442	1543	1442	1543

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	102	-102	102	-102
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	699	699	699	699
Inplane Stress Summary	811	606	811	606

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			7	7	-7	-7
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			7	7	-7	-7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			1442	1543	1442	1543

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			13	13	13	13
Outplane Bending (P)			31	-31	31	-31
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			0	0	0	0
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			699	699	699	699
Outplane Stress Summary			743	681	743	681

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			9	9	9	9
Inplane Bending (P)			102	-102	102	-102
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			0	0	0	0
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			1398	1561	1398	1561
Inplane Stress Summary			1510	1468	1511	1468

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-7	-7	7	7
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			1510	1469	1511	1468

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			13	13	13	13



Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525

Hoop Stress Summary	342	538	342	538
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	4	4	4	4
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	164	164	164	164

Axial Stress Summary	205	131	205	131
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	5	5	-5	-5
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	342	538	342	538
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525

Hoop Stress Summary	342	538	342	538
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	4	4	4	4
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	164	164	164	164

Axial Stress Summary	205	131	205	131
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	342	538	342	538

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1398	1561	1398	1561	1398	1561	1398	1561
Circ. Pl (SUS)	13	13	13	13	9	9	9	9
Circ. Q (SUS)	31	-31	31	-31	102	-102	102	-102
Long. Pm (SUS)	699	699	699	699	699	699	699	699
Long. Pl (SUS)	9	9	9	9	13	13	13	13
Long. Q (SUS)	102	-102	102	-102	31	-31	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1398	1561	1398	1561	1398	1561	1398	1561
Pm+Pl (SUS)	1411	1574	1411	1574	1407	1570	1407	1570
Pm+Pl+Q (Total)	1442	1543	1442	1543	1509	1468	1509	1468

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1561.00	1650.41	Passed
Pm+Pl (SUS)	1574.06	2475.61	Passed
Pm+Pl+Q (TOTAL)	1543.05	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	329	525	329	525	329	525	329	525
Circ. Pl (SUS)	13	13	13	13	13	13	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	164	164	164	164	164	164	164	164
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	37	-37	37	-37	37	-37	37	-37



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	329.0	525.0	329.0	525.0	329.0	525.0	329.0	525.0
Pm+Pl (SUS)	342.1	538.1	342.1	538.1	342.1	538.1	342.1	538.1
Pm+Pl+Q (Total)	342.2	538.1	342.2	538.1	342.2	538.1	342.2	538.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	525.00	1568.83	Passed
Pm+Pl (SUS)	538.07	2353.25	Passed
Pm+Pl+Q (TOTAL)	538.06	4706.50	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: NOZZLE N2

From: 40

Pressure for Reinforcement Calculations	P	163.278	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		132.92	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

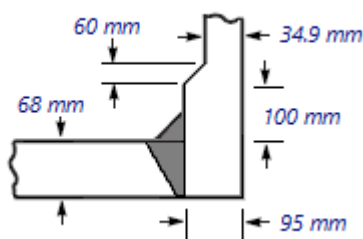
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1568.83	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.44	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	732.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: NOZZLE N2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.250	in.
Actual Thickness Used in Calculation	1.375	in.



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 390.35 \cdot (\exp(163.278/(1568.835 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 21.4077 + 1.5 + 0.0 = 22.9077 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1568.835 \cdot 1.0 \cdot \ln((2 \cdot 33.425 + 390.35) / 390.35) - 0.0 \\ &= 247.997 - 0.0 = 247.9969 \text{ kgf/cm}^2 \end{aligned}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 220.46 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 488.67 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 64.104 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 30.556 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 495.6,	Allowable	: 1568.8 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3568.5 kgf/cm ²	Passed
Occasional	: 510.3,	Allowable	: 2086.6 kgf/cm ²	Passed
Shear	: 36.5,	Allowable	: 1098.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1568.8/1650.4, 1] \\ &= 0.951 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 95.0 - 1.5 \\ &= 93.500 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 34.925 - 1.5 \\ &= 33.425 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 1203.0/66.5 \\ &= 18.090 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1200.0/2 + 1.5 \\ &= 601.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{601.5 \cdot 66.5}, 2 \cdot 195.175) \end{aligned}$$



= 199.999 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 66.5, 0.0] + \sqrt{(195.175 \cdot 93.5)} \\ &= 135.088 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 732.0 \\ &= 732.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(66.5 + 0.0) \\ &= 532.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[135.088, 732.0, 532.0] + 66.5 \\ &= 201.588 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 66.5 + ((0.0 \cdot 1.0) / 200.0) \\ &= 66.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 195.18 + 93.5) / (\sqrt{(1203.0 + 66.5) \cdot 66.5})) \\ &= 1.665 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 66.5 \cdot 200.0 \cdot \max((1.665/5)^{0.85}, 1) \\ &= 133.000 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 155.678 + 26.644 \\ &= 182.321 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 133.0 + 0.951(182.321 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 306.810 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 93.5 / \ln[1 + 93.5/195.175] \\ &= 238.883 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 66.5 / \ln[1 + 66.5/601.5] \\ &= 634.169 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.408 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 163.278 \cdot 238.883 \cdot 201.588 \\ &= 78628.1 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Force acting on the Shell [fS]:} \\ &= P \cdot Rxs(LR + tn) \\ &= 163.278 \cdot 634.169(200.0 + 93.5) \\ &= 303906.1 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Discontinuity Force from Internal Pressure [fY]:} \\ &= P \cdot Rxs \cdot Rnc \\ &= 163.278 \cdot 634.169 \cdot 195.175 \\ &= 202095.4 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Area Resisting Internal Pressure [Ap]:} \\ &= (fS + fY + fN)/P \\ &= (303906.1 + 202095.4 + 78628.1)/163.278 \\ &= 3580.6 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax1]:} \\ &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.612 / (2 \cdot 3580.582/306.81 - 634.169/66.5) \\ &= 179.3 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax2]:} \\ &= S[t/Rxs] \\ &= 1650.408[66.5/634.169] \\ &= 173.1 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure [Pmax]:} \\ &= \min(Pmax1, Pmax2) \\ &= \min(179.335, 173.065) \\ &= 173.065 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Average Primary Membrane Stress [\sigma Avg]:} \\ &= (fN + fS + fY) / AT \\ &= (78628.141 + 303906.094 + 202095.391)/306.81 \\ &= 1905.515 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{General Primary Membrane Stress [\sigma Circ]:} \\ &= P \cdot Rxs / teff \\ &= 163.278 \cdot 634.169/66.5 \\ &= 1557.1 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Local Primary Membrane Stress [PL]:} \\ &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1905.515 - 1557.081, 1557.081) \\ &= 2253.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2253.95	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	173.06	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

$$\begin{aligned} &\text{Discontinuity Force Factor [ky]:} \\ &= (Rnc + tn) / Rnc \\ &= (195.175 + 93.5)/195.175 \\ &= 1.479 \text{ For set-in Nozzles} \end{aligned}$$

$$\begin{aligned} &\text{Weld Length of Nozzle to Shell Weld [Lt]:} \\ &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (195.175 + 93.5) \\ &= 453.450 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Weld Throat Dimensions, (0.7071 \cdot Leg Dimensions) [L41T, L42T, L43T]:} \\ &= 7.071, 0.000, 0.000, \text{ mm} \end{aligned}$$

Required Weld Throat Dimensions :



tc, L41T = 7.071, must be $\geq \min(0.7 * 93.500, 6.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(202095 \cdot 1.48, 1.5 \cdot 1568.8(182.321 + 0.0), \pi/4 \cdot 163.3 \cdot 195.18^2 \cdot 1.48^2)$
 $= 106865.023 \text{ kgf}$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 106865.023 / (453.45(0.49 \cdot 7.071 + 0.6 \cdot 66.5 + 0.49 \cdot 0.0))$
 $= 543.463 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : 8 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 14 °C
Temperature reduction per Graph 3.13 : 7 °C
The computed stress ratio is : 0.867

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (57.635 \cdot 1.0) / (68.0 - 1.5 - 0.0)$
 $= 0.867$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 8 °C

The Drop for this Nozzle is : 74.0088 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 874.0089 mm

Input Echo, WRC297 Item 1, Description: NOZZLE N2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1200.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1650.41	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	387.350	mm
Nozzle Thickness	95.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 1682.44	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1568.83	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	163.28	kgf/cm ²
Radial Load	P	-5736.17	kgf
Circumferential Shear	Vc	4588.95	kgf
Longitudinal Shear	Vl	4588.95	kgf
Circumferential Moment	Mc	6.19	kgf-m.
Longitudinal Moment	Ml	6.19	kgf-m.



Torsional Moment Mt 7.22 kgf-m.

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, D_v/T_v (19.647) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_n (6.077) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 577.35 + 2 \cdot 1.65 \cdot \sqrt{(634.75(68.0 - 1.5))}$
 $= 1255.344 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1269.500 mm
 Nozzle Outside Diameter (d) = 577.350 mm
 Vessel Thickness used (T) = 66.500 mm
 Nozzle Thickness used (t) = 93.500 mm
 $T / t = 0.711$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.987$$

Nr/P = 0.076
 Mr/P = 0.132
 M0/P = 0.040
 N0/P = 0.101
 MrD/Mc = 0.269
 NrDL/Mc = 0.118
 M0d/Mc = 0.088
 N0DL/Mc = 0.063
 MrD/Ml = 0.113
 NrDL/Ml = 0.068
 M0D/Ml = 0.039
 N0DL/Ml = 0.126

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	31	-31	31	-31
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1399	1562	1399	1562
Outplane Stress Summary	1443	1544	1443	1544

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	102	-102	102	-102
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	699	699	699	699



Inplane Stress Summary 812 606 812 606

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE Au Al Bu Bl
 (Shear stress normal to Top Top Bottom Bottom
 longitudinal plane) Outside Inside Outside Inside

Outplane Shear (Vc) 7 7 -7 -7
 Outplane Shear (Vl) 0 0 0 0
 Outplane Shear (Mt) 0 0 0 0

Shear Stress Summary 7 7 -7 -7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE Au Al Bu Bl
 (Stress Intensities Top Top Bottom Bottom
 Outside Inside Outside Inside

Two * Max Shear Stress 1443 1544 1443 1544

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses Normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Membrane (P) 13 13 13 13
 Outplane Bending (P) 31 -31 31 -31
 Outplane Membrane (Mc) 0 0 0 0
 Outplane Bending (Mc) 0 0 0 0
 Outplane Membrane (ML) 0 0 0 0
 Outplane Bending (ML) 0 0 0 0
 Normal Pressure Stress 699 699 699 699

Outplane Stress Summary 743 681 743 681

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses parallel to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Inplane Membrane (P) 9 9 9 9
 Inplane Bending (P) 102 -102 102 -102
 Inplane Membrane (Mc) 0 0 0 0
 Inplane Bending (Mc) 0 0 0 0
 Inplane Membrane (ML) 0 0 0 0
 Inplane Bending (ML) 0 0 0 0
 Inplane Pressure Stress 1399 1562 1399 1562

Inplane Stress Summary 1511 1470 1512 1469

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Shear stress normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Shear (Vc) 0 0 0 0
 Outplane Shear (Vl) -7 -7 7 7
 Torsional Shear (Mt) 0 0 0 0

Shear Stress Summary -7 -7 7 7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stress Intensities Left Left Right Right
 Outside Inside Outside Inside

Two * Max Shear Stress 1511 1470 1512 1469

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE Au Al Bu Bl



(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525
Hoop Stress Summary	342	538	342	538

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	164	164	164	164
Axial Stress Summary	205	131	205	131

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	5	5	-5	-5

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	343	538	343	538

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525
Hoop Stress Summary	342	538	342	538

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0

Axial Pressure Stress	164	164	164	164
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Axial Stress Summary	205	132	206	131
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	343	538	343	538

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1399	1562	1399	1562	1399	1562	1399	1562
Circ. Pl (SUS)	13	13	13	13	9	9	9	9
Circ. Q (SUS)	31	-31	31	-31	102	-102	102	-102
Long. Pm (SUS)	699	699	699	699	699	699	699	699
Long. Pl (SUS)	9	9	9	9	13	13	13	13
Long. Q (SUS)	102	-102	102	-102	31	-31	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1399	1562	1399	1562	1399	1562	1399	1562
Pm+Pl (SUS)	1412	1575	1412	1575	1408	1571	1408	1571
Pm+Pl+Q (Total)	1443	1544	1443	1544	1510	1469	1510	1469

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1562.00	1650.41	Passed
Pm+Pl (SUS)	1575.06	2475.61	Passed
Pm+Pl+Q (TOTAL)	1544.05	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	329	525	329	525	329	525	329	525
Circ. Pl (SUS)	13	13	13	13	13	13	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	164	164	164	164	164	164	164	164
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	37	-37	37	-37	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	329.0	525.0	329.0	525.0	329.0	525.0	329.0	525.0
Pm+Pl (SUS)	342.1	538.1	342.1	538.1	342.1	538.1	342.1	538.1
Pm+Pl+Q (Total)	342.2	538.1	342.2	538.1	342.2	538.1	342.2	538.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	525.00	1568.83	Passed
Pm+Pl (SUS)	538.07	2353.25	Passed
Pm+Pl+Q (TOTAL)	538.06	4706.50	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz N9

Parent: NOZZLE N2

Pressure for Reinforcement Calculations	P	163.313	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-350 LF2	+N
Parent Allowable Stress at Temperature	Sv	1568.83	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1682.44	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	387.35	mm
Shell Finished (Minimum) Thickness	t	34.9250	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

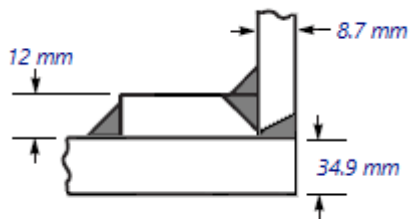
Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N9

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 45.85 \cdot (\exp(163.313/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.5963 + 1.5 + 0.0 = 4.0963 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 7.238 + 45.85) / 45.85) - 0.0$$

$$= 417.66 - 0.0 = 417.6602 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm
 Min. required shell thickness, internal p + c : 22.933 mm

Nozzle neck minimum required thickness : 4.920 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 4.920 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1568.8, 1]$$

$$= 0.970$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1568.8, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.738 - 1.5$$

$$= 7.238 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 390.35/33.425$$

$$= 11.678 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 387.35/2 + 1.5$$

$$= 195.175 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, L_r is $\min[LR1, LR2, LR3]$:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(195.175 \cdot 33.425) + 50.0}$$

$$= 130.770 \text{ mm}$$



Limit [LR2]:

$$= \sqrt{((R_{eff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((195.175 + 33.425) \cdot (33.425 + 12.0))}$$

$$= 101.903 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 22.925$$

$$= 45.850 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(130.77, 101.903, 45.85)$$

$$= 45.850 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 53.087 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 33.425, 12.0] + \sqrt{(22.925 \cdot 7.238)}$$

$$= 24.881 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(33.425 + 12.0)$$

$$= 363.400 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3]$$

$$= \min[24.881, 200.0, 363.4]$$

$$= 24.881 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / (LR - t_n))$$

$$= 33.425 + ((5.502 \cdot 1.0) / (53.087 - 7.238))$$

$$= 45.425 \text{ mm}$$

Parameter [λ]:

$$= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})))$$

$$= \min(12, (2 \cdot 22.92 + 7.238) / (\sqrt{(390.35 + 45.425) \cdot 45.425}))$$

$$= 0.377$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 33.425 \cdot 53.087 \cdot \max((0.377/5)^{0.85}, 1)$$

$$= 17.744 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 7.238 \cdot 24.881$$

$$= 1.801 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}_{42}^2 - \text{Area Cut by Reinforcement Limit}$$

$$= 0.5 \cdot 12.0^2 - 0.397$$

$$= 0.323 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}_{41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$



Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e , (LR - t_n) \cdot t_e) \\ &= \min(50.0 \cdot 12.0, (53.087 - 7.238) \cdot 12.0) \\ &= 5.502 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 17.744 + 0.97(1.801 + 0.0) + 0.5 + 0.323 + 0.0 + 1.0(5.502) \\ &= 25.817 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 7.238 / \ln[1 + 7.238/22.925] \\ &= 26.378 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 45.425 / \ln[1 + 45.425/195.175] \\ &= 217.096 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1568.835 \cdot 1.0 \\ &= 2353.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(LH + t) \\ &= 163.313 \cdot 26.378(24.881 + 33.425) \\ &= 2511.8 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 163.313 \cdot 217.096 \cdot 53.087 \\ &= 18821.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 163.313 \cdot 217.096 \cdot 22.925 \\ &= 8127.9 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (18821.9 + 8127.9 + 2511.8)/163.313 \\ &= 180.4 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p/AT - R_{xs}/t_{eff}) \\ &= 2353.252 / (2 \cdot 180.4/25.817 - 217.096/45.425) \\ &= 255.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 1568.835[33.425/217.096] \\ &= 241.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(255.891, 241.544) \\ &= 241.544 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (2511.789 + 18821.928 + 8127.933)/25.817 \\ &= 1141.194 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 163.313 \cdot 217.096/45.425 \\ &= 780.5 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1141.194 - 780.51, 780.51) \\ &= 1501.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2353.25	kgf/cm ²
Local Primary Membrane Stress	PL	1501.88	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	241.54	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (22.925 + 7.238) \\ &= 47.379 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (22.925 + 7.238 + 50.0) \\ &= 125.919 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 7.238, 6.000) \text{ Passed} \\ &tf1, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 33.425) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(8128 \cdot 1.0, 1.5 \cdot 1522.2(1.801 + 0.0), \pi/4 \cdot 163.3 \cdot 22.92^2 \cdot 1.0^2) \\ &= 674.102 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 674.1 \cdot 33.42 \cdot 1569 / (33.425 \cdot 1569 + 12.0 \cdot 1650) \\ &= 489.302 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 674.1 \cdot 12.0 \cdot 1650 / (33.425 \cdot 1569 + 12.0 \cdot 1650) \\ &= 184.799 \text{ kgf} \end{aligned}$$

Shear Stress [τ1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 489.302 / (47.379 \cdot (0.6 \cdot 8.738 + 0.49 \cdot 0.0)) \\ &= 196.991 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 184.799 / (47.379 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071)) \\ &= 36.573 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ3]:

$$\begin{aligned} &= fwp / (LTP \cdot (0.49 \cdot L42T)) \\ &= 184.799 / (125.919 \cdot (0.49 \cdot 8.485)) \\ &= 35.298 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(196.991, 36.573, 35.298) \\ &= 197.0 \text{ must be less than or equal to } 1568.8 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A



This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-52 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	53 °C
The computed stress ratio is	:	0.390

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$
$$= (2.398 \cdot 1.0) / (7.645 - 1.5 - 0.0)$$
$$= 0.390$$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-52 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -38 °C

The Drop for this Nozzle is : 2.3631 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 237.2881 mm

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Input, Nozzle Desc: Noz N8

From: 50

Pressure for Reinforcement Calculations	P	23.574	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		241.92	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

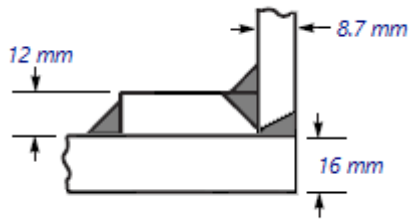
Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N8

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 45.85 \cdot (\exp(23.574/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.3578 + 1.5 + 0.0 = 1.8578 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 7.238 + 45.85) / 45.85) - 0.0$$

$$= 417.66 - 0.0 = 417.6602 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm
 Min. required shell thickness, internal p + c : 10.153 mm

Nozzle neck minimum required thickness : 4.920 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 4.920 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.4, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.738 - 1.5$$

$$= 7.238 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1203.0/14.5$$

$$= 82.966 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1200.0/2 + 1.5$$

$$= 601.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(601.5 \cdot 14.5) + 50.0}$$

$$= 143.390 \text{ mm}$$



Limit [LR2]:

$$\begin{aligned} &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\ &= \sqrt{((601.5 + 14.5) \cdot (14.5 + 12.0))} \\ &= 127.765 \text{ mm} \end{aligned}$$

Limit [LR3]:

$$\begin{aligned} &= 2 \cdot R_n \\ &= 2 \cdot 22.925 \\ &= 45.850 \text{ mm} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(143.39, 127.765, 45.85) \\ &= 45.850 \text{ mm} \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 53.087 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 14.5, 12.0] + \sqrt{(22.925 \cdot 7.238)} \\ &= 24.881 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 150.0 \\ &= 150.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(14.5 + 12.0) \\ &= 212.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[24.881, 150.0, 212.0] \\ &= 24.881 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 14.5 + ((5.502 \cdot 1) / (53.087 - 7.238)) \\ &= 26.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))) \\ &= \min(12, (2 \cdot 22.92 + 7.238) / (\sqrt{((1203.0 + 26.5) \cdot 26.5)})) \\ &= 0.294 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 14.5 \cdot 53.087 \cdot \max((0.294/5)^{0.85}, 1) \\ &= 7.698 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 7.238 \cdot 24.881 \\ &= 1.801 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}42^2 - \text{Area Cut by Reinforcement Limit} \\ &= 0.5 \cdot 12.0^2 - 0.397 \\ &= 0.323 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$



Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e , (LR - t_n) \cdot t_e) \\ &= \min(50.0 \cdot 12.0, (53.087 - 7.238) \cdot 12.0) \\ &= 5.502 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 7.698 + 0.922(1.801 + 0.0) + 0.5 + 0.323 + 0.0 + 1 \cdot (5.502) \\ &= 15.683 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 7.238 / \ln[1 + 7.238/22.925] \\ &= 26.378 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 26.5 / \ln[1 + 26.5/601.5] \\ &= 614.654 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.408 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(LH + t) \\ &= 23.574 \cdot 26.378(24.881 + 14.5) \\ &= 244.9 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.574 \cdot 614.654 \cdot 53.087 \\ &= 7692.3 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 23.574 \cdot 614.654 \cdot 22.925 \\ &= 3321.8 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (7692.3 + 3321.8 + 244.9)/23.574 \\ &= 477.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p/AT - R_{xs}/t_{eff}) \\ &= 2475.612 / (2 \cdot 477.601/15.683 - 614.654/26.5) \\ &= 65.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 1650.408[14.5/614.654] \\ &= 38.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(65.647, 38.934) \\ &= 38.934 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (244.888 + 7692.265 + 3321.775)/15.683 \\ &= 717.890 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 23.574 \cdot 614.654/26.5 \\ &= 546.8 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 717.89 - 546.787, 546.787)$$

$$= 889.0 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	888.99	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	38.93	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (22.925 + 7.238)$$

$$= 47.379 \text{ mm}$$

Weld Length of Pad to Shell Weld [LTp]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (22.925 + 7.238 + 50.0)$$

$$= 125.919 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 7.238, 6.000) \text{ Passed}$$

$$t_f1, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 14.500) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(3322 \cdot 1.0, 1.5 \cdot 1522.2(1.801 + 0.0), \pi/4 \cdot 23.6 \cdot 22.92^2 \cdot 1.0^2)$$

$$= 97.305 \text{ kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 97.3 \cdot 14.5 \cdot 1650 / (14.5 \cdot 1650 + 12.0 \cdot 1650)$$

$$= 53.243 \text{ kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 97.3 \cdot 12.0 \cdot 1650 / (14.5 \cdot 1650 + 12.0 \cdot 1650)$$

$$= 44.063 \text{ kgf}$$

Shear Stress [τ1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 53.243 / (47.379 \cdot (0.6 \cdot 8.738 + 0.49 \cdot 0.0))$$

$$= 21.435 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 44.063 / (47.379 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071))$$

$$= 8.720 \text{ kgf/cm}^2$$

Shear Stress [τ3]:

$$= fwp / (L_{Tp} \cdot (0.49 \cdot L_{42T}))$$

$$= 44.063 / (125.919 \cdot (0.49 \cdot 8.485))$$

$$= 8.416 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(21.435, 8.72, 8.416)$$

$$= 21.4 \text{ must be less than or equal to } 1650.4 \text{ kgf/cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :



Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-8 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.12	:	10 °C
The computed stress ratio is	:	0.762

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (8.0 \cdot 1.0) / (12.0 - 1.5 - 0.0) \\ &= 0.762 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.054

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.332 \cdot 1.0) / (7.645 - 1.5 - 0.0) \\ &= 0.054 \end{aligned}$$

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	-8 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-8 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	:	-85 °C
---------------------------------	---	--------

The Drop for this Nozzle is : 0.7586 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 166.7586 mm

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Input, Nozzle Desc: Noz N4

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		507.92	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.2748	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	373.0500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

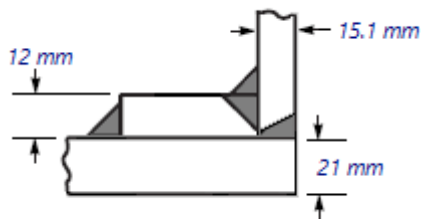
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		9.562	in.
Actual Thickness Used in Calculation		0.594	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 248.875 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 1.9322 + 3.0 + 0.0 = 4.9322 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 12.088 + 248.875) / 248.875) - 0.0$$

$$= 141.117 - 0.0 = 141.1172 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.110 mm
Min. required shell thickness, internal p + c :	17.981 mm
Min. required shell thickness, external p + c :	13.840 mm

Nozzle neck minimum required thickness	: 11.110 mm
--	-------------

Minimum nozzle neck thickness 13.208 mm , is greater than required thickness 11.110 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	129.4,	Allowable :	1522.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3825.4 kgf/cm ²	Passed
Occasional :	137.4,	Allowable :	2024.6 kgf/cm ²	Passed
Shear :	36.3,	Allowable :	1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.4, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 15.088 - 3.0$$

$$= 12.088 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:



$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 2300.0/2 + 1.5 \\
 &= 1151.500 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned}
 \text{Limit [LR1]:} \\
 &= \sqrt{(R_{eff} \cdot t) + W} \\
 &= \sqrt{(1151.5 \cdot 19.5) + 50.0} \\
 &= 199.847 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR2]:} \\
 &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\
 &= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))} \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR3]:} \\
 &= 2 \cdot R_n \\
 &= 2 \cdot 124.437 \\
 &= 248.875 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Final value of [LR]:} \\
 &= \min(LR1, LR2, LR3) \\
 &= \min(199.847, 192.059, 248.875) \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Because the nozzle is set-on, add } t_n \text{ to the result [LR]:} \\
 &= 204.146 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(124.437 \cdot 12.088)} \\
 &= 50.783 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH2]:} \\
 &= L_{pr1} \\
 &= 230.0 \\
 &= 230.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + t_e) \\
 &= 8(19.5 + 12.0) \\
 &= 252.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] \\
 &= \min[50.783, 230.0, 252.0] \\
 &= 50.783 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A5 \cdot f_{rp}) / (LR - t_n)) \\
 &= 19.5 + ((6.0 \cdot 1.) / (204.146 - 12.088)) \\
 &= 22.624 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Parameter } [\lambda]: \\
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(Di + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 124.44 + 12.088) / (\sqrt{(2303.0 + 22.624) \cdot 22.624})) \\
 &= 1.138
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 \text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 19.5 \cdot 204.146 \cdot \max((1.138/5)^{0.85}, 1) \\
 &= 39.809 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= t_n \cdot LH
 \end{aligned}$$



$$= 12.088 \cdot 50.783$$
$$= 6.138 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}42^2$$
$$= 0.5 \cdot 12.0^2$$
$$= 0.720 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}41^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e , (LR - t_n) \cdot t_e)$$
$$= \min(50.0 \cdot 12.0, (204.146 - 12.088) \cdot 12.0)$$
$$= 6.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$
$$= 39.809 + 0.922(6.138 + 0.0) + 0.5 + 0.72 + 0.0 + 1 \cdot (6.0)$$
$$= 52.690 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 12.088 / \ln[1 + 12.088/124.437]$$
$$= 130.388 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 22.624 / \ln[1 + 22.624/1151.5]$$
$$= 1162.774 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn}(LH + t)$$
$$= 23.454 \cdot 130.388(50.783 + 19.5)$$
$$= 2149.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.454 \cdot 1162.774 \cdot 204.146$$
$$= 55674.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 23.454 \cdot 1162.774 \cdot 124.437$$
$$= 33936.2 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (55674.1 + 33936.2 + 2149.3)/23.454$$
$$= 3912.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff})$$
$$= 2475.612 / (2 \cdot 3912.324/52.69 - 1162.774/22.624)$$
$$= 25.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$
$$= 1650.408[19.5/1162.774]$$
$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(25.494, 27.678)$$



$$= 25.494 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (2149.345 + 55674.094 + 33936.168) / 52.69 \\ &= 1741.495 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 23.454 \cdot 1162.774 / 22.624 \\ &= 1205.4 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1741.495 - 1205.431, 1205.431) \\ &= 2277.6 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2277.56	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	25.49	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (124.437 + 12.088) \\ &= 214.453 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (124.437 + 12.088 + 50.0) \\ &= 292.993 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 12.088, 6.000) \text{ Passed} \\ &tf1, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 19.500) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(33936 \cdot 1.0, 1.5 \cdot 1522.2(6.138 + 0.0), \pi/4 \cdot 23.5 \cdot 124.44^2 \cdot 1.0^2) \\ &= 2852.387 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 2852.4 \cdot 19.5 \cdot 1650 / (19.5 \cdot 1650 + 12.0 \cdot 1650) \\ &= 1765.765 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 2852.4 \cdot 12.0 \cdot 1650 / (19.5 \cdot 1650 + 12.0 \cdot 1650) \\ &= 1086.622 \text{ kgf} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 1765.765 / (214.453 \cdot (0.6 \cdot 14.275 + 0.49 \cdot 0.0)) \\ &= 96.135 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 1086.622 / (214.453 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071)) \\ &= 47.511 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:



$$\begin{aligned} &= fwp / (LTP \cdot (0.49 \cdot L42T)) \\ &= 1086.622 / (292.993 \cdot (0.49 \cdot 8.485)) \\ &= 89.200 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(96.135, 47.511, 89.2) \\ &= 96.1 \text{ must be less than or equal to } 1650.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 12.088 / \ln[1 + 12.088/124.437] \\ &= 130.388 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 22.624 / \ln[1 + 22.624/1151.5] \\ &= 1162.774 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (L_H + t) \\ &= 1.055 \cdot 130.388 (50.783 + 19.5) \\ &= 96.6 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1162.774 \cdot 204.146 \\ &= 2503.4 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1162.774 \cdot 124.437 \\ &= 1525.9 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (2503.4 + 1525.9 + 96.6) / 1.055 \\ &= 3912.3 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 3912.324 / 52.69 - 1162.774 / 22.624) \\ &= 25.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1650.408 [19.5 / 1162.774] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(25.494, 27.678) \\ &= 25.494 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (96.645 + 2503.374 + 1525.932) / 52.69 \\ &= 78.306 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 1162.774 / 22.624 \\ &= 54.2 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 78.306 - 54.202, 54.202) \\ &= 102.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	102.41	kgf/cm ²
Maximum Allowable External Pressure	Pmax	25.49	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.12	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (15.235 \cdot 1.0) / (12.0 - 1.5 - 0.0) \\ &= 1.000 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is -104 °C
The computed stress ratio is 0.176

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (1.791 \cdot 1.0) / (13.202 - 3.0 - 0.0) \\ &= 0.176 \end{aligned}$$

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	1 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	1 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -85 °C



The Drop for this Nozzle is : 8.1326 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 259.1326 mm

Input Echo, WRC297 Item 1, Description: Noz N4 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2300.000	mm	
Shell Thickness	21.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.41	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	242.875	mm	
Nozzle Thickness	15.0876	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-333 6		
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	373.050	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	23.45	kgf/cm ²
Radial Load	P	-683.24	kgf
Circumferential Shear	Vc	1019.77	kgf
Longitudinal Shear	Vl	1019.77	kgf
Circumferential Moment	Mc	0.76	kgf-m.
Longitudinal Moment	Ml	1.02	kgf-m.
Torsional Moment	Mt	1.22	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$
 $= 770.917 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	31.500	mm
Nozzle Thickness used	(t) =	12.088	mm
T / t	=	2.606	
d / t	=	22.589	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.007$$

Nr/P	=	0.096
Mr/P	=	0.115
M0/P	=	0.091
N0/P	=	0.198
MrD/Mc	=	0.191



NrDL/Mc = 0.138
 MØd/Mc = 0.177
 NØDL/Mc = 0.106
 MrD/Ml = 0.152
 NrDL/Ml = 0.116
 MØD/Ml = 0.135
 NØDL/Ml = 0.241

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	845	869	845	869
Outplane Stress Summary	896	845	897	845

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	422	422	422	422
Inplane Stress Summary	476	382	476	381

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	896	845	897	845

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	422	422	422	422
Outplane Stress Summary	473	399	474	398



Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	845	869	845	869
Inplane Stress Summary	899	828	899	828

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	899	828	899	828

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	276	304	276	304
Hoop Stress Summary	289	317	290	318

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	186	-186	186	-186
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	138	138	138	138
Axial Stress Summary	330	-40	333	-42

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	19	19	-19	-19
Shear due to (Vl)	0	0	0	0



Shear due to Torsion	0	0	0	0
----------------------	---	---	---	---

Shear Stress Summary	19	19	-19	-19
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	338	360	340	363
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	276	304	276	304

Hoop Stress Summary	289	318	289	318
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	186	-186	186	-186
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	138	138	138	138

Axial Stress Summary	330	-40	333	-42
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-19	-19	19	19
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-19	-19	19	19
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	338	360	340	363
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1161.25(21.0 - 1.5))}$$

$$= 769.636 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2322.500 mm
Nozzle Outside Diameter	(d) =	373.050 mm
Vessel Thickness used	(T) =	19.500 mm
Nozzle Thickness used	(t) =	62.088 mm



T / t = 0.314
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.753$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.075
 Mr/P = 0.158
 M0/P = 0.046
 N0/P = 0.093
 MrD/Mc = 0.292
 NrDL/Mc = 0.092
 M0d/Mc = 0.091
 N0DL/Mc = 0.069
 MrD/Ml = 0.125
 NrDL/Ml = 0.084
 M0D/Ml = 0.039
 N0DL/Ml = 0.115

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1373	1396	1373	1396
Outplane Stress Summary	1439	1363	1439	1363

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	170	-170	170	-170
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	686	686	686	686
Inplane Stress Summary	869	530	870	529

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	8	8	-8	-8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1439	1363	1439	1363

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	686	686	686	686
Outplane Stress Summary	752	653	753	653

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	170	-170	170	-170
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1373	1396	1373	1396
Inplane Stress Summary	1556	1240	1558	1238

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-8	-8	8	8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1556	1240	1558	1239

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	845	869	845	869	845	869	845	869
Circ. Pl (SUS)	13	13	13	13	6	6	6	6
Circ. Q (SUS)	37	-37	37	-37	47	-47	47	-47
Long. Pm (SUS)	422	422	422	422	422	422	422	422
Long. Pl (SUS)	6	6	6	6	13	13	13	13
Long. Q (SUS)	47	-47	47	-47	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	845.0	869.0	845.0	869.0	845.0	869.0	845.0	869.0
Pm+Pl (SUS)	858.1	882.1	858.1	882.1	851.1	875.1	851.1	875.1

Pm+Pl+Q (Total)	895.1	845.1	895.1	845.1	898.1	828.1	898.1	828.1
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Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	869.00	1650.41	Passed
Pm+Pl (SUS)	882.11	2475.61	Passed
Pm+Pl+Q (TOTAL)	898.11	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Circ. Pl (SUS)	16	16	16	16	13	13	13	13
Circ. Q (SUS)	49	-49	49	-49	170	-170	170	-170
Long. Pm (SUS)	686	686	686	686	686	686	686	686
Long. Pl (SUS)	13	13	13	13	16	16	16	16
Long. Q (SUS)	170	-170	170	-170	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Pm+Pl (SUS)	1389	1412	1389	1412	1386	1409	1386	1409
Pm+Pl+Q (Total)	1438	1363	1438	1363	1556	1239	1556	1239

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1396.00	1650.41	Passed
Pm+Pl (SUS)	1412.09	2475.61	Passed
Pm+Pl+Q (TOTAL)	1556.08	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	276	304	276	304	276	304	276	304
Circ. Pl (SUS)	13	13	13	13	13	13	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	138	138	138	138	138	138	138	138
Long. Pl (SUS)	6	6	6	6	6	6	6	6
Long. Q (SUS)	185	-185	187	-187	185	-185	187	-187
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19



Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	276.0	304.0	276.0	304.0	276.0	304.0	276.0	304.0
Pm+Pl (SUS)	291.4	319.1	291.4	319.1	291.4	319.1	291.4	319.1
Pm+Pl+Q (Total)	336.6	360.0	338.3	362.0	336.6	360.0	338.3	362.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	304.00	1522.22	Passed
Pm+Pl (SUS)	319.06	2283.33	Passed
Pm+Pl+Q (TOTAL)	362.00	4566.66	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: Noz N5

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		767.92	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.2748	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	473.0500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

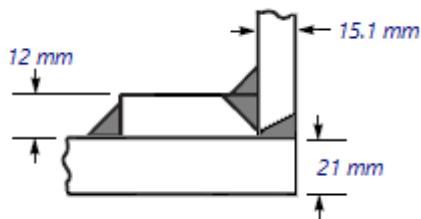
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		9.562	in.
Actual Thickness Used in Calculation		0.594	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 248.875 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 1.9322 + 3.0 + 0.0 = 4.9322 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 12.088 + 248.875) / 248.875) - 0.0$$

$$= 141.117 - 0.0 = 141.1172 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.110 mm
Min. required shell thickness, internal p + c :	17.981 mm
Min. required shell thickness, external p + c :	13.840 mm

Nozzle neck minimum required thickness : 11.110 mm

Minimum nozzle neck thickness 13.208 mm , is greater than required thickness 11.110 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	129.4,	Allowable :	1522.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3825.4 kgf/cm ²	Passed
Occasional :	137.4,	Allowable :	2024.6 kgf/cm ²	Passed
Shear :	36.3,	Allowable :	1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.4, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 15.088 - 3.0$$

$$= 12.088 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:



$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 2300.0/2 + 1.5 \\
 &= 1151.500 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned}
 \text{Limit [LR1]:} \\
 &= \sqrt{(R_{eff} \cdot t) + W} \\
 &= \sqrt{(1151.5 \cdot 19.5) + 100.0} \\
 &= 249.847 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR2]:} \\
 &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\
 &= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))} \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR3]:} \\
 &= 2 \cdot R_n \\
 &= 2 \cdot 124.437 \\
 &= 248.875 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Final value of [LR]:} \\
 &= \min(LR1, LR2, LR3) \\
 &= \min(249.847, 192.059, 248.875) \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Because the nozzle is set-on, add } t_n \text{ to the result [LR]:} \\
 &= 204.146 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(124.437 \cdot 12.088)} \\
 &= 50.783 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH2]:} \\
 &= L_{pr1} \\
 &= 230.0 \\
 &= 230.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + t_e) \\
 &= 8(19.5 + 12.0) \\
 &= 252.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] \\
 &= \min[50.783, 230.0, 252.0] \\
 &= 50.783 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A5 \cdot f_{rp}) / (LR - t_n)) \\
 &= 19.5 + ((12.0 \cdot 1) / (204.146 - 12.088)) \\
 &= 25.748 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Parameter } [\lambda]: \\
 &= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(Di + t_{eff}) \cdot t_{eff}}))) \\
 &= \min(12, (2 \cdot 124.44 + 12.088) / (\sqrt{(2303.0 + 25.748) \cdot 25.748})) \\
 &= 1.066
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 \text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 19.5 \cdot 204.146 \cdot \max((1.066/5)^{0.85}, 1) \\
 &= 39.809 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= t_n \cdot LH
 \end{aligned}$$



$$= 12.088 \cdot 50.783$$
$$= 6.138 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}42^2$$
$$= 0.5 \cdot 12.0^2$$
$$= 0.720 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}41^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, (LR - t_n) \cdot t_e)$$
$$= \min(100.0 \cdot 12.0, (204.146 - 12.088) \cdot 12.0)$$
$$= 12.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$
$$= 39.809 + 0.922(6.138 + 0.0) + 0.5 + 0.72 + 0.0 + 1.(12.0)$$
$$= 58.690 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 12.088 / \ln[1 + 12.088/124.437]$$
$$= 130.388 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 25.748 / \ln[1 + 25.748/1151.5]$$
$$= 1164.328 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn}(LH + t)$$
$$= 23.454 \cdot 130.388(50.783 + 19.5)$$
$$= 2149.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.454 \cdot 1164.328 \cdot 204.146$$
$$= 55748.5 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 23.454 \cdot 1164.328 \cdot 124.437$$
$$= 33981.5 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (55748.5 + 33981.5 + 2149.3)/23.454$$
$$= 3917.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff})$$
$$= 2475.612 / (2 \cdot 3917.431/58.69 - 1164.328/25.748)$$
$$= 28.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$
$$= 1650.408[19.5/1164.328]$$
$$= 27.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(28.044, 27.641)$$



$$= 27.641 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (2149.345 + 55748.508 + 33981.527) / 58.69 \\ &= 1565.500 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 23.454 \cdot 1164.328 / 25.748 \\ &= 1060.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1565.5 - 1060.591, 1060.591) \\ &= 2070.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2070.41	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.64	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (124.437 + 12.088) \\ &= 214.453 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (124.437 + 12.088 + 100.0) \\ &= 371.533 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 12.088, 6.000) \text{ Passed} \\ &tf1, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 19.500) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(33982 \cdot 1.0, 1.5 \cdot 1522.2(6.138 + 0.0), \pi/4 \cdot 23.5 \cdot 124.44^2 \cdot 1.0^2) \\ &= 2852.387 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 2852.4 \cdot 19.5 \cdot 1650 / (19.5 \cdot 1650 + 12.0 \cdot 1650) \\ &= 1765.765 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 2852.4 \cdot 12.0 \cdot 1650 / (19.5 \cdot 1650 + 12.0 \cdot 1650) \\ &= 1086.622 \text{ kgf} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 1765.765 / (214.453 \cdot (0.6 \cdot 14.275 + 0.49 \cdot 0.0)) \\ &= 96.135 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 1086.622 / (214.453 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071)) \\ &= 47.511 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:



$$\begin{aligned} &= fwp / (LTP \cdot (0.49 \cdot L42T)) \\ &= 1086.622 / (371.533 \cdot (0.49 \cdot 8.485)) \\ &= 70.344 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(96.135, 47.511, 70.344) \\ &= 96.1 \text{ must be less than or equal to } 1650.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 12.088 / \ln[1 + 12.088/124.437] \\ &= 130.388 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 25.748 / \ln[1 + 25.748/1151.5] \\ &= 1164.328 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (L_H + t) \\ &= 1.055 \cdot 130.388 (50.783 + 19.5) \\ &= 96.6 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1164.328 \cdot 204.146 \\ &= 2506.7 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1164.328 \cdot 124.437 \\ &= 1528.0 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (2506.7 + 1528.0 + 96.6) / 1.055 \\ &= 3917.4 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 3917.431 / 58.69 - 1164.328 / 25.748) \\ &= 28.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1650.408 [19.5 / 1164.328] \\ &= 27.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(28.044, 27.641) \\ &= 27.641 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (96.645 + 2506.72 + 1527.972) / 58.69 \\ &= 70.392 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 1164.328 / 25.748 \\ &= 47.7 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 70.392 - 47.689, 47.689) \\ &= 93.1 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	93.10	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.64	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.12	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (15.235 \cdot 1.0) / (12.0 - 1.5 - 0.0) \\ &= 1.000 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is -104 °C
The computed stress ratio is 0.176

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (1.791 \cdot 1.0) / (13.202 - 3.0 - 0.0) \\ &= 0.176 \end{aligned}$$

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	1 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	1 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -85 °C



The Drop for this Nozzle is : 8.1326 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 259.1326 mm

Input Echo, WRC297 Item 1, Description: Noz N5 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2300.000	mm	
Shell Thickness	21.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.41	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	242.875	mm	
Nozzle Thickness	15.0876	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-333 6		
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	473.050	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	23.45	kgf/cm ²
Radial Load	P	-683.24	kgf
Circumferential Shear	Vc	1019.77	kgf
Longitudinal Shear	Vl	1019.77	kgf
Circumferential Moment	Mc	0.76	kgf-m.
Longitudinal Moment	Ml	1.02	kgf-m.
Torsional Moment	Mt	1.22	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$
 $= 770.917 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	31.500	mm
Nozzle Thickness used	(t) =	12.088	mm
T / t	=	2.606	
d / t	=	22.589	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.007$$

Nr/P	=	0.096
Mr/P	=	0.115
M0/P	=	0.091
N0/P	=	0.198
MrD/Mc	=	0.191



NrDL/Mc = 0.138
 MØd/Mc = 0.177
 NØDL/Mc = 0.106
 MrD/Ml = 0.152
 NrDL/Ml = 0.116
 MØD/Ml = 0.135
 NØDL/Ml = 0.241

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	845	869	845	869
Outplane Stress Summary	896	845	897	845

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	422	422	422	422
Inplane Stress Summary	476	382	476	381

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	896	845	897	845

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	422	422	422	422
Outplane Stress Summary	473	399	474	398



Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	845	869	845	869
Inplane Stress Summary	899	828	899	828

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	899	828	899	828

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	276	304	276	304
Hoop Stress Summary	289	317	290	318

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	186	-186	186	-186
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	138	138	138	138
Axial Stress Summary	330	-40	333	-42

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	19	19	-19	-19
Shear due to (Vl)	0	0	0	0



Shear due to Torsion	0	0	0	0
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Shear Stress Summary	19	19	-19	-19
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	338	360	340	363
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	276	304	276	304

Hoop Stress Summary	289	318	289	318
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	186	-186	186	-186
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	138	138	138	138

Axial Stress Summary	330	-40	333	-42
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-19	-19	19	19
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-19	-19	19	19
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	338	360	340	363
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1161.25(21.0 - 1.5))}$$

$$= 769.636 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2322.500 mm
Nozzle Outside Diameter	(d) =	473.050 mm
Vessel Thickness used	(T) =	19.500 mm
Nozzle Thickness used	(t) =	112.088 mm



T / t = 0.174
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.223$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.070
 Mr/P = 0.135
 M0/P = 0.040
 N0/P = 0.068
 MrD/Mc = 0.284
 NrDL/Mc = 0.096
 M0d/Mc = 0.087
 N0DL/Mc = 0.076
 MrD/Ml = 0.103
 NrDL/Ml = 0.088
 M0D/Ml = 0.031
 N0DL/Ml = 0.089

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	43	-43	43	-43
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1373	1396	1373	1396
Outplane Stress Summary	1428	1365	1428	1365

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	145	-145	145	-145
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	686	686	686	686
Inplane Stress Summary	844	554	844	553

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1428	1365	1428	1365

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	43	-43	43	-43
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	686	686	686	686
Outplane Stress Summary	741	655	742	655

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	145	-145	145	-145
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1373	1396	1373	1396
Inplane Stress Summary	1530	1264	1531	1263

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1530	1264	1531	1263

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	845	869	845	869	845	869	845	869
Circ. Pl (SUS)	13	13	13	13	6	6	6	6
Circ. Q (SUS)	37	-37	37	-37	47	-47	47	-47
Long. Pm (SUS)	422	422	422	422	422	422	422	422
Long. Pl (SUS)	6	6	6	6	13	13	13	13
Long. Q (SUS)	47	-47	47	-47	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	845.0	869.0	845.0	869.0	845.0	869.0	845.0	869.0
Pm+Pl (SUS)	858.1	882.1	858.1	882.1	851.1	875.1	851.1	875.1

Pm+Pl+Q (Total)	895.1	845.1	895.1	845.1	898.1	828.1	898.1	828.1
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Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	869.00	1650.41	Passed
Pm+Pl (SUS)	882.11	2475.61	Passed
Pm+Pl+Q (TOTAL)	898.11	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Circ. Pl (SUS)	12	12	12	12	12	12	12	12
Circ. Q (SUS)	43	-43	43	-43	145	-145	145	-145
Long. Pm (SUS)	686	686	686	686	686	686	686	686
Long. Pl (SUS)	12	12	12	12	12	12	12	12
Long. Q (SUS)	145	-145	145	-145	43	-43	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Pm+Pl (SUS)	1385	1408	1385	1408	1385	1408	1385	1408
Pm+Pl+Q (Total)	1428	1365	1428	1365	1530	1263	1530	1263

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1396.00	1650.41	Passed
Pm+Pl (SUS)	1408.07	2475.61	Passed
Pm+Pl+Q (TOTAL)	1530.06	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	276	304	276	304	276	304	276	304
Circ. Pl (SUS)	13	13	13	13	13	13	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	138	138	138	138	138	138	138	138
Long. Pl (SUS)	6	6	6	6	6	6	6	6
Long. Q (SUS)	185	-185	187	-187	185	-185	187	-187
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19



Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	276.0	304.0	276.0	304.0	276.0	304.0	276.0	304.0
Pm+Pl (SUS)	291.4	319.1	291.4	319.1	291.4	319.1	291.4	319.1
Pm+Pl+Q (Total)	336.6	360.0	338.3	362.0	336.6	360.0	338.3	362.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	304.00	1522.22	Passed
Pm+Pl (SUS)	319.06	2283.33	Passed
Pm+Pl+Q (TOTAL)	362.00	4566.66	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: Noz N3

From: 70

Pressure for Reinforcement Calculations	P	23.510	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
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Distance from Bottom/Left Tangent		657.92	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.82	deg
Diameter		10.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.2748	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	473.0500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		100.0000	mm

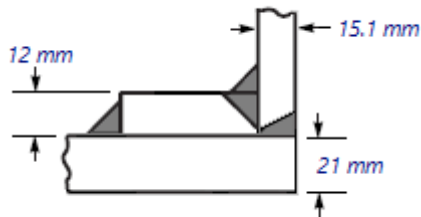
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		9.562	in.
Actual Thickness Used in Calculation		0.594	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 248.875 \cdot (\exp(23.51/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 1.9368 + 3.0 + 0.0 = 4.9368 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 12.088 + 248.875) / 248.875) - 0.0$$

$$= 141.117 - 0.0 = 141.1172 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.110 mm
Min. required shell thickness, internal p + c :	18.021 mm
Min. required shell thickness, external p + c :	13.840 mm

Nozzle neck minimum required thickness	: 11.110 mm
--	-------------

Minimum nozzle neck thickness 13.208 mm , is greater than required thickness 11.110 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	129.8,	Allowable :	1522.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3825.1 kgf/cm ²	Passed
Occasional :	137.7,	Allowable :	2024.6 kgf/cm ²	Passed
Shear :	36.3,	Allowable :	1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.4, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 15.088 - 3.0$$

$$= 12.088 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:



$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 2300.0/2 + 1.5 \\
 &= 1151.500 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned}
 \text{Limit [LR1]:} \\
 &= \sqrt{(R_{eff} \cdot t) + W} \\
 &= \sqrt{(1151.5 \cdot 19.5) + 100.0} \\
 &= 249.847 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR2]:} \\
 &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\
 &= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))} \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR3]:} \\
 &= 2 \cdot R_n \\
 &= 2 \cdot 124.437 \\
 &= 248.875 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Final value of [LR]:} \\
 &= \min(LR1, LR2, LR3) \\
 &= \min(249.847, 192.059, 248.875) \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Because the nozzle is set-on, add } t_n \text{ to the result [LR]:} \\
 &= 204.146 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(124.437 \cdot 12.088)} \\
 &= 50.783 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH2]:} \\
 &= L_{pr1} \\
 &= 230.0 \\
 &= 230.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + t_e) \\
 &= 8(19.5 + 12.0) \\
 &= 252.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] \\
 &= \min[50.783, 230.0, 252.0] \\
 &= 50.783 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A5 \cdot f_{rp}) / (LR - t_n)) \\
 &= 19.5 + ((12.0 \cdot 1) / (204.146 - 12.088)) \\
 &= 25.748 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Parameter } [\lambda]: \\
 &= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))) \\
 &= \min(12, ((2 \cdot 124.44 + 12.088) / (\sqrt{(2303.0 + 25.748) \cdot 25.748}))) \\
 &= 1.066
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 \text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 19.5 \cdot 204.146 \cdot \max((1.066/5)^{0.85}, 1) \\
 &= 39.809 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= t_n \cdot LH
 \end{aligned}$$



$$= 12.088 \cdot 50.783$$
$$= 6.138 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg42}^2$$
$$= 0.5 \cdot 12.0^2$$
$$= 0.720 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, (LR - t_n) \cdot t_e)$$
$$= \min(100.0 \cdot 12.0, (204.146 - 12.088) \cdot 12.0)$$
$$= 12.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$
$$= 39.809 + 0.922(6.138 + 0.0) + 0.5 + 0.72 + 0.0 + 1.(12.0)$$
$$= 58.690 \text{ cm}^2$$

Angle [Theta1]:

$$= \cos^{-1}[\text{Dx}/\text{Reff}]$$
$$= \cos^{-1}[600.0/1151.5]$$
$$= 1.023$$

Angle [Theta2]:

$$= \cos^{-1}[(\text{Dx} + \text{Rn})/\text{Reff}]$$
$$= \cos^{-1}[(600.0 + 124.437)/1151.5]$$
$$= 0.890$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/\text{Rn}]$$
$$= 12.088 / \ln[1 + 12.088/124.437]$$
$$= 130.388 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{\text{eff}} / \ln[1 + t_{\text{eff}}/\text{Reff}]$$
$$= 25.748 / \ln[1 + 25.748/1151.5]$$
$$= 1164.328 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot \text{Rxn} (LH + t / \sin[\theta]) \text{ (eqn. 4.5.73)}$$
$$= 23.51 \cdot 130.388 (50.783 + 19.5/\sin[90.0])$$
$$= 2154.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.51 \cdot 1164.328 \cdot 204.146$$
$$= 55881.7 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot \text{Rxs} \cdot \text{Rnc}$$
$$= 23.51 \cdot 1164.328 \cdot 124.437$$
$$= 34062.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (55881.7 + 34062.7 + 2154.5)/23.51$$
$$= 3917.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/t_{\text{eff}})$$
$$= 2475.612 / (2 \cdot 3917.431/58.69 - 1164.328/25.748)$$



= 28.0 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= S[t/Rxs]
= 1650.408[19.5/1164.328]
= 27.6 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
= min(28.044, 27.641)
= 27.641 kgf/cm²

Average Primary Membrane Stress [σ Avg]:

= (fN + fS + fY) / AT
= (2154.482 + 55881.742 + 34062.738)/58.69
= 1569.242 kgf/cm²

General Primary Membrane Stress [σ Circ]:

= P • Rxs / teff
= 23.51 • 1164.328/25.748
= 1063.1 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σ Avg - σ Circ, σ Circ)
= max(2 • 1569.242 - 1063.125, 1063.125)
= 2075.4 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allow Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2075.36	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.64	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= 1.000 For set-on Nozzles

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

= $\pi/2 \cdot \sqrt{((R_{nc}+T_n)^2 + (R_n+T_n)^2)/2}$
= $\pi/2 \cdot \sqrt{((124.437+12.088)^2 + (124.437+12.088)^2)/2}$
= 214.453 mm

Weld Length of Pad to Shell Weld for non-radial Nozzles [LTp]:

= $\pi/2 \cdot \sqrt{((R_{nc}+T_n+W)^2 + (R_n+T_n+W)^2)/2}$
= $\pi/2 \cdot \sqrt{((124.437+12.088+100.0)^2 + (124.437+12.088+100.0)^2)/2}$
= 371.533 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 8.485, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be >= min(0.7 • 12.088, 6.000) Passed
tf1, L42T = 8.485, must be >= min(0.6 • 12.000, 0.6 • 19.500) Passed

Weld Load Value [fwlds]:

= min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot R_n^2 \cdot ky^2$)
= min(34063•1.0, 1.5•1522.2(6.138+0.0), $\pi/4 \cdot 23.5 \cdot 124.44^2 \cdot 1.0^2$)
= 2859.204 kgf

Discontinuity Force [fws]:

= fwlds • t • S/(t • S + te • Sp)
= 2859.2•19.5•1650/(19.5•1650+12.0•1650)
= 1769.985 kgf

Discontinuity Force [fwp]:

= fwlds • te • Sp / (t • S + te • Sp)
= 2859.2•12.0•1650/(19.5•1650+12.0•1650)
= 1089.219 kgf

Shear Stress [τ 1]:



$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 1769.985 / (214.453 \cdot (0.6 \cdot 14.275 + 0.49 \cdot 0.0)) \\ &= 96.364 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 1089.219 / (214.453 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071)) \\ &= 47.625 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (L_{TP} \cdot (0.49 \cdot L42T)) \\ &= 1089.219 / (371.533 \cdot (0.49 \cdot 8.485)) \\ &= 70.512 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(96.364, 47.625, 70.512) \\ &= 96.4 \text{ must be less than or equal to } 1650.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[Dx/Reff] \\ &= \cos^{-1}[600.0/1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(Dx+Rn)/Reff] \\ &= \cos^{-1}[(600.0 + 124.437)/1151.5] \\ &= 0.890 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 12.088 / \ln[1 + 12.088/124.437] \\ &= 130.388 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 25.748 / \ln[1 + 25.748/1151.5] \\ &= 1164.328 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn(LH + t / \sin[\theta]) \text{ (eqn. 4.5.73)} \\ &= 1.055 \cdot 130.388(50.783 + 19.5 / \sin[90.0]) \\ &= 96.6 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1164.328 \cdot 204.146 \\ &= 2506.7 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 1.055 \cdot 1164.328 \cdot 124.437 \\ &= 1528.0 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (2506.7 + 1528.0 + 96.6) / 1.055 \\ &= 3917.4 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.612 / (2 \cdot 3917.431/58.69 - 1164.328/25.748) \\ &= 28.0 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.408[19.5/1164.328] \\ &= 27.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(28.044, 27.641) \\ &= 27.641 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (96.645 + 2506.72 + 1527.972) / 58.69 \\ &= 70.392 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.055 \cdot 1164.328 / 25.748 \\ &= 47.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 70.392 - 47.689, 47.689) \\ &= 93.1 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	93.10	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.64	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.12	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E*) / (tnom - ci - cext) \\ &= (15.272 \cdot 1.0) / (12.0 - 1.5 - 0.0) \\ &= 1.000 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated



Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.176

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (1.795 \cdot 1.0) / (13.202 - 3.0 - 0.0)$
 $= 0.176$

Gov. MDMT of the Nozzle : -46 °C
 Gov. MDMT of the Reinforcement Pad : 1 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 1 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -85 °C

The Drop for this Nozzle is : 97.8773 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 352.4112 mm

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: Noz N3 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	21.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1650.41	kgf/cm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	242.875	mm
Nozzle Thickness	15.0876	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	473.050	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:
*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	23.51	kgf/cm ²
Radial Load	P	-683.24	kgf
Circumferential Shear	Vc	1019.77	kgf
Longitudinal Shear	Vl	1019.77	kgf
Circumferential Moment	Mc	0.76	kgf-m.
Longitudinal Moment	Ml	1.02	kgf-m.
Torsional Moment	Mt	1.22	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$$

$$= 770.917 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	31.500 mm
Nozzle Thickness used	(t) =	12.088 mm
T / t	=	2.606
d / t	=	22.589

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.007$$

Nr/P	=	0.096
Mr/P	=	0.115
M0/P	=	0.091
N0/P	=	0.198
MrD/Mc	=	0.191
NrDL/Mc	=	0.138
M0d/Mc	=	0.177
N0DL/Mc	=	0.106
MrD/M1	=	0.152
NrDL/M1	=	0.116
M0D/M1	=	0.135
N0DL/M1	=	0.241

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	847	871	847	871
Outplane Stress Summary	898	847	899	847

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	423	423	423	423
Inplane Stress Summary	477	383	477	382

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	0	0	0	0
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Shear Stress Summary	7	7	-7	-7
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	898	847	899	847

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	37	-37	37	-37
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	423	423	423	423
Outplane Stress Summary	474	400	475	399

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	847	871	847	871
Inplane Stress Summary	901	830	901	830

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	901	830	901	830

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



	276	305	276	305
Hoop Pressure Stress	276	305	276	305

	290	318	290	318
Hoop Stress Summary	290	318	290	318

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	6	6	6	6
Axial Bending (P)	186	-186	186	-186
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	138	138	138	138

Axial Stress Summary	330	-40	333	-42
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	19	19	-19	-19
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	19	19	-19	-19
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	338	360	341	363
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	276	305	276	305

Hoop Stress Summary	290	318	290	318
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	186	-186	186	-186
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	138	138	138	138

Axial Stress Summary	330	-40	333	-42
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
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Shear due to (V1)	-19	-19	19	19
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-19	-19	19	19

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	338	360	341	363

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1161.25(21.0 - 1.5))}$
 $= 769.636 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2322.500 mm
Nozzle Outside Diameter	(d) =	473.050 mm
Vessel Thickness used	(T) =	19.500 mm
Nozzle Thickness used	(t) =	112.088 mm
T / t	=	0.174
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.223$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.070
Mr/P	=	0.135
M0/P	=	0.040
N0/P	=	0.068
MrD/Mc	=	0.284
NrDL/Mc	=	0.096
M0d/Mc	=	0.087
N0DL/Mc	=	0.076
MrD/Ml	=	0.103
NrDL/Ml	=	0.088
M0D/Ml	=	0.031
N0DL/Ml	=	0.089

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	43	-43	43	-43
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1376	1400	1376	1400
Outplane Stress Summary	1431	1369	1431	1369

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	145	-145	145	-145
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	688	688	688	688
Inplane Stress Summary	845	555	846	555

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1431	1369	1432	1369

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	43	-43	43	-43
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	688	688	688	688
Outplane Stress Summary	743	657	743	657

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	145	-145	145	-145
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1376	1400	1376	1400
Inplane Stress Summary	1533	1267	1535	1266

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	1533	1267	1535	1266



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	847	871	847	871	847	871	847	871
Circ. Pl (SUS)	13	13	13	13	6	6	6	6
Circ. Q (SUS)	37	-37	37	-37	47	-47	47	-47
Long. Pm (SUS)	423	423	423	423	423	423	423	423
Long. Pl (SUS)	6	6	6	6	13	13	13	13
Long. Q (SUS)	47	-47	47	-47	37	-37	37	-37
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	847.0	871.0	847.0	871.0	847.0	871.0	847.0	871.0
Pm+Pl (SUS)	860.1	884.1	860.1	884.1	853.1	877.1	853.1	877.1
Pm+Pl+Q (Total)	897.1	847.1	897.1	847.1	900.1	830.1	900.1	830.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	871.00	1650.41	Passed
Pm+Pl (SUS)	884.11	2475.61	Passed
Pm+Pl+Q (TOTAL)	900.11	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1376	1400	1376	1400	1376	1400	1376	1400
Circ. Pl (SUS)	12	12	12	12	12	12	12	12
Circ. Q (SUS)	43	-43	43	-43	145	-145	145	-145
Long. Pm (SUS)	688	688	688	688	688	688	688	688
Long. Pl (SUS)	12	12	12	12	12	12	12	12
Long. Q (SUS)	145	-145	145	-145	43	-43	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1376	1400	1376	1400	1376	1400	1376	1400
Pm+Pl (SUS)	1388	1412	1388	1412	1388	1412	1388	1412
Pm+Pl+Q (Total)	1431	1369	1431	1369	1533	1267	1533	1267

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	1400.00	1650.41	Passed
Pm+Pl (SUS)	1412.07	2475.61	Passed
Pm+Pl+Q (TOTAL)	1533.06	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	276	305	276	305	276	305	276	305
Circ. Pl (SUS)	13	13	13	13	13	13	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	138	138	138	138	138	138	138	138
Long. Pl (SUS)	6	6	6	6	6	6	6	6
Long. Q (SUS)	185	-185	187	-187	185	-185	187	-187
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	276.0	305.0	276.0	305.0	276.0	305.0	276.0	305.0
Pm+Pl (SUS)	291.4	320.1	291.4	320.1	291.4	320.1	291.4	320.1
Pm+Pl+Q (Total)	336.6	361.0	338.3	363.0	336.6	361.0	338.3	363.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	305.00	1522.22	Passed
Pm+Pl (SUS)	320.05	2283.33	Passed
Pm+Pl+Q (TOTAL)	362.99	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz N6

From: 70

Pressure for Reinforcement Calculations	P	23.684	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1057.92	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	270.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

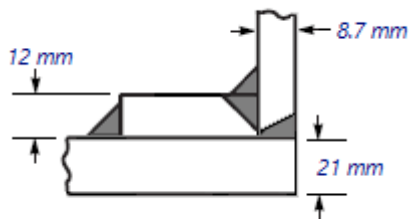
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N6

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		1.687	in.
Actual Thickness Used in Calculation		0.344	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 48.85 \cdot (\exp(23.684/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 0.383 + 3.0 + 0.0 = 3.383 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 5.738 + 48.85) / 48.85) - 0.0$
 $= 321.183 - 0.0 = 321.1826 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.420 mm
Min. required shell thickness, internal p + c	18.144 mm
Min. required shell thickness, external p + c	13.840 mm

Nozzle neck minimum required thickness	: 6.420 mm
--	------------

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 6.420 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	40.4,	Allowable :	1522.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3914.4 kgf/cm ²	Passed
Occasional :	56.9,	Allowable :	2024.6 kgf/cm ²	Passed
Shear :	81.2,	Allowable :	1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.4, 1]$
 $= 0.922$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1650.4/1650.4, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 8.738 - 3.0$
 $= 5.738 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2303.0/19.5$
 $= 118.103 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:



$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 2300.0/2 + 1.5 \\
 &= 1151.500 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned}
 \text{Limit [LR1]:} \\
 &= \sqrt{(R_{eff} \cdot t) + W} \\
 &= \sqrt{(1151.5 \cdot 19.5) + 50.0} \\
 &= 199.847 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR2]:} \\
 &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\
 &= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))} \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR3]:} \\
 &= 2 \cdot R_n \\
 &= 2 \cdot 24.425 \\
 &= 48.850 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Final value of [LR]:} \\
 &= \min(LR1, LR2, LR3) \\
 &= \min(199.847, 192.059, 48.85) \\
 &= 48.850 \text{ mm}
 \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:
 = 54.587 mm

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(24.425 \cdot 5.738)} \\
 &= 23.838 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH2]:} \\
 &= L_{pr1} \\
 &= 270.0 \\
 &= 270.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + t_e) \\
 &= 8(19.5 + 12.0) \\
 &= 252.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] \\
 &= \min[23.838, 270.0, 252.0] \\
 &= 23.838 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A5 \cdot frp) / (LR - t_n)) \\
 &= 19.5 + ((5.862 \cdot 1) / (54.587 - 5.738)) \\
 &= 31.500 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Parameter } [\lambda]: \\
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 24.42 + 5.738) / (\sqrt{(2303.0 + 31.5) \cdot 31.5})) \\
 &= 0.201
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 \text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 19.5 \cdot 54.587 \cdot \max((0.201/5)^{0.85}, 1) \\
 &= 10.645 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= t_n \cdot LH
 \end{aligned}$$



$$= 5.738 \cdot 23.838$$
$$= 1.368 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg42}^2 - \text{Area Cut by Reinforcement Limit}$$
$$= 0.5 \cdot 12.0^2 - 0.275$$
$$= 0.445 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, (LR - t_n) \cdot t_e)$$
$$= \min(50.0 \cdot 12.0, (54.587 - 5.738) \cdot 12.0)$$
$$= 5.862 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$
$$= 10.645 + 0.922(1.368 + 0.0) + 0.5 + 0.445 + 0.0 + 1.(5.862)$$
$$= 18.713 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 5.738 / \ln[1 + 5.738/24.425]$$
$$= 27.193 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 31.5 / \ln[1 + 31.5/1151.5]$$
$$= 1167.181 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn}(LH + t)$$
$$= 23.684 \cdot 27.193(23.838 + 19.5)$$
$$= 279.1 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.684 \cdot 1167.181 \cdot 54.587$$
$$= 15089.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 23.684 \cdot 1167.181 \cdot 24.425$$
$$= 6751.9 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$
$$= (15089.8 + 6751.9 + 279.1)/23.684$$
$$= 934.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff})$$
$$= 2475.612 / (2 \cdot 934.002/18.713 - 1167.181/31.5)$$
$$= 39.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$
$$= 1650.408[19.5/1167.181]$$
$$= 27.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(39.44, 27.573)$$



$$= 27.573 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (279.112 + 15089.807 + 6751.869) / 18.713 \\ &= 1182.092 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_s / t_{eff} \\ &= 23.684 \cdot 1167.181 / 31.5 \\ &= 877.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1182.092 - 877.57, 877.57) \\ &= 1486.6 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1486.61	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.57	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (24.425 + 5.738) \\ &= 47.379 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (24.425 + 5.738 + 50.0) \\ &= 125.919 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 7.071, \text{ must be } \geq \min(0.7 \cdot 5.738, 6.000) \text{ Passed} \\ t_{f1}, L42T &= 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 19.500) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_Y \cdot k_Y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_Y^2) \\ &= \min(6752 \cdot 1.0, 1.5 \cdot 1522.2(1.368 + 0.0), \pi/4 \cdot 23.7 \cdot 24.42^2 \cdot 1.0^2) \\ &= 110.971 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 111.0 \cdot 19.5 \cdot 1650 / (19.5 \cdot 1650 + 12.0 \cdot 1650) \\ &= 68.696 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 111.0 \cdot 12.0 \cdot 1650 / (19.5 \cdot 1650 + 12.0 \cdot 1650) \\ &= 42.275 \text{ kgf} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 68.696 / (47.379 \cdot (0.6 \cdot 8.738 + 0.49 \cdot 0.0)) \\ &= 27.657 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\ &= 42.275 / (47.379 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071)) \\ &= 8.366 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:



$$\begin{aligned} &= fwp / (LrP \cdot (0.49 \cdot L42T)) \\ &= 42.275 / (125.919 \cdot (0.49 \cdot 8.485)) \\ &= 8.075 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(27.657, 8.366, 8.075) \\ &= 27.7 \text{ must be less than or equal to } 1650.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 5.738 / \ln[1 + 5.738/24.425] \\ &= 27.193 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 31.5 / \ln[1 + 31.5/1151.5] \\ &= 1167.181 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn(LH + t) \\ &= 1.055 \cdot 27.193(23.838 + 19.5) \\ &= 12.4 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1167.181 \cdot 54.587 \\ &= 671.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 1.055 \cdot 1167.181 \cdot 24.425 \\ &= 300.6 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (671.9 + 300.6 + 12.4) / 1.055 \\ &= 934.0 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/t_{eff}) \\ &= 2475.612 / (2 \cdot 934.002/18.713 - 1167.181/31.5) \\ &= 39.4 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.408[19.5/1167.181] \\ &= 27.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(39.44, 27.573) \\ &= 27.573 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (12.428 + 671.924 + 300.649) / 18.713 \\ &= 52.637 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / t_{eff} \\ &= 1.055 \cdot 1167.181/31.5 \\ &= 39.1 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 52.637 - 39.077, 39.077) \\ &= 66.2 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	66.20	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.57	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.12	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (15.385 \cdot 1.0) / (12.0 - 1.5 - 0.0) \\ &= 1.000 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.076

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.355 \cdot 1.0) / (7.645 - 3.0 - 0.0) \\ &= 0.076 \end{aligned}$$

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	1 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	1 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -104 °C



The Drop for this Nozzle is : 0.3956 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 291.3956 mm

Input Echo, WRC297 Item 1, Description: Noz N6 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2300.000	mm	
Shell Thickness	21.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.41	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	42.850	mm	
Nozzle Thickness	8.7376	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-333 6		
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	160.325	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	23.68	kgf/cm ²
Radial Load	P	-132.57	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	0.03	kgf-m.
Longitudinal Moment	Ml	0.04	kgf-m.
Torsional Moment	Mt	0.05	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (2.873) is < 5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (6.904) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$
 $= 558.192 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500	mm
Nozzle Outside Diameter	(d) =	60.325	mm
Vessel Thickness used	(T) =	31.500	mm
Nozzle Thickness used	(t) =	5.738	mm
T / t	=	5.490	
d / t	=	10.514	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.222$$

$$Nr/P = 0.207$$

Mr/P = 0.129
 MØ/P = 0.316
 NØ/P = 0.250
 MrD/Mc = 0.112
 NrDL/Mc = 0.155
 MØd/Mc = 0.304
 NØDL/Mc = -0.131
 MrD/Ml = 0.112
 NrDL/Ml = 0.159
 MØD/Ml = 0.299
 NØDL/Ml = -0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	854	877	854	877
Outplane Stress Summary	882	855	882	855

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	427	427	427	427
Inplane Stress Summary	440	419	440	419

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	6	6	-6	-6
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	6	6	-6	-6

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	882	855	882	855

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	427	427	427	427

Outplane Stress Summary	455	405	455	404
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	854	877	854	877

Inplane Stress Summary	867	870	867	870
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-6	-6	6	6
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-6	-6	6	6
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	867	870	867	870
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	108	136	108	136

Hoop Stress Summary	111	139	111	139
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	13	13	13	13
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	54	54	54	54

Axial Stress Summary	127	6	128	6
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	37	37	-37	-37
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	37	37	-37	-37

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	157	152	158	152

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	108	136	108	136
Hoop Stress Summary	111	139	111	139

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	54	54	54	54
Axial Stress Summary	127	6	128	6

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-37	-37	37	37
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-37	-37	37	37

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	157	152	158	152

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{1161.25(21.0 - 1.5)}$$

$$= 556.911 \text{ mm}$$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 2322.500 mm
 Nozzle Outside Diameter (d) = 160.325 mm
 Vessel Thickness used (T) = 19.500 mm
 Nozzle Thickness used (t) = 55.738 mm
 T / t = 0.350
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.753$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.121
 Mr/P = 0.266
 M0/P = 0.078
 N0/P = 0.141
 MrD/Mc = 0.312
 NrDL/Mc = 0.072
 M0d/Mc = 0.097
 N0DL/Mc = 0.027
 MrD/Ml = 0.231
 NrDL/Ml = 0.088
 M0D/Ml = 0.073
 N0DL/Ml = 0.094

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1386	1410	1386	1410
Outplane Stress Summary	1407	1399	1407	1399

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	693	693	693	693
Inplane Stress Summary	753	641	753	641

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	4	4	-4	-4

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 1407 1399 1407 1399

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	693	693	693	693
Outplane Stress Summary	714	681	714	681

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1386	1410	1386	1410
Inplane Stress Summary	1446	1359	1446	1358

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-4	-4	4	4

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1446	1359	1446	1358

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	854	877	854	877	854	877	854	877
Circ. Pl (SUS)	3	3	3	3	2	2	2	2
Circ. Q (SUS)	25	-25	25	-25	10	-10	10	-10
Long. Pm (SUS)	427	427	427	427	427	427	427	427
Long. Pl (SUS)	2	2	2	2	3	3	3	3
Long. Q (SUS)	10	-10	10	-10	25	-25	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-6	-6	6	6
Shear Q (SUS)	0	0	0	0	0	0	0	0

Pm (SUS)	854.0	877.0	854.0	877.0	854.0	877.0	854.0	877.0
Pm+Pl (SUS)	857.1	880.1	857.1	880.1	856.1	879.1	856.1	879.1
Pm+Pl+Q (Total)	882.1	855.1	882.1	855.1	866.1	869.1	866.1	869.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	877.00	1650.41	Passed
Pm+Pl (SUS)	880.08	2475.61	Passed
Pm+Pl+Q (TOTAL)	882.08	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1386	1410	1386	1410	1386	1410	1386	1410
Circ. Pl (SUS)	4	4	4	4	4	4	4	4
Circ. Q (SUS)	16	-16	16	-16	55	-55	55	-55
Long. Pm (SUS)	693	693	693	693	693	693	693	693
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	55	-55	55	-55	16	-16	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1386	1410	1386	1410	1386	1410	1386	1410
Pm+Pl (SUS)	1390	1414	1390	1414	1390	1414	1390	1414
Pm+Pl+Q (Total)	1406	1398	1406	1398	1445	1359	1445	1359

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1410.00	1650.41	Passed
Pm+Pl (SUS)	1414.02	2475.61	Passed
Pm+Pl+Q (TOTAL)	1445.02	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	108	136	108	136	108	136	108	136
Circ. Pl (SUS)	3	3	3	3	3	3	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	54	54	54	54	54	54	54	54
Long. Pl (SUS)	13	13	13	13	13	13	13	13



Long. Q (SUS)	60	-60	60	-60	60	-60	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	37	37	-37	-37	-37	-37	37	37
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	108.0	136.0	108.0	136.0	108.0	136.0	108.0	136.0
Pm+Pl (SUS)	132.0	154.6	132.0	154.6	132.0	154.6	132.0	154.6
Pm+Pl+Q (Total)	156.9	151.3	156.9	151.3	156.9	151.3	156.9	151.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	1522.22	Passed
Pm+Pl (SUS)	154.62	2283.33	Passed
Pm+Pl+Q (TOTAL)	156.85	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz N7

From: 70

Pressure for Reinforcement Calculations	P	23.667	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	1000.0000	mm
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Distance from Bottom/Left Tangent		657.92	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		-58.65	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	270.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000	mm

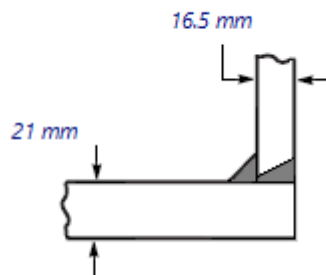
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N7

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(23.667/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.445 + 3.0 + 0.0 = 3.445 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 13.5 + 56.8) / 56.8) - 0.0$$

$$= 591.986 - 0.0 = 591.9862 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c :	18.132 mm
Min. required shell thickness, external p + c :	13.840 mm

Nozzle neck minimum required thickness : 7.800 mm

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 7.800 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	15.8,	Allowable :	1522.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3939.1 kgf/cm ²	Passed
Occasional :	20.1,	Allowable :	2024.6 kgf/cm ²	Passed
Shear :	24.0,	Allowable :	1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.5 - 3.0$$

$$= 13.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2300.0/2 + 1.5$$

$$= 1151.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:



$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot Rn) \\ &= \min(\sqrt{(1151.5 \cdot 19.5)}, 2 \cdot 28.4) \\ &= 56.800 \text{ mm} \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:
= 70.300 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, te] + \sqrt{(Rn \cdot tn)} \\ &= \min[1.5 \cdot 19.5, 0.0] + \sqrt{(28.4 \cdot 13.5)} \\ &= 19.581 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= Lpr1 \\ &= 270.0 \\ &= 270.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + te) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[19.581, 270.0, 156.0] \\ &= 19.581 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - tn)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (70.3 - 13.5)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 28.4 + 13.5) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.330 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 70.3 \cdot \max((0.33/5)^{0.85}, 1) \\ &= 13.708 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 13.5 \cdot 19.581 \\ &= 2.643 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 13.708 + 0.922(2.643 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.647 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[Dx/\text{Reff}] \\ &= \cos^{-1}[1000.0/1151.5] \\ &= 0.519 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(Dx + Rn)/\text{Reff}] \\ &= \cos^{-1}[(1000.0 + 28.4)/1151.5] \\ &= 0.467 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:



$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 13.5/\ln[1 + 13.5/28.4] \\ &= 34.714 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 19.5/\ln[1 + 19.5/1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.408 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (LH + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 23.667 \cdot 34.714 (19.581 + 19.5/\sin[90.0]) \\ &= 321.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.667 \cdot 1161.225 \cdot 70.3 \\ &= 19320.4 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 23.667 \cdot 1161.225 \cdot 30.025 \\ &= 8251.6 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (19320.4 + 8251.6 + 321.1) / 23.667 \\ &= 1178.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 1178.561 / 16.647 - 1161.225 / 19.5) \\ &= 30.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 1650.408[19.5/1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(30.173, 27.715) \\ &= 27.715 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (321.075 + 19320.432 + 8251.62) / 16.647 \\ &= 1675.611 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 23.667 \cdot 1161.225 / 19.5 \\ &= 1409.4 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1675.611 - 1409.378, 1409.378) \\ &= 1941.8 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1941.84	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.71	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

= 1.000 For set-on Nozzles

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc} + T_n)^2 + (R_n + T_n)^2)/2} \\ &= \pi/2 \cdot \sqrt{((30.025 + 13.5)^2 + (28.4 + 13.5)^2)/2} \\ &= 67.104 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 13.500, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(8252 \cdot 1.0, 1.5 \cdot 1522.2(2.643 + 0.0), \pi/4 \cdot 23.7 \cdot 28.4^2 \cdot 1.0^2) \\ &= 149.924 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 149.924 / (67.104(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 64.200 < \text{or} = 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[1000.0 / 1151.5] \\ &= 0.519 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(1000.0 + 28.4) / 1151.5] \\ &= 0.467 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 13.5 / \ln[1 + 13.5 / 28.4] \\ &= 34.714 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (L_H + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 1.055 \cdot 34.714 (19.581 + 19.5 / \sin[90.0]) \\ &= 14.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 70.3 \\ &= 860.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 30.025 \\ &= 367.7 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (860.9 + 367.7 + 14.3) / 1.055 \\ &= 1178.6 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.612 / (2 \cdot 1178.561/16.647 - 1161.225/19.5) \\ &= 30.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.408[19.5/1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(30.173, 27.715) \\ &= 27.715 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (14.307 + 860.917 + 367.691) / 16.647 \\ &= 74.665 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.055 \cdot 1161.225/19.5 \\ &= 62.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 74.665 - 62.802, 62.802) \\ &= 86.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	86.53	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.71	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.031

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E*) / (tnom - ci - cext) \\ &= (0.412 \cdot 1.0) / (16.5 - 3.0 - 0.0) \\ &= 0.031 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 81.1214 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 392.5260 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: Noz N7 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	21.0000	mm
Shell Stress Concentration Factor	1.000	



Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.41	kgf/cm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		50.800	mm
Nozzle Thickness		16.5000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	23.67	kgf/cm ²
Radial Load	P	-132.57	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	0.03	kgf-m.
Longitudinal Moment	Ml	0.04	kgf-m.
Torsional Moment	Mt	0.05	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.990) is < 5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (5.079) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 83.8 + 2 \cdot 1.65 \cdot \sqrt{(1161.25(21.0 - 1.5))}$
 $= 580.386 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2322.500 mm
Nozzle Outside Diameter	(d) =	83.800 mm
Vessel Thickness used	(T) =	19.500 mm
Nozzle Thickness used	(t) =	13.500 mm
T / t	=	1.444
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.394$$

Nr/P	=	0.207
Mr/P	=	0.271
M0/P	=	0.120
N0/P	=	0.178
MrD/Mc	=	0.264
NrDL/Mc	=	0.168
M0d/Mc	=	0.145
N0DL/Mc	=	-0.112
MrD/Ml	=	0.249
NrDL/Ml	=	0.165
M0D/Ml	=	0.135
N0DL/Ml	=	-0.044

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE Au Al Bu Bl



(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1385	1409	1385	1409
Outplane Stress Summary	1416	1390	1417	1390

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	692	692	692	692
Inplane Stress Summary	756	643	756	643

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1417	1390	1417	1390

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	692	692	692	692
Outplane Stress Summary	724	674	724	673

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1385	1409	1385	1409
Inplane Stress Summary	1449	1360	1449	1359

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	1449	1360	1449	1359

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	48	76	48	76
Hoop Stress Summary	54	82	54	82

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	72	-72	72	-72
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	101	-44	101	-44

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	11	11	-11	-11
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	103	129	104	129

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	48	76	48	76
Hoop Stress Summary	54	82	54	82

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	72	-72	72	-72
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	101	-44	101	-44

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-11	-11	11	11
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-11	-11	11	11

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	103	129	104	129

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1385	1409	1385	1409	1385	1409	1385	1409
Circ. Pl (SUS)	6	6	6	6	7	7	7	7
Circ. Q (SUS)	25	-25	25	-25	56	-56	56	-56
Long. Pm (SUS)	692	692	692	692	692	692	692	692
Long. Pl (SUS)	7	7	7	7	6	6	6	6
Long. Q (SUS)	56	-56	56	-56	25	-25	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1385	1409	1385	1409	1385	1409	1385	1409
Pm+Pl (SUS)	1391	1415	1391	1415	1392	1416	1392	1416
Pm+Pl+Q (Total)	1416	1390	1416	1390	1448	1360	1448	1360



Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1409.00	1650.41	Passed
Pm+Pl (SUS)	1416.07	2475.61	Passed
Pm+Pl+Q (TOTAL)	1448.07	4951.22	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	76	48	76	48	76	48	76
Circ. Pl (SUS)	6	6	6	6	6	6	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	72	-72	72	-72	72	-72	72	-72
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	48.0	76.0	48.0	76.0	48.0	76.0	48.0	76.0
Pm+Pl (SUS)	58.0	84.2	58.0	84.2	58.0	84.2	58.0	84.2
Pm+Pl+Q (Total)	102.5	127.9	102.5	127.9	102.5	127.9	102.5	127.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	76.00	1522.22	Passed
Pm+Pl (SUS)	84.15	2283.33	Passed
Pm+Pl+Q (TOTAL)	127.91	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz K2A

From: 70

Pressure for Reinforcement Calculations	P	23.510	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		461.42	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B
Material UNS Number		K03006
Material Specification/Type		Smls. pipe
Allowable Stress at Temperature	Sn	1522.22 kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66 kgf/cm ²

Diameter Basis (for tr calc only)		Inside
Layout Angle		30.82 deg
Diameter		2.0000 in.

Size and Thickness Basis		Actual
Actual Thickness	tn	16.5000 mm

Flange Material		SA-350 LF2
Flange Type		Long Weld Neck

Corrosion Allowance	can	1.5000 mm
---------------------	-----	-----------

Outside Projection	ho	200.0000 mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000 mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000 mm

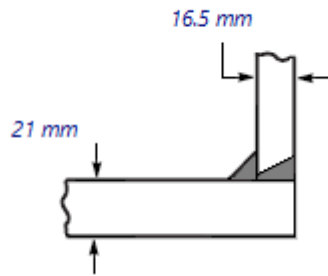
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K2A

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000 in.
Actual Thickness Used in Calculation		0.650 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(23.51/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.4187 + 1.5 + 0.0 = 1.9187 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$$

$$= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	18.021 mm
Min. required shell thickness, external p + c :	13.840 mm

Nozzle neck minimum required thickness	: 6.300 mm
--	------------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.5 - 1.5$$

$$= 15.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2300.0/2 + 1.5$$

$$= 1151.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 68.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 19.5, 0.0] + \sqrt{26.9 \cdot 15.0}$$

$$= 20.087 \text{ mm}$$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 200.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.695 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$



$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t / \sin[\theta]) \quad (\text{eqn. 4.5.73})$$
$$= 23.51 \cdot 33.848 (20.087 + 19.5 / \sin[90.0])$$
$$= 315.0 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.51 \cdot 1161.225 \cdot 68.8$$
$$= 18782.7 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 23.51 \cdot 1161.225 \cdot 26.9$$
$$= 7343.8 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (18782.7 + 7343.8 + 315.0) / 23.51$$
$$= 1124.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 2475.612 / (2 \cdot 1124.692 / 16.695 - 1161.225 / 19.5)$$
$$= 32.9 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$
$$= 1650.408 [19.5 / 1161.225]$$
$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(32.928, 27.715)$$
$$= 27.715 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (315.022 + 18782.711 + 7343.822) / 16.695$$
$$= 1583.797 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 23.51 \cdot 1161.225 / 19.5$$
$$= 1400.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 1583.797 - 1400.025, 1400.025)$$
$$= 1767.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1767.57	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.71	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + T_n)^2 + (R_n + T_n)^2) / 2}$$
$$= \pi/2 \cdot \sqrt{((26.9 + 15.0)^2 + (26.9 + 15.0)^2) / 2}$$
$$= 65.816 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7344 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.613 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 133.613 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.335 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 1.055 \cdot 33.848(20.087 + 19.5 / \sin[90.0]) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 1124.692 / 16.695 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / R_{xs}] \\ &= 1650.408[19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:



= min(Pmax1, Pmax2)
= min(32.928, 27.715)
= 27.715 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:
= (fN + fS + fY) / AT
= (14.131 + 842.547 + 329.426)/16.695
= 71.045 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.055 • 1161.225/19.5
= 62.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 71.045 - 62.802, 62.802)
= 79.3 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	79.29	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.71	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.388 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 26.8883 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.4222 mm

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Input, Nozzle Desc: Noz K2B

From: 70

Pressure for Reinforcement Calculations	P	23.628	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		461.42	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B
Material UNS Number		K03006
Material Specification/Type		Smls. pipe
Allowable Stress at Temperature	Sn	1522.22 kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66 kgf/cm ²

Diameter Basis (for tr calc only)		Inside
Layout Angle		-30.82 deg
Diameter		2.0000 in.

Size and Thickness Basis		Actual
Actual Thickness	tn	16.5000 mm

Flange Material		SA-350 LF2
Flange Type		Long Weld Neck

Corrosion Allowance	can	1.5000 mm
---------------------	-----	-----------

Outside Projection	ho	200.0000 mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000 mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000 mm

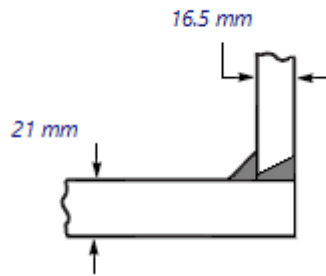
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K2B

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000 in.
Actual Thickness Used in Calculation		0.650 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(23.628/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.4208 + 1.5 + 0.0 = 1.9208 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$$

$$= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c	18.104 mm
Min. required shell thickness, external p + c	13.840 mm

Nozzle neck minimum required thickness	: 6.300 mm
--	------------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.5 - 1.5$$

$$= 15.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2300.0/2 + 1.5$$

$$= 1151.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 68.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 19.5, 0.0] + \sqrt{26.9 \cdot 15.0}$$

$$= 20.087 \text{ mm}$$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 200.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.695 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$



$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn(LH + t / \sin[\text{theta}]) \text{ (eqn. 4.5.73)}$$
$$= 23.628 \cdot 33.848(20.087 + 19.5/\sin[90.0])$$
$$= 316.6 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.628 \cdot 1161.225 \cdot 68.8$$
$$= 18876.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 23.628 \cdot 1161.225 \cdot 26.9$$
$$= 7380.6 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (18876.8 + 7380.6 + 316.6)/23.628$$
$$= 1124.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$
$$= 2475.612 / (2 \cdot 1124.692/16.695 - 1161.225/19.5)$$
$$= 32.9 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1650.408[19.5/1161.225]$$
$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(32.928, 27.715)$$
$$= 27.715 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ Avg]:

$$= (fN + fS + fY) / AT$$
$$= (316.6 + 18876.82 + 7380.617)/16.695$$
$$= 1591.732 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ Circ]:

$$= P \cdot Rxs / teff$$
$$= 23.628 \cdot 1161.225/19.5$$
$$= 1407.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma\text{Avg} - \sigma\text{Circ}, \sigma\text{Circ})$$
$$= \max(2 \cdot 1591.732 - 1407.04, 1407.04)$$
$$= 1776.4 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1776.42	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.71	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((Rnc+Tn)^2 + (Rn+Tn)^2)/2 }$$
$$= \pi/2 \cdot \sqrt{((26.9+ 15.0)^2 + (26.9+ 15.0)^2)/2 }$$
$$= 65.816 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7381 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.6 \cdot 26.9^2 \cdot 1.0^2) \\ &= 134.282 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 134.282 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.628 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 1.055 \cdot 33.848(20.087 + 19.5 / \sin[90.0]) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 1124.692 / 16.695 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1650.408 [19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:



= min(Pmax1, Pmax2)
= min(32.928, 27.715)
= 27.715 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:
= (fN + fS + fY) / AT
= (14.131 + 842.547 + 329.426)/16.695
= 71.045 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.055 • 1161.225/19.5
= 62.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 71.045 - 62.802, 62.802)
= 79.3 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	79.29	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.71	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.39 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 26.8883 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.4222 mm

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Input, Nozzle Desc: Noz K3

From: 70

Pressure for Reinforcement Calculations	P	23.510	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		767.92	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.82	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000	mm

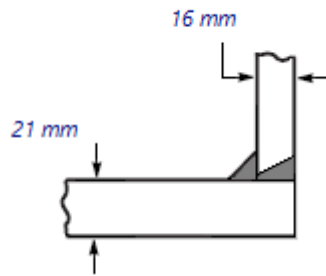
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		1.500	in.
Actual Thickness Used in Calculation		0.630	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 41.1 \cdot (\exp(23.51/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.3198 + 1.5 + 0.0 = 1.8198 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 14.5 + 41.1) / 41.1) - 0.0$$

$$= 812.735 - 0.0 = 812.7354 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c	18.021 mm
Min. required shell thickness, external p + c	13.840 mm

Nozzle neck minimum required thickness	: 6.020 mm
--	------------

Minimum nozzle neck thickness 16.002 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 1.5$$

$$= 14.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2300.0/2 + 1.5$$

$$= 1151.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 20.55)$$

$$= 41.100 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 55.600 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 19.5, 0.0] + \sqrt{20.55 \cdot 14.5}$$

$$= 17.262 \text{ mm}$$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[17.262, 200.0, 156.0] \\ &= 17.262 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (55.6 - 14.5)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 20.55 + 14.5) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.261 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 55.6 \cdot \max((0.261/5)^{0.85}, 1) \\ &= 10.842 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 14.5 \cdot 17.262 \\ &= 2.503 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 10.842 + 0.922(2.503 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 13.651 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 20.55) / 1151.5] \\ &= 1.002 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 14.5 / \ln[1 + 14.5 / 20.55] \\ &= 27.158 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$



$$= 1.5 \cdot 1650.408 \cdot 1.0$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn(LH + t / \sin[\text{theta}]) \text{ (eqn. 4.5.73)}$$
$$= 23.51 \cdot 27.158(17.262 + 19.5/\sin[90.0])$$
$$= 234.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.51 \cdot 1161.225 \cdot 55.6$$
$$= 15179.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 23.51 \cdot 1161.225 \cdot 20.55$$
$$= 5610.2 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (15179.1 + 5610.2 + 234.7)/23.51$$
$$= 894.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$
$$= 2475.612 / (2 \cdot 894.257/13.651 - 1161.225/19.5)$$
$$= 34.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1650.408[19.5/1161.225]$$
$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(34.638, 27.715)$$
$$= 27.715 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ Avg]:

$$= (fN + fS + fY) / AT$$
$$= (234.719 + 15179.053 + 5610.243)/13.651$$
$$= 1540.158 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ Circ]:

$$= P \cdot Rxs / teff$$
$$= 23.51 \cdot 1161.225/19.5$$
$$= 1400.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma\text{Avg} - \sigma\text{Circ}, \sigma\text{Circ})$$
$$= \max(2 \cdot 1540.158 - 1400.025, 1400.025)$$
$$= 1680.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1680.29	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.71	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((Rnc+Tn)^2 + (Rn+Tn)^2)/2 }$$
$$= \pi/2 \cdot \sqrt{((20.55+ 14.5)^2 + (20.55+ 14.5)^2)/2 }$$
$$= 55.056 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 14.500, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(5610 \cdot 1.0, 1.5 \cdot 1522.2(2.503 + 0.0), \pi/4 \cdot 23.5 \cdot 20.55^2 \cdot 1.0^2) \\ &= 77.977 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 77.977 / (55.056(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 40.698 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 20.55) / 1151.5] \\ &= 1.002 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 14.5 / \ln[1 + 14.5 / 20.55] \\ &= 27.158 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t / \sin[\theta]) \text{ (eqn. 4.5.73)} \\ &= 1.055 \cdot 27.158(17.262 + 19.5 / \sin[90.0]) \\ &= 10.5 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 55.6 \\ &= 680.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 20.55 \\ &= 251.7 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (680.9 + 251.7 + 10.5) / 1.055 \\ &= 894.3 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 894.257 / 13.651 - 1161.225 / 19.5) \\ &= 34.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1650.408 [19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:



= min(Pmax1, Pmax2)
= min(34.638, 27.715)
= 27.715 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:
= (fN + fS + fY) / AT
= (10.529 + 680.896 + 251.662)/13.651
= 69.088 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.055 • 1161.225/19.5
= 62.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 69.088 - 62.802, 62.802)
= 75.4 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	75.37	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.71	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.020

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.296 • 1.0)/(16.0 - 1.5 - 0.0)
= 0.020

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 22.3156 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 246.8496 mm

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Input, Nozzle Desc: Noz K4

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		597.92	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

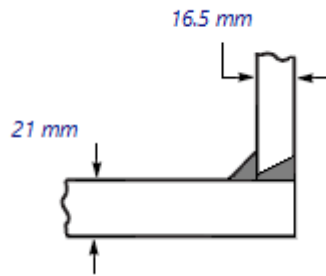
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 53.8 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 0.4177 + 1.5 + 0.0 = 1.9177 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
 $= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	17.981 mm
Min. required shell thickness, external p + c :	13.840 mm

Nozzle neck minimum required thickness :	6.300 mm
--	----------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.4, 1]$
 $= 0.922$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.5 - 1.5$
 $= 15.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2303.0/19.5$
 $= 118.103 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 2300.0/2 + 1.5$
 $= 1151.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$
 $= 53.800 \text{ mm}$

Because the nozzle is set-on, add tn to the result [LR]:



= 68.800 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 19.5, 0.0] + \sqrt{(26.9 \cdot 15.0)} \\ &= 20.087 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 230.0 \\ &= 230.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 230.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.695 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 19.5 / \ln[1 + 19.5/1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.408 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn(LH + t) \\ &= 23.454 \cdot 33.848(20.087 + 19.5) \\ &= 314.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.454 \cdot 1161.225 \cdot 68.8 \\ &= 18737.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 23.454 \cdot 1161.225 \cdot 26.9 \\ &= 7326.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (18737.9 + 7326.3 + 314.3)/23.454 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.612 / (2 \cdot 1124.692/16.695 - 1161.225/19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.408[19.5/1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(32.928, 27.715) \\ &= 27.715 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (314.271 + 18737.93 + 7326.313)/16.695 \\ &= 1580.021 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 23.454 \cdot 1161.225/19.5 \\ &= 1396.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1580.021 - 1396.687, 1396.687) \\ &= 1763.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1763.35	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.71	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (26.9 + 15.0) \\ &= 65.816 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:



$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7326 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.294 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 133.294 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.196 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t) \\ &= 1.055 \cdot 33.848(20.087 + 19.5) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 1124.692 / 16.695 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 1650.408 / 19.5 / 1161.225 \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(32.928, 27.715) \\ &= 27.715 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (14.131 + 842.547 + 329.426) / 16.695 \\ &= 71.045 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 1161.225 / 19.5 \\ &= 62.8 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 71.045 - 62.802, 62.802) \\ &= 79.3 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	79.29	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.71	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.026

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.387 \cdot 1.0) / (16.5 - 1.5 - 0.0) \\ &= 0.026 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.7635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.7635 mm

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Input, Nozzle Desc: Noz K5

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6661.6660	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		677.92	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
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Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

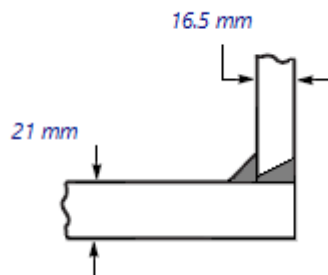
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 53.8 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 0.4177 + 1.5 + 0.0 = 1.9177 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
 $= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	17.981 mm
Min. required shell thickness, external p + c :	13.840 mm

Nozzle neck minimum required thickness	: 6.300 mm
--	------------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.4, 1]$
 $= 0.922$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.5 - 1.5$
 $= 15.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2303.0/19.5$
 $= 118.103 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 2300.0/2 + 1.5$
 $= 1151.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$
 $= 53.800 \text{ mm}$

Because the nozzle is set-on, add tn to the result [LR]:



= 68.800 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 19.5, 0.0] + \sqrt{(26.9 \cdot 15.0)} \\ &= 20.087 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 230.0 \\ &= 230.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 230.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.695 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 19.5 / \ln[1 + 19.5/1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.408 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn(LH + t) \\ &= 23.454 \cdot 33.848(20.087 + 19.5) \\ &= 314.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.454 \cdot 1161.225 \cdot 68.8 \\ &= 18737.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 23.454 \cdot 1161.225 \cdot 26.9 \\ &= 7326.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (18737.9 + 7326.3 + 314.3)/23.454 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.612 / (2 \cdot 1124.692/16.695 - 1161.225/19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.408[19.5/1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(32.928, 27.715) \\ &= 27.715 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (314.271 + 18737.93 + 7326.313)/16.695 \\ &= 1580.021 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 23.454 \cdot 1161.225/19.5 \\ &= 1396.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1580.021 - 1396.687, 1396.687) \\ &= 1763.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1763.35	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.71	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (26.9 + 15.0) \\ &= 65.816 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:



$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7326 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.294 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 133.294 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.196 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.408 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t) \\ &= 1.055 \cdot 33.848(20.087 + 19.5) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2475.612 / (2 \cdot 1124.692 / 16.695 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 1650.408 / 19.5 / 1161.225 \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(32.928, 27.715) \\ &= 27.715 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (14.131 + 842.547 + 329.426) / 16.695 \\ &= 71.045 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 1161.225 / 19.5 \\ &= 62.8 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 71.045 - 62.802, 62.802) \\ &= 79.3 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	79.29	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.71	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.026

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.387 \cdot 1.0) / (16.5 - 1.5 - 0.0) \\ &= 0.026 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.7635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.7635 mm

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Input, Nozzle Desc: Noz K1A

From: 80

Pressure for Reinforcement Calculations	P	23.504	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	17.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	650.0000	mm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm

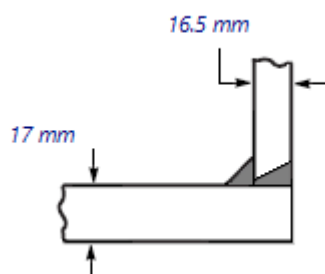
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K1A

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.





Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 53.8 \cdot (\exp(23.504/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 0.4186 + 1.5 + 0.0 = 1.9186 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
 $= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	21.610 mm
Min. required shell thickness, external p + c :	8.416 mm

Nozzle neck minimum required thickness :	6.300 mm
--	----------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.4, 1]$
 $= 0.922$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.5 - 1.5$
 $= 15.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
 $= 0.9/6 \cdot 2303.0 [2 + (2303.0/(2 \cdot 576.5))^2]$
 $= 2069.107 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{2069.107 \cdot 15.5}, 2 \cdot 26.9)$
 $= 53.800 \text{ mm}$

Because the nozzle is set-on, add tn to the result [LR]:
 $= 68.800 \text{ mm}$

Intermediate value [Cn]:
 $= \min(((t + t_e)/t_n)^{0.35}, 1)$
 $= \min(((15.5 + 0.0)/15.0)^{0.35}, 1)$
 $= \min(1.012, 1)$
 $= 1.000$

Angle for non-radial nozzles in Elliptical Heads [theta]:
 $= \text{atan}((h/R) \cdot (DR/(\sqrt{R^2 - DR^2})))$
 $= \text{atan}((576.5/1151.5) \cdot (650.0/(\sqrt{1151.5^2 - 650.0^2})))$
 $= 0.330 \text{ radians}$

Intermediate value [Cp]:
 $= \exp((0.35 \cdot d_i - X_o)/(16t))$
 $= \exp((0.35 \cdot 2303.0 - 689.641)/(16 \cdot 15.5))$
 $= 1.599$



Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1}) \\ &= \min(0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 15.0)}, 200.0) \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot f_{rp}) / (LR - t_n)) \\ &= 15.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 15.500 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 15.5 \cdot 68.8 \\ &= 10.664 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43} \\ &= 10.664 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 13.943 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 15.5 / \ln[1 + 15.5 / 2069.107] \\ &= 2076.859 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.408 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(LH + t) \\ &= 23.504 \cdot 33.848(20.087 + 15.5) \\ &= 283.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot LR) / 2 \\ &= (23.504 \cdot 2076.859 \cdot 68.8) / 2 \\ &= 16792.2 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (23.504 \cdot 2076.859 \cdot 28.557) / 2 \\ &= 6970.1 \text{ kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (16792.2 + 6970.1 + 283.1) / 23.504 \\ &= 1023.0 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff})) \\ &= 2475.612 / (2 \cdot 1023.034/13.943 - 2076.859/(2 \cdot 15.5)) \\ &= 31.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot \text{S}/[\text{t}/\text{Rxs}] \\ &= 2 \cdot 1650.408[15.5/2076.859] \\ &= 24.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(31.043, 24.635) \\ &= 24.635 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (283.118 + 16792.164 + 6970.076) / 13.943 \\ &= 1724.541 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= \text{P} \cdot \text{Rxs} / (2 \cdot \text{teff}) \\ &= 23.504 \cdot 2076.859 / (2 \cdot 15.5) \\ &= 1574.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1724.541 - 1574.661, 1574.661) \\ &= 1874.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1874.42	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	24.63	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{(((\text{Rnc} + \text{Tn})^2 + (\text{Rn} + \text{Tn})^2)/2)} \\ &= \pi/2 \cdot \sqrt{(((28.557 + 15.0)^2 + (26.9 + 15.0)^2)/2)} \\ &= 67.131 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(6970 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.578 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 133.578 / (67.131(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 57.178 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= \text{tn} / \ln[1 + \text{tn}/\text{Rn}] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$



$$= 15.5 / \ln[1 + 15.5 / 2069.107]$$
$$= 2076.859 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 1650.408$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t)$$
$$= 1.055 \cdot 33.848 (20.087 + 15.5)$$
$$= 12.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= (P \cdot R_{xs} \cdot LR) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 68.8) / 2$$
$$= 753.5 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 28.557) / 2$$
$$= 312.7 \text{ kgf}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (753.5 + 312.7 + 12.7) / 1.055$$
$$= 1023.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$
$$= 2475.612 / (2 \cdot 1023.034 / 13.943 - 2076.859 / (2 \cdot 15.5))$$
$$= 31.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / R_{xs}]$$
$$= 2 \cdot 1650.408 [15.5 / 2076.859]$$
$$= 24.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(31.043, 24.635)$$
$$= 24.635 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$
$$= (12.703 + 753.451 + 312.742) / 13.943$$
$$= 77.379 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$
$$= 1.055 \cdot 2076.859 / (2 \cdot 15.5)$$
$$= 70.7 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 77.379 - 70.654, 70.654)$$
$$= 84.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	84.10	kgf/cm ²
Maximum Allowable External Pressure	Pmax	24.63	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated



Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.388 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 14.2524 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.1376 mm

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Input, Nozzle Desc: Noz K1B

From: 80

Pressure for Reinforcement Calculations	P	23.634	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	17.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	650.0000	mm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm

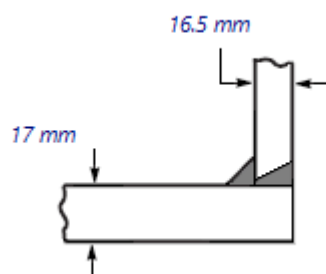
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K1B

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.





Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 53.8 \cdot (\exp(23.634/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 0.4209 + 1.5 + 0.0 = 1.9209 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
 $= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	21.708 mm
Min. required shell thickness, external p + c :	8.416 mm

Nozzle neck minimum required thickness :	6.300 mm
--	----------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.4, 1]$
 $= 0.922$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.5 - 1.5$
 $= 15.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
 $= 0.9/6 \cdot 2303.0 [2 + (2303.0/(2 \cdot 576.5))^2]$
 $= 2069.107 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{2069.107 \cdot 15.5}, 2 \cdot 26.9)$
 $= 53.800 \text{ mm}$

Because the nozzle is set-on, add tn to the result [LR]:
 $= 68.800 \text{ mm}$

Intermediate value [Cn]:
 $= \min(((t + t_e)/t_n)^{0.35}, 1)$
 $= \min(((15.5 + 0.0)/15.0)^{0.35}, 1)$
 $= \min(1.012, 1)$
 $= 1.000$

Angle for non-radial nozzles in Elliptical Heads [theta]:
 $= \text{atan}((h/R) \cdot (DR/(\sqrt{R^2 - DR^2})))$
 $= \text{atan}((576.5/1151.5) \cdot (650.0/(\sqrt{1151.5^2 - 650.0^2})))$
 $= 0.330 \text{ radians}$

Intermediate value [Cp]:
 $= \exp((0.35 \cdot d_i - X_o)/(16t))$
 $= \exp((0.35 \cdot 2303.0 - 689.641)/(16 \cdot 15.5))$
 $= 1.599$



Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1}) \\ &= \min(0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 15.0)}, 200.0) \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot f_{rp}) / (LR - t_n)) \\ &= 15.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 15.500 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 15.5 \cdot 68.8 \\ &= 10.664 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 \\ &= 10.664 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 13.943 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 15.5 / \ln[1 + 15.5 / 2069.107] \\ &= 2076.859 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.408 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(LH + t) \\ &= 23.634 \cdot 33.848(20.087 + 15.5) \\ &= 284.7 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot LR) / 2 \\ &= (23.634 \cdot 2076.859 \cdot 68.8) / 2 \\ &= 16885.0 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (23.634 \cdot 2076.859 \cdot 28.557) / 2 \\ &= 7008.6 \text{ kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (16885.0 + 7008.6 + 284.7) / 23.634 \\ &= 1023.0 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff})) \\ &= 2475.612 / (2 \cdot 1023.034/13.943 - 2076.859/(2 \cdot 15.5)) \\ &= 31.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot \text{S}/[\text{Rxs}] \\ &= 2 \cdot 1650.408[15.5/2076.859] \\ &= 24.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(31.043, 24.635) \\ &= 24.635 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (284.683 + 16885.0 + 7008.61) / 13.943 \\ &= 1734.075 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= \text{P} \cdot \text{Rxs} / (2 \cdot \text{teff}) \\ &= 23.634 \cdot 2076.859 / (2 \cdot 15.5) \\ &= 1583.4 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1734.075 - 1583.367, 1583.367) \\ &= 1884.8 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	1884.78	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	24.63	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{(((\text{Rnc} + \text{Tn})^2 + (\text{Rn} + \text{Tn})^2)/2)} \\ &= \pi/2 \cdot \sqrt{(((28.557 + 15.0)^2 + (26.9 + 15.0)^2)/2)} \\ &= 67.131 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(7009 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.6 \cdot 26.9^2 \cdot 1.0^2) \\ &= 134.317 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 134.317 / (67.131(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 57.494 < \text{or} = \text{to } 1650.408 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= \text{tn} / \ln[1 + \text{tn}/\text{Rn}] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$



$$= 15.5 / \ln[1 + 15.5 / 2069.107]$$
$$= 2076.859 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 1650.408$$
$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t)$$
$$= 1.055 \cdot 33.848 (20.087 + 15.5)$$
$$= 12.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= (P \cdot R_{xs} \cdot LR) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 68.8) / 2$$
$$= 753.5 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 28.557) / 2$$
$$= 312.7 \text{ kgf}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (753.5 + 312.7 + 12.7) / 1.055$$
$$= 1023.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$
$$= 2475.612 / (2 \cdot 1023.034 / 13.943 - 2076.859 / (2 \cdot 15.5))$$
$$= 31.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / R_{xs}]$$
$$= 2 \cdot 1650.408 [15.5 / 2076.859]$$
$$= 24.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(31.043, 24.635)$$
$$= 24.635 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$
$$= (12.703 + 753.451 + 312.742) / 13.943$$
$$= 77.379 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$
$$= 1.055 \cdot 2076.859 / (2 \cdot 15.5)$$
$$= 70.7 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 77.379 - 70.654, 70.654)$$
$$= 84.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	84.10	kgf/cm ²
Maximum Allowable External Pressure	Pmax	24.63	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated



Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.39 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 14.2524 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.1376 mm

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Input, Nozzle Desc: MH1

From: 80

Pressure for Reinforcement Calculations	P	23.569	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1650.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	17.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	21.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	710.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.2000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

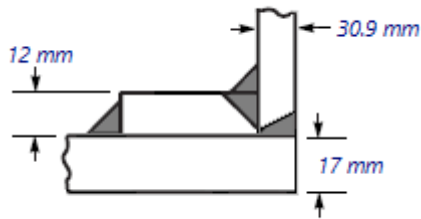
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	21.564	in.
Actual Thickness Used in Calculation	1.218	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 550.726 \cdot (\exp(23.569/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.2967 + 1.5 + 0.0 = 5.7967 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 29.437 + 550.726) / 550.726) - 0.0$$

$$= 154.606 - 0.0 = 154.6062 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	21.659 mm
Min. required shell thickness, external p + c :	8.416 mm

Nozzle neck minimum required thickness	: 9.840 mm
--	------------

Minimum nozzle neck thickness 27.076 mm , is greater than required thickness 9.840 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.4, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.4, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 30.937 - 1.5$$

$$= 29.437 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$$

$$= 0.9/6 \cdot 2303.0 [2 + (2303.0/(2 \cdot 576.5))^2]$$

$$= 2069.107 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, L_r is $\min[LR1, LR2, LR3]$:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(2069.107 \cdot 15.5) + 50.2}$$

$$= 229.284 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((Reff + t) \cdot (t + t_e))}$$

$$= \sqrt{((2069.107 + 15.5) \cdot (15.5 + 12.0))}$$



$$= 239.430 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 275.363$$

$$= 550.726 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(229.284, 239.43, 550.726)$$

$$= 229.284 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 258.721 \text{ mm}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((15.5 + 12.0)/29.437)^{0.35}, 1)$$

$$= \min(0.976, 1)$$

$$= 0.976$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot d_i - X_o) / (16t))$$

$$= \exp((0.35 \cdot 2303.0 - 304.8) / (16 \cdot 15.5))$$

$$= 7.547$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1})$$

$$= \min(12.0 + 0.976 \cdot \sqrt{(275.363 \cdot 29.437)}, 305.0)$$

$$= 99.913 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot f_{rp}) / (LR - t_n))$$

$$= 15.5 + ((6.024 \cdot 1) / (258.721 - 29.437))$$

$$= 18.127 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 15.5 \cdot 258.721$$

$$= 40.102 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 29.437 \cdot 99.913$$

$$= 29.412 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}_{42}^2$$

$$= 0.5 \cdot 12.0^2$$

$$= 0.720 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}_{41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, (LR - t_n) \cdot t_e)$$

$$= \min(50.2 \cdot 12.0, (258.721 - 29.437) \cdot 12.0)$$

$$= 6.024 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 + f_{rp}(A5)$$

$$= 40.102 + 0.922(29.412 + 0.0) + 0.5 + 0.72 + 0.0 + 1 \cdot (6.024)$$

$$= 74.473 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$



$$= 29.437 / \ln[1 + 29.437 / 275.363]$$

$$= 289.832 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$$

$$= 18.127 / \ln[1 + 18.127 / 2069.107]$$

$$= 2078.158 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1650.408 \cdot 1.0$$

$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t)$$

$$= 23.569 \cdot 289.832 (99.913 + 15.5)$$

$$= 7883.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= (P \cdot R_{xs} \cdot LR) / 2$$

$$= (23.569 \cdot 2078.158 \cdot 258.721) / 2$$

$$= 63360.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (23.569 \cdot 2078.158 \cdot 275.363) / 2$$

$$= 67436.3 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (63360.8 + 67436.3 + 7883.9) / 23.569$$

$$= 5884.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 2475.612 / (2 \cdot 5884.061 / 74.473 - 2078.158 / (2 \cdot 18.127))$$

$$= 24.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / t / R_{xs}$$

$$= 2 \cdot 1650.408 / 15.5 / 2078.158$$

$$= 24.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(24.585, 24.619)$$

$$= 24.585 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (7883.916 + 63360.844 + 67436.312) / 74.473$$

$$= 1862.168 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 23.569 \cdot 2078.158 / (2 \cdot 18.127)$$

$$= 1351.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1862.168 - 1351.002, 1351.002)$$

$$= 2373.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2373.33	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	24.58	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

= 1.000 For set-on Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

$= \pi/2 \cdot (R_n + t_n)$
 $= \pi/2 \cdot (275.363 + 29.437)$
 $= 478.779 \text{ mm}$

Weld Length of Pad to Shell Weld [LtP]:

$= \pi/2 \cdot (R_n + t_n + W)$
 $= \pi/2 \cdot (275.363 + 29.437 + 50.2)$
 $= 557.633 \text{ mm}$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 8.485, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 29.437, 6.000)$ Passed
tf1, L42T = 8.485, must be $\geq \min(0.6 \cdot 12.000, 0.6 \cdot 15.500)$ Passed

Weld Load Value [fwelds]:

$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(67436 \cdot 1.0, 1.5 \cdot 1522.2(30.036 + 0.0), \pi/4 \cdot 23.6 \cdot 275.36^2 \cdot 1.0^2)$
 $= 14035.904 \text{ kgf}$

Discontinuity Force [fws]:

$= \text{fwelds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$
 $= 14035.9 \cdot 15.5 \cdot 1650 / (15.5 \cdot 1650 + 12.0 \cdot 1650)$
 $= 7911.153 \text{ kgf}$

Discontinuity Force [fwp]:

$= \text{fwelds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$
 $= 14035.9 \cdot 12.0 \cdot 1650 / (15.5 \cdot 1650 + 12.0 \cdot 1650)$
 $= 6124.750 \text{ kgf}$

Shear Stress [τ_1]:

$= \text{fws} / (L_T \cdot (0.6 \cdot \text{tw}_1 + 0.49 \cdot L_{43T}))$
 $= 7911.153 / (478.779 \cdot (0.6 \cdot 15.5 + 0.49 \cdot 0.0))$
 $= 177.673 \text{ kgf/cm}^2$

Shear Stress [τ_2]:

$= \text{fwp} / (L_T \cdot (0.6 \cdot \text{tw}_2 + 0.49 \cdot L_{41T}))$
 $= 6124.75 / (478.779 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071))$
 $= 119.950 \text{ kgf/cm}^2$

Shear Stress [τ_3]:

$= \text{fwp} / (L_{TP} \cdot (0.49 \cdot L_{42T}))$
 $= 6124.75 / (557.633 \cdot (0.49 \cdot 8.485))$
 $= 264.169 \text{ kgf/cm}^2$

Maximum Shear Stress in the Welds:

$= \max(\tau_1, \tau_2, \tau_3)$
 $= \max(177.673, 119.95, 264.169)$
 $= 264.2$ must be less than or equal to 1650.4 kgf/cm²

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$= t_n / \ln[1 + t_n/R_n]$
 $= 29.437 / \ln[1 + 29.437/275.363]$
 $= 289.832 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:

$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 18.127 / \ln[1 + 18.127/2069.107]$
 $= 2078.158 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:

$= 1.5 \cdot S$
 $= 1.5 \cdot 1650.408$



$$= 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn(LH + t) \\ &= 1.055 \cdot 289.832(99.913 + 15.5) \\ &= 352.8 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= (P \cdot Rxs \cdot LR) / 2 \\ &= (1.055 \cdot 2078.158 \cdot 258.721) / 2 \\ &= 2835.1 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= (P \cdot Rxs \cdot Rnc) / 2 \\ &= (1.055 \cdot 2078.158 \cdot 275.363) / 2 \\ &= 3017.5 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (2835.1 + 3017.5 + 352.8) / 1.055 \\ &= 5884.1 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff)) \\ &= 2475.612 / (2 \cdot 5884.061 / 74.473 - 2078.158 / (2 \cdot 18.127)) \\ &= 24.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S[t/Rxs] \\ &= 2 \cdot 1650.408[15.5/2078.158] \\ &= 24.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(24.585, 24.619) \\ &= 24.585 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (352.77 + 2835.112 + 3017.47) / 74.473 \\ &= 83.324 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / (2 \cdot teff) \\ &= 1.055 \cdot 2078.158 / (2 \cdot 18.127) \\ &= 60.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 83.324 - 60.451, 60.451) \\ &= 106.2 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	106.20	kgf/cm ²
Maximum Allowable External Pressure	Pmax	24.58	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 1 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 1 °C
Temperature reduction per Graph 3.12 : 0 °C
The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (13.726 • 1.0)/(12.0 - 1.5 - 0.0)
= 1.000

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 2 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : 7 °C
Temperature reduction per Graph 3.12 : 5 °C
The computed stress ratio is : 0.886

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (13.726 • 1.0)/(17.0 - 1.5 - 0.0)
= 0.886

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the Nozzle : 2 °C
Gov. MDMT of the Reinforcement Pad : 1 °C
Gov. MDMT of the nozzle to shell joint welded assembly : 2 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -85 °C

The Drop for this Nozzle is : 22.4143 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 344.4142 mm

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Input Echo, Tubesheet Number 1, Description: U-TUBE SHEET

Shell Data:

Main Shell Description: SH.1

Shell Maximum Design Pressure	Psd,max	23.45	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	23.45	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1200.000	mm
Shell Temperature for Internal Pressure	Ts	80.00	°C
Shell Material		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1650.41	kgf/cm ²
Shell Allowable Stress at Ambient		1784.41	kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	163.16	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	163.16	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	68.0000	mm
Channel Internal Corrosion Allowance	cac	1.5000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1200.000	mm
Channel Design Temperature	TEMPC	80.00	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1650.41	kgf/cm ²
Channel Allowable Stress at Ambient		1784.41	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1066	
Tube Wall Thickness	et	2.5000	mm
Tube Outside Diameter	D	25.0000	mm
Total Straight Tube Length	Lt	8297.00	mm
Straight Tube Length (bet. inner tubsht faces) L		7987.00	mm
Design Temperature of the Tubes		80.00	°C
Tube Material		SA-334 1	
Tube Material UNS Number		K03008	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1308.09	kgf/cm ²
Tube Allowable Stress At Ambient		1407.13	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1962.14	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	32.0000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.5000	mm
Groove Weld Leg	ag	2.5000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	584.500	mm
Largest Center-to-Center Tube Distance	Ul	47.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	8.0000	mm

Tubesheet Data:



Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 80.00 °C
 Tubesheet Material SA-350 LF2

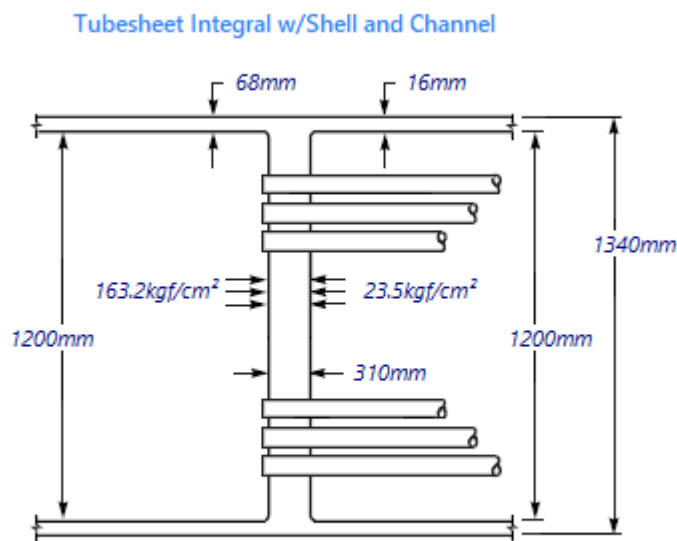
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 1568.83 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1682.44 kgf/cm²
 Thickness of Tubesheet h 310.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 1.5000 mm
 Tubesheet Outside Diameter A 1340.000 mm

Area of the Untubed Lanes AL 549.4 cm²

Junction Stress Reduction option Analyze as Simply Supported
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²



TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci$ Equation 4.3.1
= $0.5 \cdot 20.0 \cdot (\exp(163.158/(1308.1 \cdot 1.0)) - 1) + 0$
= 1.3284 + 0
= 1.3284 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{Ro \cdot t}$
= $8571.5 / \sqrt{12.5 \cdot 2.5}$
= 1533.317

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
= $0.55 / (25.0/2.5)$
= 0.0550

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t/Do)$
= $1.6 \cdot 0.055 \cdot 2032634 (2.5/25.0)$
= 17887.176 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy
= $17887.2/1962.1$
= 9.116

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
= $1962.1/3866.9$
= 0.507

Ratio [AE]:

= Fhe / E
= $17887.2/2032634$
= 0.008800

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
= $(\ln(0.507) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
= 0.148

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
= $1962.1 (1 + 0.002) / \ln(1 + 0.002)^{0.147778}$
= 4926.163 kgf/cm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f^2 - R)$
= $0.6 (1.0 - 0.507)$
= 0.296

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
= $(3866.9 \cdot \exp(0.296)) / (0.296^{0.296})$
= 7450.251 kgf/cm²

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$



$$= 1.5 \cdot 0.5071.5 - 0.5 \cdot 0.5072.5 - 0.5073.5$$

$$= 0.357$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	1933	-2.084	0.312E-05	0.287E-05	0.912E-05	-0.792E-05	0.130E+06	0.605E-02
2	966	-4.924	0.573E-07	0.573E-07	0.175E-05	-0.175E-05	0.165E+07	0.821E-02
3	1450	-3.504	0.594E-06	0.592E-06	0.459E-05	-0.456E-05	0.586E+06	0.632E-02
4	1691	-2.794	0.145E-05	0.142E-05	0.663E-05	-0.644E-05	0.282E+06	0.279E-02
5	1812	-2.439	0.215E-05	0.207E-05	0.782E-05	-0.733E-05	0.192E+06	0.624E-03
6	1752	-2.617	0.177E-05	0.172E-05	0.721E-05	-0.690E-05	0.233E+06	0.128E-02
7	1782	-2.528	0.195E-05	0.189E-05	0.751E-05	-0.712E-05	0.212E+06	0.382E-03
8	1797	-2.484	0.205E-05	0.198E-05	0.766E-05	-0.723E-05	0.202E+06	0.107E-03
9	1789	-2.506	0.200E-05	0.193E-05	0.759E-05	-0.717E-05	0.207E+06	0.141E-03
10	1793	-2.495	0.203E-05	0.196E-05	0.762E-05	-0.720E-05	0.204E+06	0.181E-04
11	1795	-2.489	0.204E-05	0.197E-05	0.764E-05	-0.721E-05	0.203E+06	0.440E-04
12	1794	-2.492	0.203E-05	0.196E-05	0.763E-05	-0.721E-05	0.204E+06	0.129E-04
13	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.720E-05	0.204E+06	0.264E-05
14	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.721E-05	0.204E+06	0.512E-05
15	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.721E-05	0.204E+06	0.124E-05
16	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.720E-05	0.204E+06	0.699E-06

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 1794.25/1962.14$$

$$= 1.729$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 1794.252/1.729$$

$$= 1037.499 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 1037.499 (2.5/25.0)$$

$$= 207.500 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.3284 mm
Allowable Internal Pressure at Corroded thickness	291.89 kgf/cm ²
Required Internal Design Pressure	163.16 kgf/cm ²
Allowable External Pressure at Corroded thickness	207.50 kgf/cm ²
Required External Design Pressure	23.45 kgf/cm ²
Required Thickness due to Shell Side pressure	0.6257 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 584.5 + 25.0$$

$$= 1194.000 \text{ mm}$$



Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (32.0 - 25.0) / 32.0$$

$$= 0.219$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1200.0 / 1194.0$$

$$= 1.0050$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1200.0 / 1194.0$$

$$= 1.0050$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 310.0$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t(E_t/E) \cdot (S_t/S) \cdot (\rho); dt - 2t)$$

$$= \max(25.0 - 2 \cdot 2.5 (2032634/2032634) \cdot (1308/1568) \cdot (0); 25.0 - 2 \cdot 2.5)$$

$$= 25.0000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 32.0 / \sqrt{ (1 - 4 \cdot \min(549.43 ; 4 \cdot 1194.0 \cdot 32.0) / (\pi \cdot 1194.0^2)) }$$

$$= 32.8152 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.8152 - 25.0) / 32.8152$$

$$= 0.2382$$

$$h/p = 9.687500 ; \mu^* = 0.238159 ; E^* = 410040 \text{ kgf/cm}^2$$

$$E^*/E = 0.201728 ; v^* = 0.409245$$

Note: As h/p (9.688) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) / t_s) }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ ((1200.0 + 16.0) / 16.0) }$$

$$= 0.0130 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - v_s^2))$$

$$= 0.013 \cdot 2032634 \cdot 16.0^3 / (6 (1 - 0.3^2))$$

$$= 19872634.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (1200.0 + 16.0) \cdot 19872634 / 310.0^3 \cdot (1 + 310.0 \cdot 0.013 + (310.0 \cdot 0.013)^2 / 2)$$

$$= 64249.5 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2 (D_s + t_s)))$$

$$= 1200.0^2 / (4 \cdot 2032634 \cdot 16.0) \cdot (1 - 1200.0 \cdot 0.3 / (2 (1200.0 + 16.0)))$$

$$= 0.94308\text{E-}02 \text{ mm/kgf/cm}^2$$

Parameter [w_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$



$$= 1.005 \cdot 19872634 \cdot 0.013 \cdot 0.009431 (1 + 310.0 \cdot 0.013)$$

$$= 12372.2480 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{((1200.0 + 68.0)68.0)}$$

$$= 0.0062 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.006 \cdot 2032634 \cdot 68.0^3 / (6(1 - 0.3^2))$$

$$= 724660992.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1200.0 + 68.0 / \cos(0)) \cdot 0.72466E+09 / 310.0^3 \cdot (1.0 + 310.0 \cdot 0.006 \cdot (1.0 + 1.0) / 2 + (310.0 \cdot 0.006)^2 / 2 \cdot 1.0)$$

$$= 881020.8 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1200.0^2 / (4 \cdot 2032634 \cdot 68.0) \cdot (1 - 1200.0 \cdot 0.3 / (2(1200.0 + 68.0)))$$

$$= 0.22348E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.005 \cdot 0.72466E+09 [0.002235 (1.0 + 0.006 \cdot 1.0 \cdot 310.0) - 0 (1.0 / 310.0 + 1.0 \cdot 0.006 / 2)]$$

$$= 29413.7246 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1340.0 / 1194.0$$

$$= 1.1223$$

Coefficient [F]:

$$= (1 - v_s) / E_s \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 410040 \cdot (64249.48 + 881020 + 2032634 \cdot \ln(1.12))$$

$$= 1.6997$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1194.0^2 / 16 \cdot [(1.005 - 1) \cdot (1.005^2 + 1) \cdot 23.454 - (1.005 - 1) \cdot (1.005^2 + 1) \cdot 163.158]$$

$$= -125736.3906 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -125736.39 + 29413.725 \cdot 163.158 - 12372.248 \cdot 23.454$$

$$= 4383173.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M_s - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= (4383173) - 1194.0^2 / 32 \cdot 1.7 (23.45 - 163.16) / (1 + 1.7)$$

$$= 5542123.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$



Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t) \\ &= 5542123 + 1194.0^2/64 \cdot (3 + 0.409) \cdot (23.45 - 163.16) \\ &= -5067407.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(5542123; -5067407.5) \\ &= 5542123.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 5542123 / (0.2382 \cdot (310.0 - 8.0)^2) \\ &= 1530.8993 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1568.83 \\ &= 3137.67 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 5544243 / (0.2382 \cdot (219.1763 - 8.0)^2) \\ &= 3132.1055 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 219.1763 + 0 + 0 \\ &= 219.1763 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(219.1763; 152.224) \\ &= 219.1763 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |23.45 - 163.16| \\ &= 139.704 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.219 \cdot 310.0 / 1194.0 \cdot \min(0.8 \cdot 1568.835; 0.533 \cdot 2349.607) \\ &= 284.504 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.219) \cdot (1194.0 / 1252.341) \cdot |23.45 - 163.16| + 0 \\ &= 152.2240 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\ &= 1200.0^2 / (4 \cdot 16.0 \cdot (1200.0 + 16.0)) \cdot 23.454 \\ &= 433.9759 \text{ kgf/cm}^2 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot \\ &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 438116.31 / 16.0^2 \cdot [0.013 \cdot 0.2207 + 6(1 - 0.409) / 410040 \cdot \end{aligned}$$



$$1194.0/310.0^3 \cdot (1 + 310.0 \cdot 0.013/2) \cdot (5542123 + 1194.0^2/32 \cdot (23.45 - 163.16)) \\ = 1010.353 \text{ kgf/cm}^2$$

Shell Membrane + Bending Stress [σ_s]:
= $|\sigma_{s,m}| + |\sigma_{s,b}|$ (Should be $\leq$ SPS,s)
= 1444.328 kgf/cm² (Should be $\leq$ 4955.82)

Axial Channel Membrane Stress [$\sigma_{c,m}$]:
= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
= $1200.0^2 / (4 \cdot 68.0 (1200.0 + 68.0) \cdot \cos(0)) \cdot 163.158$
= 681.213 kgf/cm²

Axial Channel Bending Stress [$\sigma_{c,b}$]:
= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
= $6 \cdot 15976031 / 68.0^2 \cdot [(0.002235 \cdot 1.0 \cdot 0.006 - 0 \cdot 1.0) \cdot 163.158 - 6((1 - 0.409) / 410040) \cdot 1194.0 / 310.0^3 \cdot (1.0 + 0.006 \cdot 310.0 / 2 \cdot 1.0) \cdot (5542123 + 1194.0^2 / 32 \cdot (23.454 - 163.158))]$
= 2557.854 kgf/cm²

Channel Membrane + Bending Stress [σ_c]:
= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq$ SPS,c)
= 3239.068 kgf/cm² (Should be $\leq$ 4955.82)

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:
 $\beta_s, k_s, \lambda_s, \delta_s$ are all = 0

Intermediate Channel side parameters for the Simply Supported condition:
 $\beta_c, k_c, \lambda_c, \delta_c$ are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:
= $(1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
= $(1 - 0.41) / 410040 \cdot (0 + 0 + 2032634 \cdot \ln(1.12))$
= 0.3378

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s, P_t) [Mts]:
= $D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$
= $1194.0^2 / 16 \cdot [(1.005 - 1) \cdot (1.005^2 + 1) \cdot 23.454 - (1.005 - 1) \cdot (1.005^2 + 1) \cdot 163.158]$
= -125736.3906 kgf/cm²•mm²

Moment acting on unperforated tubesheet rim [M*]:
= $M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$
= -125736.39 + 0 • 163.158 - 0 • 23.454
= -125736.3906 kgf/cm²•mm²

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:
= $M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$
= $1477697 + 1194.0^2 / 64 \cdot (3 + 0.409) \cdot (23.45 - 163.16)$



$$= -9131834.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= |M_o| \\ &= |-9131834.0| \\ &= 9131834.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 9131834 / (0.2382 (310.0 - 8.0)^2) \\ &= 2522.4839 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1568.83 \\ &= 3137.67 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 9131834 / (0.2382 (278.7825 - 8.0)^2) \\ &= 3137.6245 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 278.7825 + 0 + 0 \\ &= 278.7825 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(HreqB; HreqS) \\ &= \max(278.7825; 152.224) \\ &= 278.7825 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$\begin{aligned} &= |Ps - Pt| \\ &= |23.45 - 163.16| \\ &= 139.704 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\ &= 4 \cdot 0.219 \cdot 310.0 / 1194.0 \cdot \min(0.8 \cdot 1568.835; 0.533 \cdot 2349.607) \\ &= 284.504 \text{ kgf/cm}^2 \end{aligned}$$

|Ps - Pt| is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (Do / \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot |Ps - Pt| + Cats + Catc \\ &= 1 / (4 \cdot 0.219) \cdot (1194.0 / 1252.341) \cdot |23.45 - 163.16| + 0 \\ &= 152.2240 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned} &= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps \\ &= 1200.0^2 / (4 \cdot 16.0 (1200.0 + 16.0)) \cdot 23.454 \\ &= 433.9759 \text{ kgf/cm}^2 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned} &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu) / E \cdot \\ &\quad Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\ &= 6 \cdot 0 / 16.0^2 \cdot [0 \cdot 0.2207 + 6(1 - 0.409) / 410040 \cdot \\ &\quad 1194.0 / 310.0^3 \cdot (1 + 310.0 \cdot 0 / 2) \cdot (1477697 + 1194.0^2 / 32 \cdot \\ &\quad (23.45 - 163.16))] \\ &= -0.000 \text{ kgf/cm}^2 \end{aligned}$$



Shell Membrane + Bending Stress [σ_s]:
= $|\sigma_{s,m}| + |\sigma_{s,b}|$ (Should be $\leq$ SPS,s)
= 433.976 kgf/cm² (Should be $\leq$ 4955.82)

Axial Channel Membrane Stress [$\sigma_{c,m}$]:
= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
= $1200.0^2 / (4 \cdot 68.0 (1200.0 + 68.0) \cdot \cos(0)) \cdot 163.158$
= 681.213 kgf/cm²

Axial Channel Bending Stress [$\sigma_{c,b}$]:
= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
= $6 \cdot 0 / 68.0^2 \cdot [(0 \cdot 1.0 \cdot 0 - 0 \cdot 1.0) \cdot 163.158 - 6((1 - 0.409) / 410040) \cdot 1194.0 / 310.0^3 \cdot (1.0 + 0 \cdot 310.0 / 2 \cdot 1.0) \cdot (1477697 + 1194.0^2 / 32 \cdot (23.454 - 163.158))]$
= 0.000 kgf/cm²

Channel Membrane + Bending Stress [σ_c]:
= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq$ SPS,c)
= 681.213 kgf/cm² (Should be $\leq$ 4955.82)

Analysis Finished.

Tube Metal Area [At]:
= $\pi (d_o - t) t$
= $\pi (25.0 - 2.5) 2.5$
= 1.767 cm²

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:
= $\pi \cdot t (d_o - t) S$
= $\pi \cdot 2.5 (25.0 - 2.5) 1308.09$
= 2311.583 kgf

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
= $0.85 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1308.09$
= 2311.5833 kgf

Weld Strength Factor [fw]:
= S / S_w where $S_w = \min(S \text{ and } S_t)$
= $1308.09 / 1308.09$
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
= $0.55 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1308.09$
= 1507.2808 kgf

Minimum Length of Weld [ar]:
= $2(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o)$
= $2(\sqrt{(0.75 \cdot 25.0)^2 + 1.07 \cdot 2.5 \cdot (25.0 - 2.5) 1.0 \cdot 1.0} - 0.75 \cdot 25.0)$
= 3.083 mm

Minimum Required Fillet Weld Leg [afr]: 1.5416 mm
Minimum Required Groove Weld Leg [agr]: 1.5416 mm

Tube-Tubesheet Jt allowable, 2311.58 is $\geq$ tube strength 2311.58 kgf

Note: This tube-tubesheet joint is a Full Strength joint



Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	2522.5 <=	3137.7 kgf/cm ²	Ok
Tubesheet shear stress	615.0 <=	1252.3 kgf/cm ²	Ok
Stress in Shell at Tubesheet	434.0 <=	4955.8 kgf/cm ²	Ok
Stress in Channel at Tubesheet	681.2 <=	4955.8 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	278.783	310.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.542	2.500	Ok
Tube-Tubesheet Groove Weld Leg :	1.542	2.500	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	300.631 ...	2946	3138	718 1252	Ps+Pt D1	Ok
D2uc	118.958 ...	423	3138	103 1252	Ps+Pt D2	Ok
D3uc	278.783 ...	2522	3138	615 1252	Ps+Pt D3	Ok
D1c	302.888 ...	2991	3138	725 1252	Ps+Pt-c D1	Ok
D2c	120.741 ...	430	3138	104 1252	Ps+Pt-c D2	Ok
D3c	280.983 ...	2561	3138	621 1252	Ps+Pt-c D3	Ok
Max:	302.8875 ... mm	0.953	0.579	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		...	4956	681	4956	Ok
D2uc		434	4956	...	4956	Ok
D3uc		434	4956	681	4956	Ok
D1c		...	4956	699	4956	Ok
D2c		481	4956	...	4956	Ok
D3c		481	4956	699	4956	Ok
Max		0.097	0.141			

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:



Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	302.888	310.000	Ok
Tube thickness :	1.328	2.500	Ok
Tube-tubesheet fillet weld leg :	1.542	2.500	Ok
Tube-tubesheet groove weld leg :	1.542	2.500	Ok

Given Shell Thickness: 16.0000 mm
 Given Channel Thickness: 68.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	171.165	1.000	171.165	1.000
Tubesheet Shear Stress	281.749	1.000	281.749	1.000
Shell Stress (Axial, Junction)	281.749	1.000	54.935	1.000
Tubesheet-Channel Junction	227.490	1.000	241.813	1.000
Tube Pressure Stress	291.891	1.000	207.499	1.000
Minimum MAWP	171.165		54.935	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	186.358	1.000	186.358	1.000
Tubesheet Shear Stress	305.770	1.000	305.770	1.000
Shell Stress (Axial, Junction)	305.770	1.000	65.506	1.000
Tubesheet-Channel Junction	255.197	1.000	289.311	1.000
Tube Pressure Stress	313.992	1.000	219.790	1.000
Minimum MAPnc	186.358		65.506	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(310.0/4, \min(310.0, 16.0))$$

$$= 77.500 \text{ mm}$$

MDMT reduction ratio:

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(23.45/65.51, 1)$$

$$= 0.358$$

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 14 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.7M- Non PWHT, the MDMT is : 40 °C
 Temperature reduction per Graph 3.12 : 26 °C
 The computed stress ratio is : 0.358

Warning: Impact Testing Is Mandatory!

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:



= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(310.0/4, min(310.0, 68.0))
= 77.500 mm

MDMT reduction ratio:
= min(pt/Tubeside MAPnc, 1)
= min(163.16/186.36, 1)
= 0.876

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : 11 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : 17 °C
Temperature reduction per Graph 3.13 : 6 °C
The computed stress ratio is : 0.876

Warning: Impact Testing Is Mandatory!

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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